Probabilistic Narrative Identification: Evidence from Tax Policy*

Tom Pesso†

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[Latest version here]

Abstract

This paper generalizes narrative identification by treating ambiguity in the historical record as identifying information. Traditional narrative approaches resolve documentary evidence through a binary classification, treating selected policy actions as fully exogenous. Instead, I interpret narrative evidence as probabilistic support for exogeneity. Identification proceeds at the level of reduced-form moments: variation in narrative support disciplines outcome–policy comovement, allowing the moment associated with exogenous policy variation to be recovered. I construct the support measure with a document-grounded language-model procedure that retrieves policy-specific passages, extracts stated motivations, and maps those motivations into exogeneity probabilities.

Empirically, I first evaluate the measure against the benchmark classifications of Romer and Romer (2010) and Cloyne (2013). I then construct a new dataset of routine U.S. federal tax proposals from 1950 to 2024. The resulting aggregate tax responses are contractionary, closely aligned with the canonical narrative tax literature, and operate mainly through investment. Finally, I construct a new narrative dataset of French VAT proposals from 1960 to 2024 and combine it with UK VAT changes from Cloyne's dataset. The pooled UK–France evidence provides the indirect-tax counterpart: VAT policy increases contract output mainly through private consumption and generate a short-lived rise in inflation, consistent with tax pass-through.

Keywords: Narrative identification; fiscal policy; tax shocks; causal inference.
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†Universitat Pompeu Fabra (UPF), Barcelona School of Economics (BSE). Contact: `tom.pesso@upf.edu`

Introduction

Estimating the macroeconomic effects of fiscal policy is difficult because policy actions respond to current and anticipated economic conditions. A large literature addresses this identification problem using restrictions on the dynamics of macroeconomic variables or external identifying information, including instruments and narrative evidence (V. Ramey, 2016). Narrative identification exploits contemporaneous historical records, including policymakers' stated motivations, to isolate policy actions that are plausibly exogenous to macroeconomic conditions. This approach has been applied across a wide range of settings, including monetary policy, taxation, government spending, and oil-market disruptions (Friedman & Schwartz, 1963; Hamilton, 1985; Mertens & Ravn, 2012; Ramey, 2011; Romer & Romer, 1989, 2010). Its implementation, however, requires researchers to judge whether the available historical evidence is sufficiently convincing to treat a given policy action as exogenous to contemporaneous and anticipated macroeconomic conditions, even when the documentary record is mixed, incomplete, or ambiguous.

Traditional narrative designs resolve this uncertainty through a binary inclusion decision: actions judged sufficiently credible are retained for identification, while the remainder are excluded. This requires ambiguity in the documentary record to be resolved before the policy actions are used for identification and therefore creates an inherent trade-off between coverage and the credibility of the exogeneity claim. A demanding standard for that claim retains only the policy actions for which the documentary evidence is most compelling but produces a sparse shock series; a more permissive standard yields a denser series at the cost of admitting actions for which the evidence supporting exogeneity is weaker and the resulting classification is more contestable.

Instead of reducing documentary evidence to a binary classification, this paper exploits variation across policy actions in the strength of narrative support for exogeneity. Conceptually, the framework retains two underlying types, an exogenous type and an endogenous policy-response type, but does not assume that the researcher observes which type a given policy action belongs to. The documentary record instead provides probabilistic narrative support p for membership in the exogenous type. Intermediate values of p therefore represent uncertainty about exogeneity rather than partial exogeneity. The identifying argument requires this support measure to be calibrated in population: among actions assigned support p, the fraction belonging to the exogenous type is p. Traditional narrative identification emerges as a special case obtained by choosing a threshold for p, which reintroduces a potentially contestable credibility cutoff for the exogeneity claim.

As shown in Section 1, exploiting variation in narrative support for identification requires one additional restriction. Conditional on an action's underlying type, narrative support p must not systematically sort actions with different outcome-policy comovement. Under calibration and this restriction, the outcome-policy moment at each level of p is a different weighted average of the moments associated with exogenous policy variation and endogenous policy responses. Variation in narrative support therefore changes these weights and identifies the moment associated with exogenous policy variation. Identification operates at the level of moments, so no individual policy action need be classified as exogenous with certainty.

The framework therefore requires a measure of narrative support for exogeneity constructed from contemporaneous policy documents. Conventional narrative identification applies broad coding principles across many policy actions, requiring repeated judgments about how the documentary evidence should be interpreted. The construction used here instead concentrates these judgments in a limited set of explicit choices that can be scrutinized and varied systematically. In practice, embedding models locate relevant passages, and open-weight large language models extract stated motivations from them. I then use a multinomial logit to map hidden-state representations of these motivations into probabilities of exogeneity. Conditional on the specified choices, the resulting measure is replicable and remains traceable to the passages on which it is based, allowing the underlying evidence to be manually audited and subjected to systematic diagnostics.

Because narrative support is estimated from text rather than observed, I treat it as generated measurement rather than raw data (Battaglia et al., 2024). Since identification operates through moments indexed by narrative support, residual measurement error enters the analysis through the identifying moments rather than as a separate classification problem. This perspective connects the framework to a broader econometric literature on measurement error, and imperfect moment conditions. The empirical analysis reports variation across calibration samples, representation models, and repeated motivation extractions as diagnostics of measurement stability. It then illustrates two possible approaches to assessing the implications of residual measurement error. Corrected projected moments provide one benchmark when the remaining errors can be approximated as classical (Evdokimov & Zeleneev, 2022). Approximate-moment sensitivity analysis provides a complementary benchmark by assessing robustness to departures from the maintained identifying moment restrictions (Armstrong & Kolesár, 2021).

I then apply the identification framework and document-based measurement procedure to the macroeconomic effects of tax policy. The empirical analysis proceeds in three stages. First, I evaluate the document-grounded support measure against established narrative classifications. Second, I use the Romer–Romer tax narrative to illustrate the sensitivity of binary identification to the credibility threshold for exogeneity, before turning to a much denser U.S. dataset where narrative support can be used directly for identification. Third, I apply the framework to VAT, where sparse policy variation and mixed motivations make binary narrative identification especially difficult.

The first stage evaluates whether the estimated support measure captures the information embedded in conventional narrative classifications. I compare the estimated probabilities with the narrative classifications of Romer and Romer (2010) and Cloyne (2013). In both samples, policy actions classified as exogenous receive higher narrative support on average, indicating that the document-grounded measure recovers a systematic signal contained in the historical record. In the Romer–Romer sample, however, actions classified as exogenous receive average support of only 0.60, indicating that the documentary evidence remains ambiguous for many actions retained by the binary classification.

To remain close to the traditional binary design and illustrate the consequences of the credibility threshold, I impose alternative cutoffs on the estimated support probabilities. The resulting macroeconomic response depends importantly on where this threshold is set: some cutoffs recover a contraction close to the Romer–Romer benchmark, while more demanding cutoffs remove influential intermediate-support episodes and reverse the medium-run estimate. I then dispense with the threshold and use variation in narrative support directly to identify the exogenous-policy moment. This relaxes the requirement that individual actions be judged sufficiently credible for binary inclusion, but the original Romer–Romer sample is too sparse to exploit variation in support precisely.[1]

[1]This does not mean that the framework generally requires more policy variation than a binary narrative design. A binary design can retain more observations by lowering the credibility threshold for exogeneity, at the cost of a weaker exogeneity claim. The proposed framework instead uses the full set of policy actions but separately estimates the

This motivates the construction of a new, much denser U.S. tax dataset. I collect more than 1,000 routine federal tax proposals from the annual budget process, largely outside the original Romer–Romer sample.[2] The resulting series contains roughly three times as many nonzero quarters as the Romer–Romer series and provides substantially more variation in narrative support. This variation allows the moment associated with exogenous policy variation to be recovered from how outcome–tax comovement changes with narrative support.

The estimates imply that proposed tax increases contract output and lower inflation, with the output contraction concentrated in investment. The estimated effects remain close to the Romer–Romer benchmark even when documentary ambiguity is retained as identifying information rather than resolved through a potentially contestable credibility threshold. A composition analysis attributes most of the contraction to individual and corporate income-tax proposals. These findings are stable across alternative motivation-extraction, representation, and calibration choices, including when these choices are varied jointly, and remain robust to departures from the maintained identifying moment restrictions.

The third stage turns to VAT, which provides a distinct form of taxation from the income-tax-heavy U.S. application. Consumption taxes operate through a different tax wedge than labor and capital income taxes (Trabandt & Uhlig, 2011), and recent evidence from temporary VAT changes points to consumer prices and household spending as important transmission channels (Bachmann et al., 2026). VAT is also a setting in which binary narrative identification is especially difficult. Policy changes are infrequent, and their documentary records often combine inflation, fiscal, structural, and harmonization motives. A stringent credibility threshold therefore retains only a small number of unambiguous episodes, whereas a more permissive threshold requires treating contestable cases as exogenous. I construct a new long-run narrative dataset of French VAT proposals from 1960 to 2024 and combine it with the UK VAT changes in Cloyne's dataset. The pooled UK–France estimates show that VAT increases raise prices on impact and contract output primarily through private consumption, providing a consumption-tax counterpart to the investment-heavy U.S. response.

Overall, the paper shows that ambiguity in the historical record need not be resolved before estimation. Instead, variation in narrative support can itself be exploited for identification. The document-grounded measurement procedure developed here makes this approach transparent and reproducible, while the new U.S. tax-proposal and French VAT-proposal datasets, together with the existing UK VAT narrative, show that the framework both recovers established findings and extends narrative methods to settings where binary credibility judgments would provide little identifying variation.

Related Literature

Narrative and Macroeconomic Identification. This paper relates first to the literature identifying policy shocks from time-series restrictions and the historical record. Structural VARs recover shocks through timing, sign, or other model-based restrictions (Blanchard & Perotti, 2002; Sims, 1980; Uhlig, 2005), while external-instrument methods use variables that are correlated with the target pol-

moments associated with exogenous policy variation and endogenous policy responses. Greater variation in narrative support therefore improves the precision with which these moments can be distinguished.

[2] The budget documents record tax proposals together with their contemporaneous rationales and reported implementation dates, following the spirit of Cloyne (2013). The analysis focuses on whether policy actions respond to macroeconomic conditions through their stated motivations. Anticipation between announcement and implementation is a distinct identification issue (Leeper et al., 2013; Ramey, 2011) and is not addressed here.

icy shock and orthogonal to other structural innovations (V. Ramey, 2016; Stock & Watson, 2018). Narrative approaches provide a distinct source of identifying information: contemporaneous records are used to construct policy-shock measures that can enter estimation directly or serve as external instruments. Foundational applications include monetary policy (Friedman & Schwartz, 1963; Romer & Romer, 1989, 2004), oil and government spending shocks (Hamilton, 1985; Ramey, 2011; Ramey & Shapiro, 1998), and tax changes (Cloyne, 2013; Romer & Romer, 2010). Recent work continues to extend narrative identification to new settings, including global shipping disruptions and historical tariff policy (Bini, 2025; den Besten & Känzig, 2026; Känzig & Raghavan, 2026). Related work combines sparse narrative shocks with additional time-series information (Barnichon & Mesters, 2025). This paper instead extracts additional identifying information from variation in the strength of documentary support across policy actions.

Text and AI in Economic Analysis. A growing literature transforms unstructured text into empirical measures of beliefs, expectations, policy intent, and economic narratives. Foundational surveys emphasize the need to define the text unit, target construct, representation, and validation design (Ash & Hansen, 2023; Gentzkow et al., 2019). Recent LLM-based work follows two broad approaches. One uses language models as scalable research assistants that produce labels, summaries, rationales, or structured fields (Clayton et al., 2025; Jamilov, 2025; Martell, 2025). The other retains model outputs such as hidden-state representations, classifier or token probabilities, and model disagreement as statistical signals that can be calibrated or aggregated into economic variables (Buckmann & Hill, 2025; Chen et al., 2026; Hartley, 2025).

This paper combines these roles. Language models recover stated motivations from fiscal documents, while a mapping estimated on a calibration sample translates model representations into policy-level probabilities of exogeneity. The resulting object is a probabilistic economic measurement rather than a model-generated label, and it remains traceable to the documentary evidence from which it is constructed.

AI-Generated Measurements and Imperfect Moments. Because narrative support is generated from text, the paper relates to recent work on inference with variables constructed from unstructured data (Battaglia et al., 2024; Carlson & Dell, 2025; Ludwig et al., 2025). This literature emphasizes that generated measurements may contain systematic error and develops corrections when validation data provide information about the target variable. Direct validation is more demanding here because the target is a continuous probability of latent exogeneity, for which existing binary expert classifications do not provide observed counterparts. I therefore consider two complementary approaches to residual measurement error. Corrected projected moments provide a benchmark when the remaining error can be approximated as classical (Evdokimov & Zeleneev, 2022), while approximate-moment sensitivity analysis assesses robustness to departures from the maintained identifying moments (Armstrong & Kolesár, 2021).

Tax Transmission and Composition. The empirical applications connect to work on the macroeconomic effects of tax changes. Narrative evidence shows that tax increases are contractionary (Cloyne, 2013; Mertens & Ravn, 2012; Romer & Romer, 2010), while subsequent work emphasizes differences across personal, corporate, and consumption taxes (Mertens & Ravn, 2013; Nguyen et al., 2021; Trabandt & Uhlig, 2011). The U.S. application extends this evidence to a denser set of routine federal tax measures and decomposes the response by tax instrument. The VAT application also relates to evidence on the identification and measurement of consumption-tax shocks (Riera-Crichton et al., 2016), VAT pass-through (Benzarti et al., 2020), and household consumption responses (Bachmann et al., 2026; Cashin & Unayama, 2016). The new French narrative proposal series, combined with UK VAT changes, links these indirect-tax margins to aggregate output, prices, and consumption over a long historical sample.

1 A Probabilistic Model of Narrative Identification

The objective of this paper is to estimate the dynamic causal effects of discretionary tax-policy actions on macroeconomic outcomes using narrative evidence from policy documents. Let $\Delta\tau_t$ denote the aggregate tax-policy change recorded in period t, and let y_{t+h} denote an outcome of interest at horizon $h \geq 0$. The structural relation of interest is

$$y_{t+h} = \beta_h \Delta\tau_t + \varepsilon_{t+h},$$

where β_h is the causal response of the outcome to exogenous tax variation and ε_{t+h} collects other macroeconomic disturbances.

A useful macroeconomic benchmark, also underlying the narrative tax classification in Cloyne (2013), is to view the recorded tax-policy measure as combining a policy innovation with a systematic policy response:

$$\Delta\tau_t = \xi_t + r(y_t, \pi_t, b_t, g_t).$$

Here $r(\cdot)$ summarizes systematic responses to economic activity, inflation, fiscal pressure, current spending decisions, and other variables entering the policy rule, while ξ_t denotes variation unrelated to those conditions. The representation is schematic: the analysis below does not specify or estimate $r(\cdot)$, nor does it require each recorded policy action to admit this additive decomposition. Its purpose is to organize the source of endogeneity that narrative evidence is intended to address.

Multiplying the structural relation by $\Delta\tau_t$ gives

$$\mathbb{E}(y_{t+h}\Delta\tau_t) = \beta_h\,\mathbb{E}(\Delta\tau_t^2) + \mathbb{E}(\Delta\tau_t\varepsilon_{t+h}).$$

Identification therefore requires $\mathbb{E}(\Delta\tau_t\varepsilon_{t+h}) = 0$ for the policy variation used in estimation. This covariance vanishes for exogenous policy variation but generally does not for policy actions generated by systematic policy responses, since the economic conditions inducing the policy action may also affect y_{t+h}.

Traditional narrative approaches use contemporaneous historical evidence to distinguish these sources of policy variation, classifying policy actions as either admissible or inadmissible for identification. This strategy has proved influential in empirical work on tax policy, as in Romer and Romer (2010), Mertens and Ravn (2013), and Cloyne (2013). Section 3.1 makes this connection explicit for the applications by mapping systematic policy responses into the motivation categories used to assess whether a policy action is plausibly orthogonal to current macroeconomic and fiscal conditions. The historical evidence underlying these classifications, however, is rarely all-or-nothing. A policy action may combine several motivations, and the available documents may support an exogenous interpretation more strongly for some actions than for others. The same issue arises for other narrative fiscal instruments, such as the defense-news shocks in Ramey and Zubairy (2018): geopolitical motivations may provide strong evidence against a domestic stabilization response without implying exogeneity with probability one.

This section treats variation in the strength of this evidence as identifying information. Narrative evidence is summarized by a continuous index $p_t \in [0, 1]$ measuring support for an exogenous interpretation of the policy action recorded in period t. Variation in p_t is used to distinguish the outcome–tax moment associated with exogenous policy variation from the moment associated with systematic policy responses. The framework therefore retains uncertainty about exogeneity rather than requiring it to be resolved before estimation: no individual policy action needs to be classified as exogenous with certainty. In the applications, p_t is constructed from policy documents using the language-model procedure in Sections 2 and 3; Subsection 1.5 considers measurement error in the resulting score.

1.1 Binary Narrative Identification

Formalizing the binary benchmark makes explicit the exogeneity restriction underlying traditional narrative identification, which maps documentary evidence into a binary inclusion decision. A policy action is retained for identification when the historical record is judged to provide sufficiently convincing evidence that it is unrelated to current or anticipated macroeconomic conditions; other policy actions are excluded. Let $\mathbb{1}\{\Delta\tau_t \text{ is classified as exogenous}\}$ denote this decision.

The binary narrative decision can be represented within the probabilistic framework introduced here as

$$\mathbb{1}\{\Delta\tau_t \text{ is classified as exogenous}\} = \mathbb{1}\{p_t \geq \bar{p}\},$$

for some evidentiary cutoff $\bar{p} \in (0, 1)$. This representation makes the credibility threshold underlying binary narrative identification explicit: continuous variation in narrative support is converted into a binary inclusion decision. Retained tax changes are then assumed to satisfy

$$\mathbb{E}[\varepsilon_{t+h}\Delta\tau_t \mid p_t \geq \bar{p}] = 0, \qquad h \geq 0.$$

The binary classification therefore translates a judgment about the strength of the documentary evidence into a zero-covariance restriction for the retained policy variation.

This representation highlights two implications of binary narrative identification. First, policy actions below the cutoff do not contribute identifying information, even when the historical record provides substantial support for an exogenous interpretation that falls short of the credibility threshold. Second, once the cutoff is crossed, residual uncertainty is set aside: retained actions are treated as providing orthogonal policy variation. The framework below relaxes both features by using variation in narrative support directly for identification.

1.2 Probabilistic Narrative Support and Policy-Response Moments

Let $\mathcal{D}_t$ denote the contemporaneous documentary record associated with the tax change in period t. I represent each tax change as belonging to one of two unobserved types, with $u_t = 1$ denoting the exogenous type and $u_t = 0$ the policy-response type.[3] In terms of the policy-rule benchmark above, the policy-response type corresponds to observations whose relevant variation reflects a systematic response to current or anticipated economic or fiscal conditions. The documentary record is used to construct a support index $p_t \in [0, 1]$ measuring the strength of the evidence that $u_t = 1$. The identifying argument requires this support score to be calibrated in population:

$$\mathbb{E}(u_t \mid p_t = p) = \Pr(u_t = 1 \mid p_t = p) = p. \tag{1}$$

Thus, among observations assigned support level p, the fraction belonging to the exogenous type is p in population.

Within-type moment stability and exclusion restriction. Calibration determines how narrative support maps into the weights placed on the exogenous and policy-response types. Identification additionally requires that narrative support does not systematically change the relevant outcome–tax moment within either type. For each horizon $h \geq 0$ and all p in the support of p_t,

$$\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = j, p_t = p) = \mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = j), \qquad j \in \{0, 1\}. \tag{2}$$

For the policy-response type, this restriction has a direct policy-rule interpretation. Narrative support may change the probability that an observation reflects exogenous policy variation rather than a systematic policy response, but it should not also systematically sort different policy responses with different outcome–tax moments. For example, suppose demand-management and spending-driven tax changes generate different outcome–tax moments. If their relative importance within the policy-response type varies systematically with p_t, then variation in narrative support would capture not only changes in the probability of exogeneity but also changes in the composition of endogenous policy responses. Equation (2) rules out this additional source of variation in the affine benchmark.[4]

The analogous restriction for the exogenous type rules out systematic sorting by tax magnitude, composition, implementation timing, anticipation, or other features that change the relevant outcome–tax moment within that type. Finally, the exogenous type satisfies the usual orthogonality restriction,

$$\mathbb{E}(\Delta\tau_t\varepsilon_{t+h} \mid u_t = 1) = 0, \qquad h \geq 0.$$

This is the macroeconomic exclusion restriction imposed on the latent policy variation that narrative evidence is intended to isolate.

Taken together, calibration, within-type moment stability, and orthogonality constitute the maintained identifying restrictions of the framework. They are not directly testable: latent type is unobserved, and the orthogonality condition additionally involves the structural disturbance ε_{t+h}.

[3] Appendix A.2 considers the stronger alternative in which a single tax-policy action contains both exogenous and policy-response components and p_t calibrates the latent exogenous share relevant for the outcome–tax moment.

[4] Appendix A shows that the policy-response type may contain several policy-response channels. The two-way representation does not require these channels to have the same moment; it requires their average policy-response moment to remain stable across narrative-support levels.

Support-indexed moments. Under calibration and within-type moment stability, conditioning the outcome–tax moment on narrative support gives:[5]

$$\mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p) = p\,\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1) + (1-p)\,\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 0). \tag{3}$$

Equation (3) is the central identifying relation. Narrative support changes the weights placed on two economically distinct moments: the moment associated with exogenous policy variation and the moment generated by systematic policy responses. As p increases, the former receives greater weight and the latter less. At $p = 1$, the policy-response component receives zero weight, and the support-indexed moment equals the exogenous-type moment. This value need not be observed in the sample; it is recovered from how the observed outcome–tax moment varies with narrative support under the maintained restrictions.

Applying the same argument to the denominator moment $\mathbb{E}(\Delta\tau_t^2 \mid p_t = p)$ and using orthogonality within the exogenous type, the causal response is

$$\beta_h = \frac{\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1)}{\mathbb{E}(\Delta\tau_t^2 \mid u_t = 1)}.$$

Thus, β_h is recovered by evaluating the support-indexed numerator and denominator moments at $p = 1$.

The within-type stability restriction does not require tax composition, institutions, or the macroeconomic environment to remain unchanged over the historical sample. It requires only that such variation is not systematically related to narrative support within latent type in a way that changes the corresponding outcome–tax moment.

1.3 Identification from Variation in Narrative Support

Identification requires variation in narrative support. If p_t were constant, the observed outcome–tax moment would provide only one weighted average of the exogenous and policy-response moments in equation (3). Variation in p_t changes the relative weights on these two latent types and allows their respective moments to be separated.

To see this most clearly, suppose p_t takes two values, p_L and p_H, with $p_H \neq p_L$. Equation (3) then implies

$$\begin{cases} \mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p_L) = p_L\,\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1) + (1-p_L)\,\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 0), \\ \mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p_H) = p_H\,\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1) + (1-p_H)\,\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 0). \end{cases}$$

These are two observable conditional moments with different weights on the same exogenous and policy-response moments. Solving the system gives

$$\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1) = \frac{(1-p_L)\,\mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p_H) - (1-p_H)\,\mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p_L)}{p_H - p_L}.$$

[5] By iterated expectations over u_t,

$$\mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p) = \Pr(u_t = 1 \mid p_t = p)\,\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1, p_t = p) + \Pr(u_t = 0 \mid p_t = p)\,\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 0, p_t = p).$$

Equation (1) gives $\Pr(u_t = 1 \mid p_t = p) = p$ and $\Pr(u_t = 0 \mid p_t = p) = 1 - p$, while equation (2) allows the conditioning on p_t to be dropped inside the two type-conditional moments.

The same system identifies the policy-response moment. Under the analogous within-type stability restriction for $\Delta\tau_t^2$, applying the same argument to the squared tax change identifies $\mathbb{E}(\Delta\tau_t^2 \mid u_t = 1)$ and hence the causal response β_h.

1.4 Moment Conditions and a GMM Formulation

Equation (3) implies an affine relation between the outcome–tax moment and narrative support. Let

$$m_h(p) \equiv \mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p)$$

denote this support-indexed moment. Under calibration and within-type moment stability,

$$m_h(p) = \underbrace{\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 0)}_{a_h} + p\,\underbrace{\left[\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1) - \mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 0)\right]}_{b_h},$$

with $\theta_h = (a_h, b_h)'$. In the population model, a_h is the policy-response moment, while $m_h(1) = a_h + b_h$ is the exogenous-type moment identified in the preceding subsection.

The affine specification implies the conditional moment

$$\mathbb{E}[y_{t+h}\Delta\tau_t - m_h(p_t; \theta_h) \mid p_t] = 0. \tag{4}$$

Multiplying equation (4) by $(1, p_t)'$ gives the two unconditional moment conditions

$$\mathbb{E}\big[\big(y_{t+h}\Delta\tau_t - m_h(p_t; \theta_h)\big)(1, p_t)'\big] = 0.$$

The resulting just-identified estimator is equivalent to the OLS projection of $y_{t+h}\Delta\tau_t$ on a constant and p_t. Applying the same projection to the squared tax change recovers the exogenous-type denominator at $p = 1$. The ratio of the two moments evaluated at $p = 1$ therefore yields β_h.

More generally, the conditional-moment representation allows the support-indexed relation to be specified as a finite-dimensional function $m_h(p; \theta_h)$. Following Chamberlain (1987), multiplication by a vector of functions $\phi(p_t)$ gives

$$\mathbb{E}\big[\big(y_{t+h}\Delta\tau_t - m_h(p_t; \theta_h)\big)\phi(p_t)\big] = 0.$$

The affine specification with $\phi(p_t) = (1, p_t)'$ is the baseline used in the paper; richer specifications provide alternative ways of representing the support-indexed moment.

In the empirical implementation, outcomes and tax changes are residualized on the controls included in each macroeconomic specification. Section 3.4 gives the corresponding sample estimator and stacked moment system. Appendix E states the rank and regularity conditions and derives the asymptotic distribution and HAC variance estimator for the resulting response ratio.

1.5 Measurement Error in Narrative Support

The identification argument above treats narrative support p_t as observed. In practice, it is a generated measure constructed from policy documents through the language-model procedure developed below. This distinction matters because errors in p_t can perturb the support-indexed moments used to recover the exogenous-type response. Recent work on variables generated from unstructured data emphasizes that such errors may reflect not only noise but also systematic features of the model, prompt, or measurement procedure (Battaglia et al., 2024; Carlson & Dell, 2025; Ludwig et al., 2025).

A natural approach in this literature is to use hand-coded validation data to characterize the measurement error and correct the resulting econometric estimates. This route is more demanding here because the target object is not a binary exogenous–endogenous classification, but a calibrated probability of latent exogeneity. Existing expert classifications provide useful external validation of whether the constructed support measure aligns with established narrative judgments, as examined in Section 4, but they do not directly reveal the continuous probability p_t.

I therefore address residual measurement error through two complementary approaches. First, when the remaining error in narrative support can be approximated as mean-zero classical measurement error, I use corrected projected moments in the spirit of Evdokimov and Zeleneev (2022). Second, I allow the projected identifying moments to hold only approximately and assess how inference changes as the permitted violation increases, following Armstrong and Kolesár (2021). These approaches operate directly on the moments through which narrative support identifies the exogenous-type response.

The empirical implementation also varies the motivation extraction, representation model, and calibration sample used to construct p_t. The resulting dispersion documents the stability of the generated support measure, while the induced changes in the projected moments provide an empirical benchmark for the magnitude of measurement perturbations considered in the sensitivity analysis. Section 3.5 implements these exercises, with the corresponding derivations reported in Appendix E.

2 Measuring Narrative Support for Exogeneity

The empirical task is to translate the qualitative evidence associated with the policy action recorded in period t into the narrative support p_t used for identification. I construct this measure in three steps: recover the stated motivations for the policy action from the relevant documentary record, measure how those motivations map into economically defined narrative categories, and aggregate the resulting evidence into a policy-level support probability. Section 3 specifies the choices used to implement this construction in the tax applications.

A policy action may have several stated motivations, some supporting an exogenous interpretation and others indicating a response to economic or fiscal conditions. The measurement construction preserves this mixed evidence rather than resolving it into a binary classification. Schematically,

$$\begin{array}{c}\text{documentary}\\ \text{record}\end{array} \longrightarrow \begin{array}{c}\text{relevant}\\ \text{passages}\end{array} \longrightarrow \begin{array}{c}\text{stated}\\ \text{motivations}\end{array} \longrightarrow \begin{array}{c}\text{motivation-level}\\ \text{support}\end{array} \longrightarrow p_t.$$

2.1 From Stated Motivations to Support Probabilities

I begin with the economically substantive step, mapping stated motivations into p_t, before explaining how those motivations are recovered from documents. Mapping stated motivations into support requires a well-defined set of economic categories against which the textual signal can be interpreted (Ash & Hansen, 2023; Gentzkow et al., 2019). Let $\mathcal{G}$ denote these categories and $\mathcal{G}_{\text{exo}} \subset \mathcal{G}$ the subset whose stated rationales support an exogenous interpretation. Section 3.1 defines the tax-specific taxonomy used in the applications. Given these categories, the empirical task is to map each stated motivation M into probability mass over $\mathcal{G}$,

$$\left(\widehat{P}_c(M)\right)_{c\in\mathcal{G}}, \qquad \widehat{P}_c(M) \geq 0, \qquad \sum_{c\in\mathcal{G}} \widehat{P}_c(M) = 1.$$

Language models are useful for this task because stated policy motivations can express subtle or overlapping economic rationales that are difficult to capture with fixed dictionaries or surface-level textual features. I use language-model representations of the motivations as inputs to a multinomial logistic model that maps these textual signals into probabilities over the predefined narrative categories. The probability mapping is calibrated using labeled synthetic motivations generated from the category definitions. The economic taxonomy therefore determines the distinctions to be measured, while the calibrated statistical model determines how strongly a stated motivation loads on each category. Section 3.2 describes the calibration samples, language-model representations, and statistical specification used in the applications. Appendix C.5 reports an alternative construction based directly on token-probability signals.

The support attached to motivation M is the probability mass assigned to the categories supporting the exogenous type:

$$p(M) = \sum_{c\in\mathcal{G}_{\text{exo}}} \widehat{P}_c(M).$$

For a policy action with several stated motivations, the baseline support p_t averages their motivation-level probabilities. This aggregation preserves mixed documentary evidence at the policy level: motivations associated with systematic policy responses reduce support for the exogenous type, while motivations associated with more orthogonal policy objectives increase it.

The 1977 French VAT reform fixes ideas and illustrates the ambiguity that motivates this probabilistic mapping. The reform was quantitatively important: it reduced the standard VAT rate from 20 percent to 17.6 percent, merged it with the intermediate rate, and carried a reported fiscal impact of 8.5 billion francs, about 0.5 percent of GDP. It therefore provides substantial policy variation for studying the macroeconomic effects of taxation, while its contemporaneous documentary record contains both short-run stabilization and structural-reform motives. The retrieved documentary evidence is condensed into the following three stated motivations:

- ***Mitigate the rise in prices after the end of the price freeze****: the measure aims to limit the inflationary rebound likely to occur at the end of the price-freeze period by reducing the tax burden on consumption.*

- ***Complete a structural reform****: the measure is the final step in a reform undertaken since 1969 to merge the standard VAT rate with the intermediate rate and reduce the number of VAT rates.*

- ***Fight inflation****: the reduction of the rate is part of the broader government programme to combat inflation by easing indirect taxation.*

Table 1 reports the estimated category probabilities. The first and third motivations load mainly on Demand Management and therefore contribute primarily to the policy-response categories. The second places most of its probability mass on the exogeneity-supporting Long Run and External categories.

Table 1: Calibrated Category Probabilities for the 1977 VAT Reform Example

	Category	M1	M2	M3	Overall policy prob.
Endo	Demand Management (DM)	0.76	0.00	0.81	0.55
	Spending-Driven (SD)	0.02	0.00	0.01	
	Supply Stimulus (SS)	0.03	0.00	0.00	
	Deficit Reduction (DR)	0.01	0.01	0.01	
Exo	Deficit Consolidation (DC)	0.10	0.09	0.13	0.45
	External (EXT)	0.08	0.41	0.03	
	Ideological (IL)	0.00	0.01	0.01	
	Long-Run (LR)	0.01	0.48	0.00	

Note: Each motivation column sums to one across the eight categories. The shaded cell marks the highest category within each motivation. The final column reports probability mass assigned to the endogenous and exogenous blocks after averaging across motivations.

Aggregating across the three motivations yields an interior policy-level support probability: 0.45 of the estimated probability mass supports the exogenous interpretation, while 0.55 supports the policy-response interpretation.

2.2 Recovering Stated Motivations from Documents

The preceding subsection takes stated motivations as given. Recovering them is difficult when the policy rationale occupies only a small part of a long documentary record. Let q_t denote a short description of the policy action recorded in period t, and let $\mathcal{D}_t$ denote the collection of passages drawn from its contemporaneous policy documents. A text representation $\psi(\cdot)$ places the policy description and document passages in a common semantic space. The baseline retrieval rule retains the K passages closest to the policy description:

$$\mathcal{R}_K(q_t; \mathcal{D}_t) = \underset{d \in \mathcal{D}_t}{\overset{K}{\arg\min}}\, \mathrm{dist}\{\psi(q_t), \psi(d)\}\,. \tag{5}$$

Retrieval therefore restricts the evidence available for motivation extraction to passages likely to contain the stated rationale for the policy action; it does not itself classify the action. Semantic-distance retrieval is the baseline implementation, but other auditable ways of locating the relevant evidence, including manually indexed documents or model-generated document maps, could be used instead.

The retrieved passages are then converted into a short set of stated motivations. The language model is instructed to recover only rationales supported by those passages and not to infer motivations absent from the documentary record. This step has two roles. First, it isolates the substantive reasons for the policy action from the retrieved text. Second, it expresses those reasons in a compact and comparable form that can be mapped into category probabilities and manually audited against the source passages. In the 1977 VAT example, this stage produces the three motivations used in the probability calculation above.

Traceability is maintained by preserving the evidence chain behind each score. Starting from p_t, one can recover the motivation-level category probabilities, the extracted motivations on which they were computed, and the passages in the original documents supporting those motivations. The calibration and aggregation choices used along this chain are recorded in the replication material. This permits manual audit of the documentary basis of the score and, conditional on the stated choices and a fixed computational environment, reproduction of the mapping. Subsection 3.3 reports the embedding models, document segmentation, retrieval depth, extraction prompts, and language models used in the applications.

2.3 Grounding and Placebo Diagnostics

Because the support score is generated from text, reproducibility and agreement across alternative constructions do not by themselves establish validity. Similar scores may reflect generic linguistic patterns or model priors rather than the passages associated with the specific policy action. Errors may enter through evidence retrieval, motivation extraction, or the subsequent probability construction. The diagnostics below examine whether the extracted motivations are grounded in the retrieved policy evidence.

The diagnostic builds on perturbation-based analyses of model behavior, which assess how outputs change when selected parts of the input are altered or withheld (Feng et al., 2018; Prabhakaran et al., 2019). The implementation is closest to Cohen-Wang et al. (2024): both vary which context sources are included and approximate the resulting changes in the model output with a linear surrogate. Here, the context sources are retrieved passages from policy documents, so the resulting contributions provide evidence linking an extracted motivation to specific parts of the documentary record.

Marginal passage contributions. For a policy action with K retrieved passages, let v_ℓ indicate whether passage ℓ is included when the extracted motivation is evaluated. I generate random passage-inclusion patterns $v = (v_1, \ldots, v_K)$ while holding all other inputs fixed and use the linear approximation

$$\text{logodds}(v) \approx b + \sum_{\ell=1}^{K} w_\ell v_\ell,$$

where the left-hand side is the sequence-level log odds assigned to regenerating the motivation. The coefficients are estimated by OLS across the random inclusion patterns and summarize the contribution of each passage. A positive w_ℓ indicates that including passage ℓ increases the model's support for the extracted motivation, while a coefficient near zero indicates limited dependence on that passage.

Placebo comparison. Positive passage contributions do not necessarily establish policy-specific grounding. For each policy action, I therefore repeat the exercise after adding semantically unrelated placebo passages drawn from the same document. I compare the contributions of the retrieved passages with the empirical distribution of contributions assigned to the placebo passages. If a motivation is grounded in the policy-specific evidence, the passages selected by retrieval should contribute more strongly than unrelated text from the same documentary source.

Table 2 reports the passage-level contributions and placebo comparisons for the 1977 French VAT reform.

Appendix B.6.1 formalizes a complementary event-level randomization diagnostic and reports its application to the French VAT proposals. These diagnostics assess whether the extracted motivations are anchored in the documentary record. They do not quantify how alternative measurements of p_t affect the estimated macroeconomic responses; that sensitivity analysis is described in Subsection 3.5.

Table 2: Passage-Level Attribution Coefficients for the 1977 VAT Reform

	Retrieved segments (1–10)									
	1	2	3	4	5	6	7	8	9	10
M_1	41.56	20.92	7.33	-0.24	-1.76	8.36	2.78	-0.69	-4.93	4.47
M_2	18.82	-4.13	15.29	-4.56	0.25	1.20	1.93	-3.47	-0.51	-3.26
M_3	2.88	8.99	1.19	0.76	-1.26	1.04	1.05	-1.40	-0.46	2.67

Note: Each entry reports the OLS estimate of the corresponding passage's marginal contribution to the sequence-level log odds assigned to regenerating the motivation. The coefficients are estimated from 64 random inclusion patterns over ten retrieved and thirty placebo passages. Shaded cells identify positive retrieved-passage coefficients that exceed every positive placebo coefficient for the same motivation. The shading is a descriptive coefficient-level comparison; Appendix B.6.1 defines a complementary event-level randomization diagnostic.

This section has defined the narrative-support score and the generic restrictions placed on its construction. Section 3 now specifies the choices used in the tax applications, including how support is constructed for individual policy actions and aggregated to the quarterly frequency used in estimation.

3 Constructing and Using Narrative Support

This section describes the common empirical implementation used in the applications. The preceding analysis indexes policy variation by period. In the applications, several individual policy actions may occur in the same quarter, so I use i to index actions and t to index quarters. The implementation first maps each policy description and its contemporaneous documents into an action-level probability, $(q_i, \mathcal{D}_i) \mapsto p_i$, following Section 2. It then aggregates action-level revenue effects and probabilities into $(\Delta\tau_t, p_t)$ [6] and estimates the response associated with exogenous policy variation in Section 1.4. The section explains the taxonomy, retrieval parameters, model and calibration choices, aggregation rules, and econometric specification used to implement these mappings. Conditional on these choices and a fixed computational environment, the measurement rule is systematic and reproducible. All applications use this common implementation unless otherwise stated.

3.1 Semantic Taxonomy for Tax-Policy Motivations

The empirical applications study tax policy. Implementing narrative identification therefore requires connecting the econometric exclusion restriction to an observable semantic criterion: which stated motivations provide documentary evidence that a tax action did not respond to contemporaneous macroeconomic or fiscal conditions? I use the tax-policy taxonomy developed by Romer and Romer (2010) and refined by Cloyne (2013) as the maintained definition of this documentary proxy.

Following Cloyne (2013), the economic distinction underlying the taxonomy can be represented schematically as

$$\Delta\tau_i = \xi_i + r(y_i, \pi_i, b_i, g_i),$$

where y_i denotes the state of economic activity, including relevant labor-market conditions; π_i is inflation; b_i captures deficit or debt pressure; and g_i captures current spending decisions. The function

[6] Appendix A.3 shows when action-level calibration carries through this aggregation and states the corresponding within-type moment-stability conditions needed for identification with the quarterly support index.

$r(\cdot)$ captures systematic tax-policy responses to these conditions, while ξ_i denotes variation motivated by objectives unrelated to them.

The Cloyne taxonomy decomposes the policy-response component into economically distinct channels. Demand management and supply stimulus correspond primarily to responses to activity or inflation; short-run deficit reduction responds to fiscal pressure; and spending-driven changes respond to current expenditure decisions. Each category therefore corresponds to a different component of the policy rule and to a potentially different source of bias. Long-run, ideological, external, and deficit-consolidation motives instead provide support for assignment to the exogenous type. Figure 1 reports the detailed category definitions and examples.

This decomposition also clarifies the within-type stability restriction in Section 1. The scalar p_i should measure support for exogeneity rather than differences in how the measurement procedure scores distinct policy-response motives. If deficit-, demand-, supply-, and spending-driven policy changes receive systematically different support scores for reasons unrelated to exogenous membership, and these channels have different outcome–tax moments, then p_i also sorts the composition of policy-response bias. The affine slope would then combine changes in the share of exogenous actions with changes in the nature of the endogenous policy response, potentially distorting the $p = 1$ endpoint. Estimating separate probabilities for the eight motives before aggregating them into p_i helps place these channels on a common scale and makes channel-specific differences in the scores observable.

Group	Sub category	Explanation and examples
Endogenous (N)	1. Demand management (DM)	• Targeting the aggregate level of demand e.g. to boost investment, consumption, growth, or curb inflation. • Specific help to households and individuals by stimulating disposable income. • Dealing with a balance of payments crisis via demand.
	2. Supply stimulus (SS)	• Certain help for businesses during a downturn (e.g. NIC cut). • Short term sector support (e.g. targeted tax cuts for a sector).
	3. Deficit reduction/BoP crisis (DR)	• Direct measures to deal with a budget or external deficit contemporaneously caused.
	4. Spending-driven (SD)	• Taxes which fund specific spending commitments.
Exogenous (X)	1. LR performance (LR)	• Measures to improve competitiveness, productivity, efficiency and long-run growth (but not taken to offset a shock). • Simplification and deregulation measures. • Long-term support for business or sectors of the economy.
	2. Ideological (IL)	• Long-term social or political goals, independent of their effect on performance and not to offset current shocks. • Some anti-avoidance measures (where no other motive is given).
	3. External (ET)	• Court rulings and enforcement of directives.
	4. Deficit consolidation (DC)	• Measures to lower inherited deficit for reasons of economic philosophy or to offset current actions in the future. • Does not include actions forced on the government, or decisions contemporaneous motivated by a current shock.

Figure 1: Narrative Classification of Tax Changes, Reproduced from Cloyne (2013)

Each extracted motivation is evaluated against the definitions of these eight motives. The definitions are stated in fixed prompts, making the classification criteria explicit and reproducible. They can be used either directly to score historical motivations or to generate labeled synthetic examples, as described next. The prompt templates are reported in Appendix C.1.

3.2 Synthetic Calibration and Probability Construction

There are two ways to use the category prompts to extract a classification signal from a language model. A direct approach examines how the model's token-probability distribution changes when a historical motivation is added to a category prompt. Appendix C.5 describes this alternative. The baseline instead uses the category definitions to generate synthetic motivations with known categories and estimates a probability mapping from their language-model representations. Buckmann and Hill (2025) show that hidden states can retain classification-relevant information that is not reliably revealed by a model's generated answer, and that simple statistical classifiers trained on these representations can match or outperform direct prompting with relatively few labeled observations. Hartley (2025) similarly treats the mapping from language-model representations to probabilities as an estimable component of economic measurement and emphasizes the value of retaining probabilistic outputs when constructing aggregate indices.

The category prompts provide a compact semantic description of what each narrative category includes and excludes. In the baseline construction, these definitions are used to sample a varied collection of motivations for each category, following approaches that generate labeled text for downstream classification (Beck et al., 2026; Ye et al., 2022). The resulting samples allow the researcher to inspect the motivations allocated to the different categories before estimating the probability mapping. By contrast, when classification is obtained directly from a prompt, the model's interpretation of the category remains largely implicit and may differ across models because their training produces different representations of the same semantic distinctions.

I implement the probability mapping defined in Section 2.1 by estimating a multinomial logit of the known synthetic category labels on the hidden-state representations. Coefficient shrinkage is selected by stratified cross-validation to limit overfitting; Appendix C.3 provides the estimation details. Applying the fitted model to a historical motivation yields probabilities over the eight categories. I sum the probabilities associated with exogenous policy variation and, when an action has several motivations, average their support as described in Section 2.1. The synthetic exercise calibrates probabilities over narrative categories. Interpreting their sum as a probability of latent exogeneity additionally requires the population calibration condition in Section 1, which the synthetic exercise cannot test because latent type is unobserved.

The synthetic calibration samples are constructed to span both clear and more ambiguous expressions of each narrative category. I generate four types of motivations. Canonical examples closely follow the category definitions, while institutional examples use the language and context typical of fiscal-policy documents. Boundary examples introduce features associated with a nearby category while preserving a clear primary motivation. Finally, mixed-motive examples include multiple stated considerations but designate one category as primary. In the latter two cases, the additional considerations remain in the text, while the primary motivation determines the label used to estimate the probability mapping.

For each of these four designs, I generate three independently seeded calibration samples. Each sample contains 70 motivations for each of the eight categories, for a total of 560 observations. This produces twelve separate calibration samples. The Romer–Romer and Cloyne observations are not used in their construction or in estimating the probability mappings and are reserved for external validation. Appendix C.2 provides the generation instructions and examples; the complete synthetic samples are included in the replication materials.

I estimate the probability mapping separately for each of the twelve calibration samples and each of the three language-model representations. A given historical motivation therefore receives 36 probability estimates. The baseline averages across these estimates, reducing dependence on a particular synthetic sample or representation model, while their dispersion provides a measure of sensitivity to these choices. Appendix D.1–D.4 reports construction audits, holdout performance, and agreement across sets and models. How these repeated measurements enter the measurement-robustness analysis is discussed below. Appendix D.5 additionally shows that reducing the hidden-state representations to 32 or 64 principal components preserves policy rankings, external-benchmark separation, and the main U.S. GDP and inflation responses.

3.3 Document Processing and Motivation Recovery

Retrieval settings. I implement the retrieval step in Section 2.2 using each policy action's description as the query against its contemporaneous document collection. I use OpenAI's *text-embedding-3-large* for the French VAT documents, OpenAI's *text-embedding-3-small* for the reconstructed Cloyne sources, and Google's *text-embedding-004* for the U.S. budget documents. The Romer–Romer validation starts from edited narratives and therefore does not require document retrieval. Documents are split into fixed-length chunks and the retrieval keeps the ten most relevant passages.

I use the French VAT corpus to assess retrieval quality in this fiscal-policy setting. Each VAT measure is manually linked to the Senate-report article discussing the corresponding budget-law provision. Retrieval performance changes little across chunk lengths and retrieval depths. Among the ten highest-ranked passages, the correct article is retrieved for between 97 and 100 percent of measures across chunk lengths from 512 to 4,096 characters. With the baseline length of 2,048 characters, the correct article is retrieved in 97 percent of cases, with an average of 5.2 of the ten passages overlapping it. Appendix B.3 reports the complete comparison. I do not have an analogous hand-linked benchmark for the U.S. and UK corpora. I expect retrieval performance to be similar, although the appropriate settings may differ because annual U.S. budget documents and UK fiscal and parliamentary records are structured differently from French Senate reports.

Motivation extraction and representation. The retrieved passages are passed to the language models with a fixed prompt asking for the government's stated motivations for the tax change. The prompt instructs the model to rely only on the retrieved excerpts, avoid adding interpretation, and return at most three motivations in a uniform bullet-point format. The dynamic fields in the prompt are the fiscal year, the tax type, and the policy description taken from the policy dataset. Appendix B.4 reports the exact extraction instructions and the settings used in the open-weight extraction sensitivity exercise.

Baseline implementation. For each policy action, the retrieval rule retains the ten passages most closely related to the policy description, and the baseline holds one extracted set of at most three motivations fixed. Let L_i denote the number of motivations retained for action i, and let $\widehat{p}_{i\ell}^{(m,k)}$ denote the support assigned to motivation ℓ using representation model m and calibration sample k. The baseline policy-level score is

$$p_i = \frac{1}{L_i} \sum_{\ell=1}^{L_i} \left(\frac{1}{36} \sum_{m=1}^{3} \sum_{k=1}^{12} \widehat{p}_{i\ell}^{(m,k)} \right).$$

Each motivation is therefore evaluated under 36 representation-by-calibration mappings before the motivation-level scores are averaged within the policy action. The baseline averages across representation and calibration choices, but not across alternative motivation extractions.

Model variation and replication. The hidden-state representation stage uses three open-weight instruction-tuned models: *Gemma-4B*, *Mistral-7B*, and *Qwen-9B*. To assess whether the results depend on motivation extraction, I separately retain twenty fixed-seed stochastic motivation draws from each of Qwen-9B, Gemma-4B, and Mistral-7B. Holding retrieval fixed, I reconstruct policy-level support for each draw and re-estimate the macroeconomic responses. These alternative extractions are retained as an end-to-end sensitivity exercise rather than included in the baseline average. All retrieved passages, prompts, outputs, model identifiers, hidden-state files, and probability-mapping settings are stored so that each support score can be traced back to the source text, extracted motivations, calibration sample, and model calls. Appendix C.4 reports the exact Hugging Face repositories, pinned revisions, and runtime environment used for replication; Appendix D.4 separates variation across draw-averaged extraction models from stochastic variation within a model.

3.4 Support-Indexed Moment Estimation

This subsection gives the sample counterpart of the support-indexed moment in Section 1.4. For each horizon, I remove predictable variation in outcomes and tax changes using the controls in the empirical specification, estimate affine numerator and denominator moments, and evaluate both at $p = 1$. The application sections specify the samples, outcomes, horizons, and controls.

Policy-to-quarter aggregation. Let A_t denote the set of policy actions assigned to quarter t. Each action has a source-reported revenue effect $\Delta\tau_i$ and a policy-level support score $p_i \in [0, 1]$. The quarterly tax-policy measure is

$$\Delta\tau_t = \sum_{i \in A_t} \Delta\tau_i.$$

When at least one action is recorded, the quarterly support index is the absolute-tax-weighted average

$$p_t = \frac{\sum_{i \in A_t} |\Delta\tau_i| p_i}{\sum_{i \in A_t} |\Delta\tau_i|}.$$

Under the conditional calibration restriction stated in Appendix A.3, p_t is the expected share of gross tax variation in the quarter that belongs to exogenous policy actions. Equivalently, it is the probability that a randomly selected unit of gross tax variation comes from an exogenous action. It is not generally the probability that the entire quarter belongs to a single exogenous type. Nevertheless, $p_t = 1$ retains the interpretation that all policy variation in the quarter belongs to the exogenous type.

I retain quarters without a recorded policy action and set $\Delta\tau_t = p_t = 0$ in the estimation sample. Because no policy action is observed, the maintained model does not give p_t a latent-type interpretation in those quarters. Setting $p_t = 0$ is therefore a convention for completing the quarterly series, rather than a statement that no-action quarters belong to the policy-response type. This convention is nevertheless empirically consequential: it determines where no-action observations enter the affine moment relation and may therefore affect the variation used to recover the moments evaluated at

$p = 1$. The fitted moment at $p = 0$ consequently reflects both no-action quarters and policy-action quarters receiving little support for exogeneity.

Appendix H assesses the importance of this convention by assigning alternative common values between zero and one to no-action quarters and by estimating the relevant moments using recorded-action quarters only.

Estimation and inference. Fix a horizon h. Let n_h denote the number of usable quarters at that horizon, and let $\widehat{E}_h$ denote the sample average over those quarters. Define the residuals

$$\widetilde{y}_{t,h}(\pi_{y,h}) = y_{t+h} - W'_{t,h}\pi_{y,h}, \qquad \widetilde{\Delta\tau}_{t,h}(\pi_{\tau,h}) = \Delta\tau_t - W'_{t,h}\pi_{\tau,h}.$$

With outcomes and tax changes understood as residualized, a_h and b_h are the empirical counterparts of the affine numerator coefficients introduced in Section 1.4:

$$\mathbb{E}\left[\widetilde{y}_{t,h}\widetilde{\Delta\tau}_{t,h} \mid p_t = p\right] = a_h + b_h p.$$

I impose the analogous affine specification on the squared residualized tax change,

$$\mathbb{E}\left[\widetilde{\Delta\tau}_{t,h}^{2} \mid p_t = p\right] = a_{D,h} + b_{D,h}p,$$

so $a_h + b_h$ is the fitted numerator at $p = 1$, while $a_{D,h} + b_{D,h}$ is the corresponding fitted denominator. Collect the parameters in

$$\eta_h = (\pi'_{y,h}, \pi'_{\tau,h}, a_h, b_h, a_{D,h}, b_{D,h})'.$$

The complete estimator can be represented by the stacked moment

$$g_{t,h}(\eta_h) = \begin{pmatrix} W_{t,h}\widetilde{y}_{t,h}(\pi_{y,h}) \\ W_{t,h}\widetilde{\Delta\tau}_{t,h}(\pi_{\tau,h}) \\ (1, p_t)'\left[\widetilde{y}_{t,h}(\pi_{y,h})\widetilde{\Delta\tau}_{t,h}(\pi_{\tau,h}) - a_h - b_h p_t\right] \\ (1, p_t)'\left[\widetilde{\Delta\tau}_{t,h}(\pi_{\tau,h})^2 - a_{D,h} - b_{D,h}p_t\right] \end{pmatrix}. \tag{6}$$

The estimator $\widehat{\eta}_h$ solves

$$\widehat{E}_h[g_{t,h}(\widehat{\eta}_h)] = 0.$$

The first two blocks define the residualization of the outcome and the tax change. The last two blocks contain four scalar support-indexed moments: an intercept and a support moment for the outcome–tax cross product, together with the corresponding two moments for the squared tax change. Because the system is just identified, solving it jointly is equivalent to residualizing first and then estimating the two affine projections by OLS.

The estimated high-support response is

$$\widehat{\beta}_h = \frac{\widehat{a}_h + \widehat{b}_h}{\widehat{a}_{D,h} + \widehat{b}_{D,h}}.$$

Inference uses HAC standard errors that allow for serial dependence in the quarterly moment conditions and applies the delta method to this ratio. Appendix E provides the rank conditions, long-run covariance estimator, and response variance.

For later reference, let q_t denote any candidate support value and define the four-dimensional support-moment block

$$g^S_{t,h}(\eta_h; q_t) = \begin{pmatrix} (1, q_t)' \left[\widetilde{y}_{t,h}\widetilde{\Delta\tau}_{t,h} - a_h - b_h q_t\right] \\ (1, q_t)' \left[\widetilde{\Delta\tau}^{\,2}_{t,h} - a_{D,h} - b_{D,h} q_t\right] \end{pmatrix}, \tag{7}$$

where the residuals are evaluated at their estimated projection coefficients. For any complete support sequence $\mathbf{q} = (q_t)$ over the estimation sample, define

$$\overline{g}^S_h(\eta_h; \mathbf{q}) = \widehat{E}_h\left[g^S_{t,h}(\eta_h; q_t)\right].$$

The baseline sequence is $\mathbf{p} = (p_t)$; an alternative evaluator, calibration sample, or motivation extraction generates a complete alternative sequence $\mathbf{p}^{(r)} = (p_t^{(r)})$.

3.5 Sensitivity to Measurement Error in Narrative Support

The document-to-support mapping in Section 2 makes narrative support a generated measure whose construction can be varied along several dimensions. In the applications, I vary three components: the motivations extracted from the retrieved passages, the language model used to represent those motivations, and the synthetic sample used to calibrate the probability mapping. The baseline support score averages across the three representation models and twelve calibration samples. For the U.S. budget application, a joint exercise retains all $3 \times 20 \times 3 \times 12 = 2{,}160$ extraction-model, extraction-draw, representation-model, and calibration-sample combinations. These are alternative measurement constructions, not independent draws from a statistical model of measurement error.

Narrative support does not enter the first two blocks of the complete moment system. Changing p_t therefore does not directly alter the moments defining the residualization coefficients. It perturbs the four support-indexed moments in $g^S_{t,h}$. The complete system nevertheless remains relevant for inference because uncertainty in the residualization coefficients propagates to the endpoint estimate.

Corrected-moment benchmark. To describe the corrected-moment benchmark, write the generated quarterly support score as $p_t = p_t^* + e_t$, where p_t^* is latent narrative support. The benchmark assumes that the residual error e_t is conditionally mean zero and orthogonal to the variables entering the support-indexed moments. Let ω_t denote its conditional variance. Under this benchmark, the corrected support-moment block is

$$g^{S,\mathrm{ME}}_{t,h} = g^S_{t,h}(\eta_h; p_t) + \omega_t \begin{pmatrix} 0 \\ b_h \\ 0 \\ b_{D,h} \end{pmatrix}. \tag{8}$$

The correction enters the two support-weighted moments because these are the conditions in which the measured support score appears both in the fitted value and as a weighting variable. Following

Evdokimov and Zeleneev (2022), this can be understood by expanding the observed moments around zero measurement error. The first-order term vanishes under mean-zero error, while the remaining distortion depends on ω_t. Because the affine support moments are quadratic in p_t, the displayed correction is exact under the additive classical-error benchmark. Appendix E.7 provides the componentwise derivation.

The repeated constructions distinguish several sources of uncertainty. Variation across calibration samples and representation models captures uncertainty in mapping a fixed motivation into a probability. For the corrected-moment benchmark, I retain the resulting 3×12 model-by-calibration probabilities and aggregate each construction to quarters using the same absolute-tax weights as for the baseline score. I estimate ω_t as the variance of their average using a two-way decomposition of model, calibration-sample, and residual dispersion. This allows constructions that share a model or a calibration sample to be related rather than treating the 36 probabilities as independent measurements. Appendix E.7 gives the exact construction. I do not incorporate extraction variation into ω_t, since doing so would additionally require treating differences in extracted motivations as mean-zero classical error. Instead, I carry extraction variation through direct re-estimation and approximate-moment diagnostics, both separately and jointly with the representation and calibration choices.

I also report extraction variation directly as an end-to-end sensitivity exercise. Holding retrieval fixed, I generate twenty motivation draws with each of Qwen-9B, Gemma-4B, and Mistral-7B, reconstruct policy-level and quarterly support, and re-estimate the GDP and inflation responses. This check asks whether the macroeconomic conclusions depend on which model extracts the stated motivations, without interpreting the alternative constructions as instruments. Appendix F.4 reports the resulting probability and endpoint comparisons.

Approximate-moment sensitivity. The classical benchmark does not cover systematic measurement errors that shift the moment conditions themselves. I therefore complement it with an approximate-moment analysis applied to $g^S_{t,h}$. Let B_h be diagonal, with entries equal to the HAC standard deviations of these four moments. The sensitivity set allows

$$\mathbb{E}[g^S_{t,h}(\eta_{h,0}; p_t)] = \frac{B_h \delta_h}{\sqrt{n_h}}, \qquad \|\delta_h\| \leq M. \tag{9}$$

The case $M = 0$ imposes exact validity of the baseline affine moments. Larger values permit progressively greater misspecification relative to the sampling variability of those moments. Following Armstrong and Kolesár (2021), I construct confidence intervals allowing for the worst-case direction of misspecification within each radius M.

Observed alternative support constructions provide an empirical scale for this exercise. Holding the baseline parameter estimate fixed, define the moment displacement generated by alternative construction r as

$$\Delta_h^{(r)} = \overline{g}^S_h(\widehat{\eta}_h; \mathbf{p}^{(r)}) - \overline{g}^S_h(\widehat{\eta}_h; \mathbf{p}),$$

and its equivalent sensitivity radius as

$$M^{(r)}_{h,\mathrm{eq}} = \sqrt{n_h} \left\| B_h^{-1} \Delta_h^{(r)} \right\|. \tag{10}$$

This radius measures the total standardized displacement across the four support moments. It indicates whether the values of M considered in the sensitivity analysis cover perturbations comparable to those generated by actual measurement choices.

The equivalent radius does not, by itself, measure the effect of a perturbation on $\widehat{\beta}_h$. That effect also depends on whether the displacement points in a direction that changes the fitted $p = 1$ numerator and denominator. I therefore report three complementary diagnostics: the equivalent radius, the endpoint relevance of the observed displacement, and the response obtained after re-estimating the model under the alternative support construction. Endpoint relevance is the absolute linearized endpoint change divided by the largest change that a perturbation with the same equivalent radius could produce. Appendix E.8 defines this directional calculation formally, while Appendix F.4 reports the empirical results.

4 External Validation of Narrative Support

This section evaluates the proposed narrative-support measure against external benchmarks. These exercises assess whether the measure reproduces the ordering implicit in expert narrative classifications; because latent type is unobserved, they do not test the population calibration condition maintained in Section 1. The empirical design separates two validation margins. First, using edited Romer–Romer narratives provided in their appendix, does the probability-construction stage assign higher support to episodes classified as exogenous in the canonical U.S. narrative series? Second, using Cloyne's UK tax data, does the full document-grounded procedure align with an independent classification when retrieval, summarization, and motivation extraction are all performed from policy documents?

Expert narrative classifications inevitably involve difficult judgment calls, particularly for episodes with mixed or ambiguous motivations. Nevertheless, because the support measure is built from the same semantic distinction between categories that underlies the narrative literature, one would expect episodes classified as exogenous to receive stronger documentary support for exogeneity on average. Establishing this relationship is what makes the probability object informative for the budget-document application that follows.

4.1 Validation using the Romer–Romer narratives

This subsection evaluates whether the common procedure in Section 3 recovers the economic content of Romer and Romer (2010) when applied to the edited narratives used in the original study. The exercise holds fixed the curated Romer–Romer historical narratives and asks whether the procedure assigns systematically higher exogeneity support to RR-exogenous episodes and whether the resulting probability-based shock series reproduces the benchmark macroeconomic response.

By conditioning on the Romer–Romer narratives, this benchmark abstracts from document retrieval while preserving the summarization, motivation decomposition, and probability aggregation stages of the framework. It therefore tests whether the probability object carries identifying content similar to the original RR narratives.

4.1.1 Setup

The Romer–Romer replication material documents 50 major federal tax changes in the United States between 1945 and 2007. For each episode, the authors provide a detailed narrative discussion of the motivations underlying the reform, drawing on contemporaneous sources such as presidential speeches, budget documents, and Congressional records. Each narrative concludes with a classification of the tax change as either exogenous or endogenous.

To construct the benchmark validation task, I extract the narrative text describing each policy episode and remove the portion in which the authors explicitly state and justify their classification. In particular, I delete the sentence reporting the final classification along with the two immediately preceding sentences, which often contain direct cues revealing the intended label (see Appendix F.1.1 for examples). The resulting edited narratives form the input to the validation exercise.

4.1.2 From edited narratives to policy-level probabilities

The edited RR narrative does not enter the benchmark as a single paragraph to be mapped directly into a binary label. Instead, each edited narrative is first summarized into a set of stated motivations. Because a single RR episode can contain several rationales, the same policy may generate multiple motivations, each of which is scored separately. These motivation-level outputs are then mapped into exogeneity probabilities and aggregated into a policy-level structural probability.

The construction combines the probability mappings estimated across calibration sets and language models as described in Subsection 3.2. Throughout this benchmark, the object of interest is therefore the policy-level narrative support measure p_i, rather than the output from any single mapping.

4.1.3 Probability validation against RR classifications

Table 3 reports the original Romer–Romer binary label and asks whether RR-exogenous episodes receive systematically higher policy-level support. Mean support rises from 0.43 for RR-endogenous episodes to 0.60 for RR-exogenous episodes, and the share above tighter thresholds also rises.

Table 3: Policy-Level Structural Probabilities by Original RR Classification

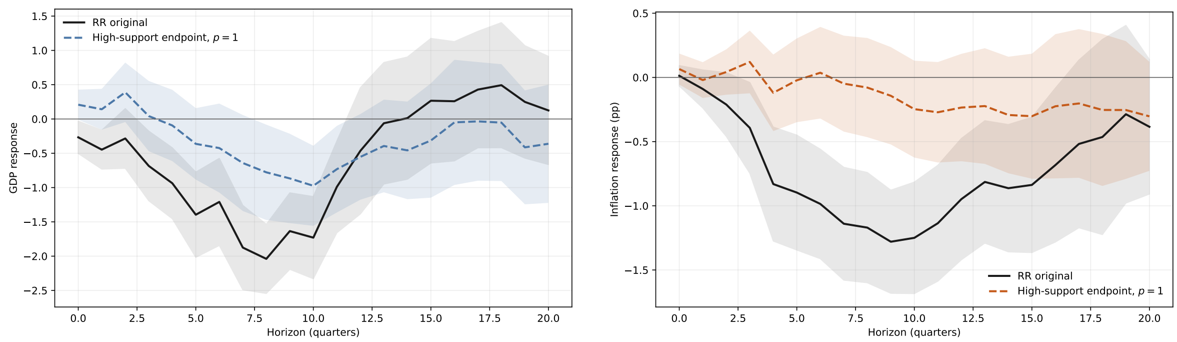
Class	N	Mean p	Share $p \geq 0.5$	Share $p \geq 0.75$
Endogenous	24	0.43	41.7%	0.0%
Exogenous	26	0.60	65.4%	30.8%
Difference (exo - endo)	–	0.18	23.7 pp	30.8 pp

Note: Entries are computed from the 50 edited RR narratives. p is the baseline support probability averaged across twelve calibration samples and three representation models.

Mean support of 0.60 among RR-exogenous episodes remains well below one and, under the maintained calibration interpretation, implies material uncertainty about latent type. Appendix F.2.5 relates intermediate probabilities to ambiguities in the original narratives, including current-macro motives, spending-package dependence, and deficit or social-insurance financing. Appendix F.2 reports probabilistic label-prediction and training-leakage diagnostics.

4.1.4 Romer–Romer through support-indexed moments

I next apply the support-indexed moment estimator from Section 1.4 to the Romer–Romer tax episodes. For each horizon, the controlled outcome–tax moment is projected on narrative support and evaluated at the high-support endpoint ($p = 1$). This assesses whether the extracted support measure preserves the identifying content of the original classification.



(a) GDP (b) Inflation

Figure 2: Romer–Romer Panel Under the Support-Indexed Moment Estimator

Note: The figure reports the RR policy panel estimated with the support-indexed moment estimator in Section 1.4. The black line uses the original RR exogenous indicator in the same moment estimator. The blue line uses the probability-based affine endpoint at $p = 1$, normalized by the affine denominator evaluated at $p = 1$. Shaded areas denote 68% HAC confidence bands.

Figure 2 compares the support-indexed moment estimator of Section 1.4 with the same estimator applied to the original RR exogenous classification. The probability-based estimates do not fully recover the benchmark RR responses. Table 3 nevertheless shows that RR-exogenous episodes receive systematically higher support than RR-endogenous episodes. Taken together, these results indicate that the score contains narrative ranking information, but that the 50 RR episodes provide too little support variation for the continuous moment-sorting approach to recover the benchmark response precisely.

Appendix F.2.3 replaces the continuous support measure with threshold indicators $\mathbb{1}\{p_i \geq \bar{p}\}$. This resolves narrative ambiguity through a binary classification rule, as in traditional narrative identification. At $\bar{p} = 0.6$, the resulting response is close to the RR benchmark; at $\bar{p} = 0.7$, the medium-run estimate changes sign. The threshold exercise is therefore not a monotone credibility–precision trade-off: changing the evidentiary standard changes which influential episodes identify the response and can overturn the economic conclusion. Appendix F.2.5 documents the narrative content of these intermediate-support episodes. This sensitivity is one reason for using support probabilities directly rather than selecting a single cutoff. Appendix F.2.2 reports the exact Romer–Romer distributed-lag replication as a complementary benchmark closest to the original published specification.

4.2 External Label Validation: Cloyne (2013) UK Tax Changes

The Romer–Romer exercise above isolates the probability-construction stage by starting from curated narrative text. I next ask whether the same probability object remains informative in a more demanding setting, where the relevant policy documents must first be reconstructed. Cloyne (2013) provides a useful benchmark for this purpose. The dataset records individual UK tax changes, their revenue effects and implementation dates, and an expert narrative classification of each change as endogenous or exogenous.

The validation starts from Cloyne's row-level tax workbook. The reconstruction preserves the 2,462 tax-change rows in the original file and maps them into 85 narrative events. For each event, I build an event-level source base from contemporaneous UK fiscal and parliamentary materials, apply the retrieval and summarization procedure described above, and extract up to three stated motivations for each policy row. These motivations are then mapped into structural exogeneity probabilities. Because Cloyne's classification is also defined at the policy-row level, it provides a direct external label against which to compare the resulting probabilities.

Table 4 asks whether rows classified as exogenous by Cloyne receive systematically higher support than rows classified as endogenous.

Table 4: Policy-Level Structural Probabilities by Cloyne Classification

Class	N	Mean p	Share $p \geq 0.5$	Share $p \geq 0.75$
Endogenous	795	0.72	86.0%	51.2%
Exogenous	1,306	0.82	97.6%	75.2%
Difference (exo - endo)	–	0.10	11.6 pp	24.0 pp

Note: Entries use 2,101 policy rows with usable Cloyne labels and complete probability inputs. No UK macroeconomic outcome enters the probability construction.

The Cloyne validation provides evidence about ranking, not calibration in levels or exact recovery of the binary labels. Cloyne-exogenous rows have average support of 0.82, compared with 0.72 for Cloyne-endogenous rows, and their share above 0.75 is 24.0 percentage points higher. Support is high in both classes, however, so a cutoff of 0.5 has little discriminatory content. Many Cloyne-endogenous rows receive high support because the reconstructed source text contains mixed package-level motivations, or because Cloyne's row-level classification incorporates dependencies within a fiscal package that are not summarized by current-macro motives alone. The relevant result is the ordering across the two groups rather than the ability of a particular cutoff to reproduce Cloyne's classification. Appendix F.2.6 reports the corresponding one-variable label-prediction exercise.

Together, the two benchmarks show that the support measure preserves expert ordering when applied both to edited narratives and to reconstructed policy documents. They also expose the limits of binary validation: the RR response is sensitive to the chosen cutoff, and the Cloyne groups overlap substantially. I now use the resulting support measure to study tax transmission and the margins through which high-support tax changes operate.

5 Macroeconomic Effects of U.S. Budget Tax Proposals

I now apply the narrative identification framework to tax proposals reported in the annual *Budget of the United States Government*. Relative to existing narrative tax series, these documents provide substantially broader coverage of federal tax-policy proposals and retain information on individual tax instruments. They also contain the contemporaneous policy discussion needed to recover the administration's stated motivations for each proposal. I use these data to estimate the macroeconomic response to proposed tax changes with high narrative support and to examine which tax instruments account for the aggregate response.

5.1 U.S. tax data and narrative support

Existing narrative coverage. The narrative tax dataset of Romer and Romer (2010) identifies major federal tax changes in the United States over the postwar period. It covers large legislative episodes, including major tax reforms, Social Security legislation, and federal highway funding acts. The resulting series provides a well-defined measure of large, infrequent tax changes, but covers only a subset of federal tax-policy actions and records their composition at a relatively aggregated tax-instrument level.

Budget-based extension. I construct a complementary dataset from the annual *Budget of the United States Government* over 1950–2024. These documents report the administration's proposed changes to federal taxation as part of the fiscal program submitted to Congress. For each fiscal year, I extract newly proposed tax measures at the highest level of detail reported in the Budget, retaining their estimated revenue effects and tax category whenever available. A proposed measure may introduce a new tax provision or modify an existing one, for example through an extension, acceleration, reduction, or expiration. The relevant observation is therefore the proposed policy change relative to prevailing tax policy, rather than whether the underlying tax provision itself is new.

The resulting data provide a much denser record of federal tax-policy proposals than the canonical narrative series and preserve the contemporaneous policy text needed to construct narrative support. For comparison, I also apply the same extraction procedure to the major tax acts identified by Romer and Romer (2010), producing a finer tax-level decomposition than that available in Mertens and Ravn (2013). The main application below, however, uses the measures proposed in the annual budget documents.

The budget-based and Romer–Romer samples differ both in coverage and in the stage of the policy process at which tax changes are recorded. Romer and Romer record enacted tax changes, whereas the budget data record proposals made by the administration to Congress. The two sources can nevertheless overlap when a proposal reported in the Budget is subsequently enacted and enters the Romer–Romer series, or when a later budget proposes a modification of a provision associated with an earlier Romer–Romer episode. A single Romer–Romer episode may also correspond to several budget proposals distinguished by tax instrument or proposed effective date. Appendix F.1.2 documents the clearest overlaps and explains why relating the two sources requires a provision-level crosswalk rather than a mechanical match by law, date, or aggregate amount.

Table 5 summarizes the proposed tax measures extracted from the annual budgets. The six tax categories follow the classifications used in the source documents rather than an ex post taxonomy imposed by the researcher. For each measure, I record the proposed effective date reported in the Budget and assign the measure to the corresponding quarter. When no specific effective quarter is stated, I assign the measure to the first quarter of the fiscal year. The resulting dataset therefore describes changes in federal taxation proposed by the administration to Congress through the annual budget process, rather than the universe of enacted federal tax legislation.

Documentary evidence. The documentary corpus consists of the same annual budget volumes. In addition to reporting the administration's proposed tax measures and their estimated revenue effects, the budgets discuss the fiscal program and the motivations underlying individual proposals. They therefore provide a contemporaneous documentary record from which to recover the stated rationale

Table 5: Distribution of Proposed Tax Measures by Category (United States)

Tax Category	**# Proposed Measures**
Individual Income Tax	371
Corporate Income Tax	361
Consumption Tax	238
Payroll / Social Insurance Tax	114
Other Receipts	89
Wealth Transfer Tax	17
Total	**1190**

for the measures in the dataset. For each proposed measure, I use its label as the query against the corresponding budget volume and retrieve the passages most closely related to the proposal. The retrieved passages are then used to recover the stated motivations and construct narrative support following the common procedure in Section 3.

Quarterly variation in narrative support. I aggregate the proposed tax measures and their policy-level support scores to the quarterly frequency using the procedure in Section 3. This produces a quarterly budget-proposal series and a corresponding narrative-support index, p_t, which enter the support-indexed moment estimator.

Figure 3 reports the distribution of narrative support across quarters with budget-tax activity. The 1,078 matched measures correspond to 134 event quarters. Support is concentrated in the interior of the unit interval: its mean is 0.58 and its median is 0.59. The budget data therefore contain substantial variation in the strength of narrative support without naturally separating into fully exogenous and fully endogenous policy episodes.

Figure 4 shows the quarterly proposed revenue effects together with narrative support. The upper panel compares the budget-proposal series with the original Romer–Romer enacted exogenous tax series. The budget series contains 133 non-zero quarters, compared with 45 in the Romer–Romer series, and the correlation between the two series is -0.03. The lower panel reports p_t for the budget event quarters. Narrative support varies substantially over time and is nearly uncorrelated with the absolute size of the net quarterly proposed revenue effect, with a correlation of -0.02.

5.2 Aggregate effects and tax composition

Figure 5 reports the estimated responses of GDP and inflation to a proposed tax increase at high narrative support. The estimates apply the support-indexed moment estimator in Section 1.4 to the budget-based tax proposals. GDP declines following the proposed tax increase, with the largest contraction around quarters 8–12. Inflation also declines at medium horizons.

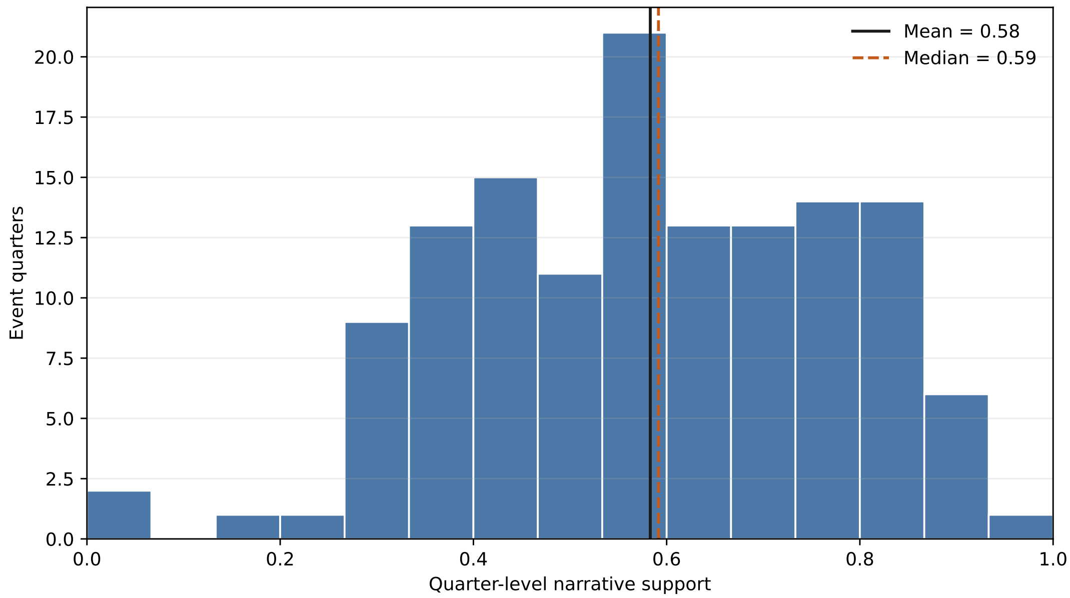


Figure 3: Distribution of Quarter-Level Narrative Support for U.S. Budget Tax Measures

Note: The figure uses U.S. budget measures with complete proposed effective dates, tax amounts, and narrative-support scores that can be matched to the quarterly macroeconomic sample. Policy-level support is aggregated to the quarter using absolute tax amounts as weights. The matched sample contains 134 quarters with budget-tax activity.

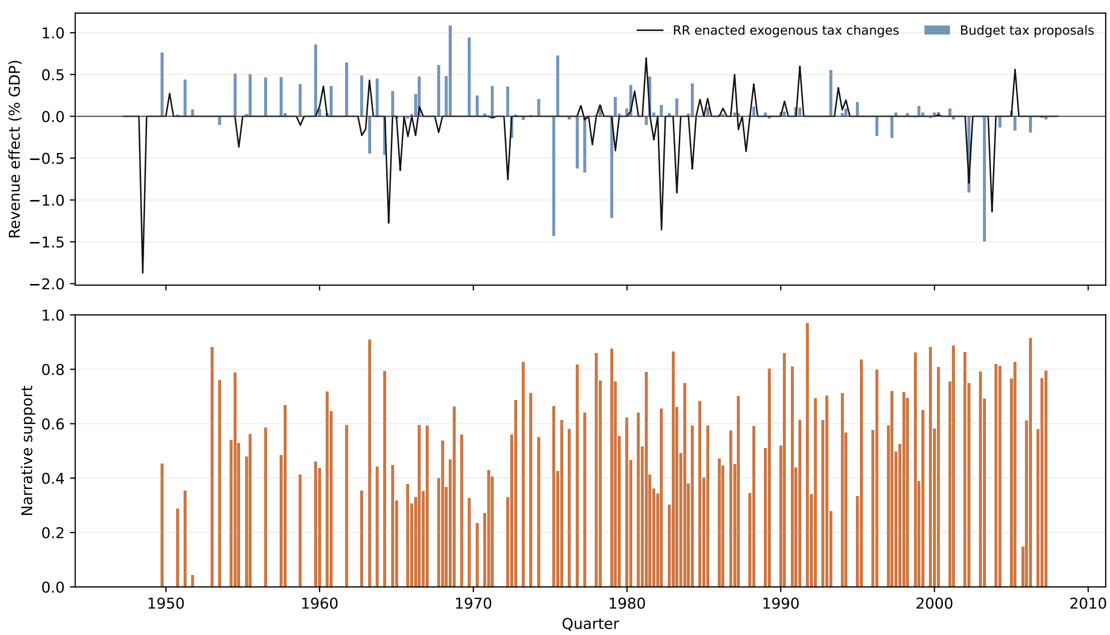
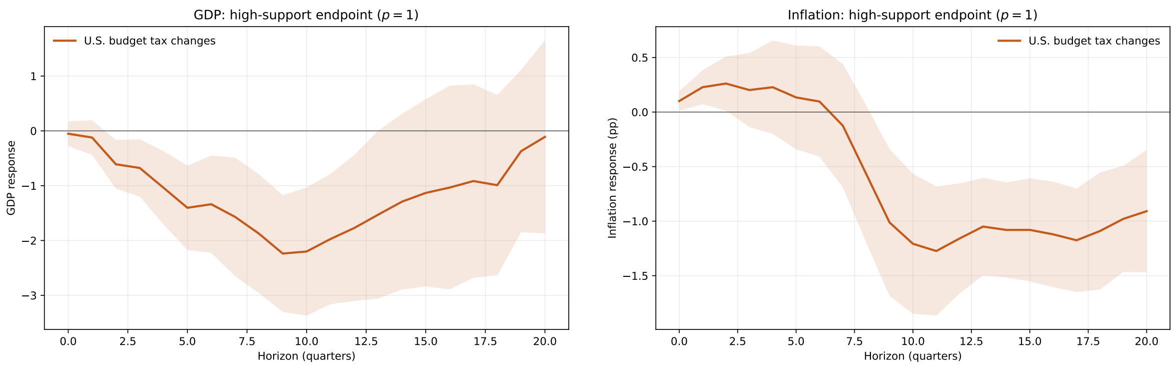


Figure 4: Quarterly Budget Tax Proposals and Narrative Support

Note: The upper panel plots proposed revenue effects for the budget series and enacted tax changes for the original Romer–Romer exogenous series, both expressed as a percentage of contemporaneous GDP. The lower panel reports the quarter-level narrative-support index p_t for quarters with budget-tax activity. "nzq" denotes the number of non-zero quarters.



(a) GDP

(b) Inflation

Figure 5: Macroeconomic Responses to U.S. Budget Tax Proposals at High Narrative Support

Note: The figure reports responses evaluated at $p = 1$ using the support-indexed moment estimator in Section 1.4. Responses are normalized by the fitted tax-variation moment evaluated at $p = 1$. Shaded areas denote 68% HAC confidence bands.

Appendix H shows that the U.S. results are preserved when no-action quarters are assigned alternative common values between zero and one or when the relevant moments are estimated using recorded-action quarters only. At horizons 4, 8, and 12, the GDP response remains negative under every valid specification. The medium-run decline in inflation and the large investment contraction are also preserved.

Table 6 examines the sensitivity of the estimates to the construction of narrative support. I vary the motivation-extraction model, extraction draw, representation model, and calibration sample separately, and then vary all four components jointly. Among the individual components, the GDP response is most sensitive to the extraction model, while variation in the calibration sample is relatively more important for inflation. Differences across representation models and repeated extraction draws are smaller.

Table 6: Sensitivity of U.S. Endpoint Estimates to Narrative-Support Construction

		GDP		Inflation	
Construction varied	N	Median	P90	Median	P90
Motivation-extraction model	3	0.256	0.321	0.061	0.088
Stochastic extraction draw	60	0.067	0.150	0.032	0.089
Representation model	3	0.114	0.129	0.023	0.040
Calibration sample	12	0.098	0.180	0.097	0.130
All stages jointly	2,160	0.268	0.606	0.145	0.327

Note: Entries report absolute changes in the re-estimated $p = 1$ endpoint relative to the baseline construction. Statistics summarize horizons 4, 8, and 12. GDP is measured in percent and inflation in percentage points. N is the number of alternative constructions at each horizon. The rows vary one stage at a time except for the final row, which combines all stages. These are sensitivity specifications, not independent draws or confidence intervals.

The joint variation changes the magnitude of the estimates but leaves the main medium-run pattern largely unchanged. The median absolute change relative to the baseline is 0.27 percentage points for GDP and 0.14 percentage points for inflation; at the 90th percentile, the corresponding changes are 0.61 and 0.33 percentage points. GDP remains negative at horizons 4, 8, and 12 across all 2,160 constructions. The inflation response is less stable at short horizons, retaining its baseline sign in 75 percent of constructions at horizon 4, but retains its baseline sign in all constructions at horizons 8 and 12.

The alternative constructions also provide an empirical scale for the approximate-moment sensitivity analysis. Each construction induces a displacement of the four identifying moments, whose standardized magnitude can be expressed as an equivalent radius M^{eq}. Joint variation across all stages generates median equivalent radii around one, but the resulting perturbations are only weakly aligned with the directions that most affect the numerator–denominator ratio: median endpoint relevance is 0.19 for GDP and 0.18 for inflation. The approximate-moment analysis is deliberately more conservative because it considers the most adverse direction within a given radius. These observed radii should therefore be interpreted as empirical scale benchmarks rather than estimates of structural misspecification. Appendix Table 23 reports the complete mapping between alternative support constructions, equivalent radii, and endpoint relevance.

The aggregate GDP response combines tax proposals across several instruments. To assess which instruments account for the contraction, I decompose it into additive tax-category contributions. Let $\Delta\tau_{k,t}$ denote the quarterly proposed revenue effect associated with instrument k, so that

$$\Delta\tau_t = \sum_k \Delta\tau_{k,t}.$$

For each instrument, I estimate the outcome–tax moment using the aggregate narrative-support index p_t and normalize the resulting contribution by the same tax-variation moment used for the aggregate response. The contribution of instrument k at horizon h is

$$\widehat{\beta}_{k,h}^{\mathrm{add}} = \frac{\widehat{a}_{k,h} + \widehat{b}_{k,h}}{\widehat{a}_{D,h}^{\mathrm{all}} + \widehat{b}_{D,h}^{\mathrm{all}}}.$$

Because the decomposition uses a common support index and normalization, the contributions add to the aggregate response:

$$\widehat{\beta}_h^{\mathrm{all}}(1) = \sum_k \widehat{\beta}_{k,h}^{\mathrm{add}}.$$

These contributions describe which tax instruments account for the estimated aggregate response in the historical sample; they should not be interpreted as separate causal effects of each instrument.

Table 7 reports the decomposition at horizons 4, 8, and 12. The medium-run GDP contraction is concentrated in individual and corporate income-tax proposals. Individual income-tax proposals account for the largest contribution, reaching −2.01 at horizon 8 and −2.17 at horizon 12. Corporate income-tax proposals contribute an additional −0.86 and −0.68, respectively. These negative contributions are partly offset by the broader consumption and excise category. Individual income-tax proposals are not more frequent than corporate income-tax proposals, but they represent a larger share of gross proposed revenue changes and receive similar average narrative support.

The consumption-tax categories contribute much less systematically. Standard excises have small estimated contributions, while the broader consumption and excise category contributes positively at the main horizons. This category combines heterogeneous excise, energy-related, and other measures and does not provide a clean measure of changes in consumption taxation. I therefore do not interpret these estimates as an indirect-tax response. The VAT application in Section 6 provides a more clearly defined setting for studying the effects of consumption taxation.

Table 7: Tax Composition of the U.S. GDP Response

Instrument	Event share	Gross tax share	Mean p	GDP $h = 4$	GDP $h = 8$	GDP $h = 12$
Individual income	30.6%	37.0%	0.72	-1.21 (0.62)	-2.01 (1.01)	-2.17 (1.12)
Corporate income	32.2%	22.7%	0.71	-0.57 (0.38)	-0.86 (0.57)	-0.68 (0.49)
Clean standard excise	8.9%	11.2%	0.52	-0.08 (0.15)	-0.24 (0.19)	-0.12 (0.12)
Nonstandard consumption/excise	11.1%	15.5%	0.53	0.70 (0.36)	0.91 (0.54)	1.10 (0.61)
Other low-support taxes	17.2%	13.6%	0.50	0.12 (0.18)	0.32 (0.25)	0.10 (0.25)

Note: Entries are additive contributions to the aggregate affine $p = 1$ budget endpoint; parentheses report HAC standard errors. The rows use the aggregate support index and common endpoint denominator and therefore sum to the aggregate response up to rounding.

5.3 Macroeconomic transmission

The tax-composition results help interpret the aggregate response. Individual and corporate income-tax proposals account for most of the medium-run GDP contraction, suggesting that private consumption and investment are the relevant expenditure margins. Figure 6 reports their responses together with GDP. Consumption declines following the proposed tax increase, while the response of investment is larger and more persistent. The contraction in output is therefore accompanied by a pronounced decline in private investment.

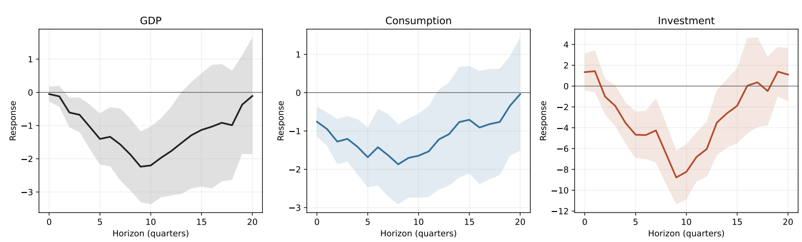


Figure 6: GDP, Consumption, and Investment Responses at High Narrative Support

Note: The figure reports responses evaluated at $p = 1$ using the same specification as Figure 5. GDP, consumption, and investment are log quantity indexes, so responses are expressed in percent. Shaded areas denote 68% HAC confidence bands.

The relative importance of investment is consistent with the composition of the underlying tax proposals. Previous evidence similarly finds that personal, corporate, and consumption taxes operate through different margins (Mertens & Ravn, 2013; Nguyen et al., 2021), while structural models distinguish labor, capital, and consumption taxation through their different behavioral effects (Trabandt & Uhlig, 2011). The U.S. estimates should therefore be interpreted as the response to a composite of tax proposals, dominated at high narrative support by individual and corporate income-tax proposals, rather than as a common effect across tax instruments. Appendix F.5 reports additional expenditure components and comparisons with the Romer–Romer responses.

This distinction also motivates the VAT application below. The heterogeneous consumption and excise measures in the U.S. budget data do not isolate a clearly defined change in consumption taxation.

The VAT application provides a more narrowly defined setting in which to study the effects of indirect taxation.

6 The Indirect-Tax Contrast: VAT in the United Kingdom and France

The U.S. budget endpoint combines high-support tax proposals. The decomposition above shows that its medium-run contraction is concentrated in income-tax margins, while the U.S. consumption/excise rows are too heterogeneous to provide a clean indirect-tax experiment. I therefore turn to VAT policy actions in the United Kingdom and France as the indirect-tax contrast. The goal is not to replace the U.S. end-to-end application, but to ask whether the same support-indexed estimator recovers the transmission pattern associated with a specific tax base. Value-added-tax policy actions are central to modern fiscal policy, but they are hard to study with binary narrative methods. VAT reforms are sparse, often proposed as part of broader tax packages, and frequently justified by several motives at once: revenue needs, price stabilization, purchasing power, simplification, and competitiveness. These are precisely the cases in which forcing a policy action into a binary exogenous/endogenous label discards useful information.

The VAT application asks whether document-based probabilities can turn this mixed narrative evidence into a macroeconomic object. The economic channel is sharp. A VAT increase raises the tax-inclusive price of consumption goods, lowers household purchasing power, and should contract domestic absorption, especially private consumption. It may also generate a short-run inflation response and an external offset through lower imports. The empirical question is therefore not only whether VAT shocks reduce output, but whether the response has the mechanism one would expect from an indirect tax.

The United Kingdom and France provide a natural two-country setting. Both have long postwar VAT or VAT-equivalent histories and rich documentary records, while the United States does not have a federal VAT. The pooled application estimates the response to a high-support VAT increase using the same probability scale and GMM logic as the U.S. application. The section therefore provides the indirect-tax counterpart to the U.S. aggregate budget endpoint: after the U.S. application shows how a broad federal tax mix transmits, the VAT exercise asks whether a specific consumption-price wedge produces the mechanism predicted for an indirect tax.

6.1 VAT policy data and documentary sources

The French VAT data are built from the *Voies et Moyens Tome I*, the revenue annex to the annual draft budget law. This source decomposes the projected change in tax revenues into mechanical growth in the tax base, accounting-scope changes, the continuing effects of previous reforms, and newly proposed measures. The application focuses on discretionary policy actions with direct macroeconomic content, rather than spontaneous revenue movements or scope transfers. In the full French corpus, the budget documents record 140 new VAT measures and 324 previous-measure entries. VAT is also quantitatively important: in the 2024 budget documents it accounts for about 38 percent of gross central-government tax revenues and about 6 percent of GDP.

The French proposals are assigned to their stated effective quarter or, when none is reported, to the first quarter of the budget year. The UK measures retain Cloyne's implementation dates for announced budget measures. Source-reported revenue effects retain their sign and are scaled by the country-specific GDP normalization used in the panel application.

The documentary corpus for the French side consists primarily of contemporaneous Senate finance reports. These reports are useful because they discuss the budget articles in detail, record the government's stated rationale, and often scrutinize the economic and fiscal consequences of the proposed measures. For the VAT corpus, each measure can be linked to the Senate-report discussion of the corresponding budget-law provision. This structure is used to audit retrieval quality in the implementation section: the document retriever locates the relevant discussion for almost all VAT measures among the top retrieved passages. The extracted motivations are then mapped into the same robust support score used in the U.S. application.

For the United Kingdom, I use the action-level tax data underlying the Cloyne application and select VAT rows from tax-type and source labels containing VAT or value-added-tax language. The UK and French probabilities are placed on the same robust hidden-state calibration scale, so the panel uses a common support measure rather than country-specific binary classifications.

Table 8 summarizes the country-level inputs. The main VAT panel is the common UK–France sample from 1955Q1 to 2007Q4. It contains enough VAT timing variation to estimate an instrument-specific response, while preserving the same quarter-level support construction across countries.

Table 8: VAT Policy Data Used in the UK–France Panel

Country	Main documentary source	Rows	nzq	Mean p	$p75$	$p90$
France	*Voies et Moyens* Tome I; Senate finance reports	122	39	0.71	0.81	0.88
United Kingdom	Cloyne action-level data; Budget, Hansard, Treasury record	168	72	0.80	0.91	0.94

Note: Rows count VAT policy actions and nzq counts nonzero VAT quarters in the common 1955Q1–2007Q4 sample. Support statistics are computed over nonzero VAT quarters after aggregating policy-level probabilities with absolute tax-change weights.

Figure 7 reports the resulting quarter-level support distribution. The object is not a country-specific binary label: event-quarter support is interior in both countries, with the UK distribution shifted toward higher support and the French distribution retaining more mass in the middle of the unit interval.

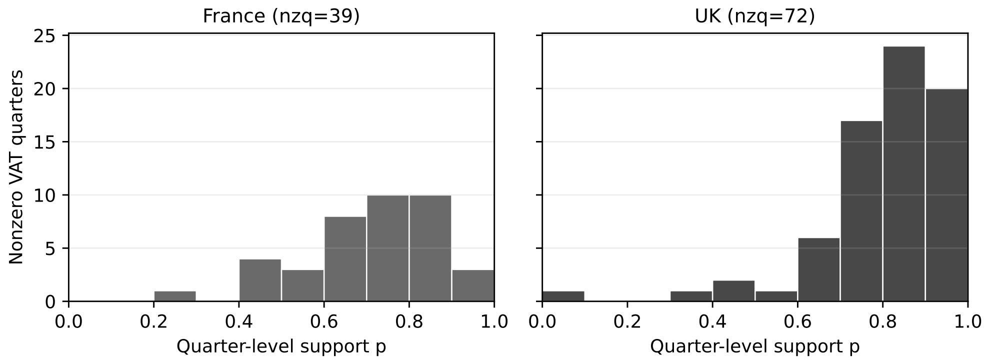


Figure 7: Quarter-Level Exogeneity Support for VAT Policy Actions

Note: The figure plots the distribution of quarter-level support p_{ct}^{VAT} for nonzero VAT quarters. Quarter-level support is the absolute-tax-weighted average of action-level support among VAT measures assigned to the same country-quarter.

The average support is high, especially in the United Kingdom. This is consistent with the nature of the VAT policy record. Many VAT changes are structural reforms of the tax system: introduction or coverage changes, rate unification, simplification, European harmonization, or long-run revenue design. They are therefore often documented as tax-design or fiscal-structure measures rather than as short-run stabilization responses. The probabilities should not be read as saying that every VAT action is an ideal isolated shock. They say that the stated motives often look closer to the exogenous/structural side of the narrative taxonomy than to contemporaneous macroeconomic management. This distinction also matters for Nguyen et al. (2021). The UK episodes they emphasize, including the VAT introduction transition in 1973Q2 and the 1979Q3 rate unification, are not assigned low support in this calibration: their quarter-level support is about 0.95 and 0.77, respectively. Their concern is therefore better viewed as a concern about anticipation, tax packages, and the isolation of the VAT component within broader consumption-tax measures, rather than as a prediction that these specific episodes should have near-zero narrative support.

Figure 8 translates the VAT policy rows into the quarterly policy series used in the estimation. The upper panels plot net quarterly VAT policy changes; the lower panels show the corresponding quarter-level exogeneity support. This is the direct UK–France analogue of the U.S. construction figure, but restricted to the VAT instrument.

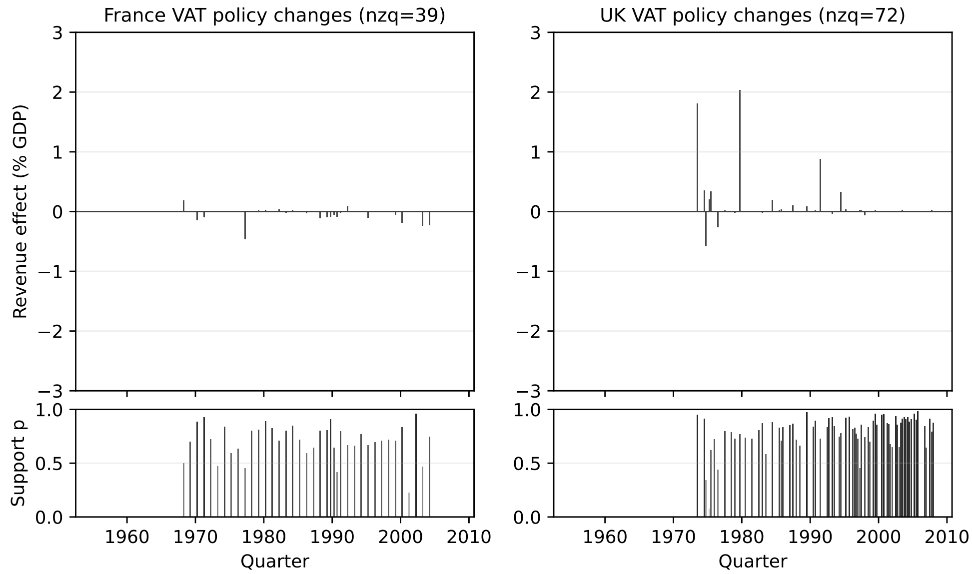


Figure 8: From VAT Measures to Quarterly VAT Policy Series in France and the United Kingdom

Note: The upper panels plot VAT policy rows aggregated to the country-quarter level, expressed as a percent of GDP. The lower panels report the corresponding absolute-tax-weighted quarter-level support. "nzq" denotes the number of nonzero VAT quarters in the common 1955Q1–2007Q4 sample.

6.2 Empirical design

The data are organized at the country-quarter level. In this subsection, $\Delta\tau_{ict}$ denotes the source-reported revenue effect for VAT policy row i in country c, quarter t, while $\Delta\tau_{ct}$ and p_{ct} denote the corresponding country-quarter VAT policy measure and support index. I aggregate them as

$$\Delta\tau_{ct} = \sum_{i\in(c,t)} \Delta\tau_{ict}, \qquad p_{ct} = \frac{\sum_{i\in(c,t)} |\Delta\tau_{ict}| p_{ict}}{\sum_{i\in(c,t)} |\Delta\tau_{ict}|}, \tag{11}$$

I retain quarters without a recorded VAT action and set $\Delta\tau_{ct} = 0$ and $p_{ct} = 0$, following the baseline convention in Section 3.4. For quarters with recorded VAT actions, p_{ct} is the absolute-tax-weighted measure of narrative support; actions with larger revenue effects receive more weight.

The estimator is the pooled version of the high-support GMM estimator used in the U.S. application. For each horizon h, I residualize the outcome and the quarterly VAT policy measure using the common macro controls and form the numerator moment $x_{ct,h} = \widetilde{y}_{ct,h}\widetilde{\Delta\tau}_{ct,h}$. I then estimate

$$x_{ct,h} = A_h + d_{c,h}\,(p_{ct} - 1) + u_{ct,h}. \tag{12}$$

The parameter A_h is the common high-support numerator endpoint. The country-specific slopes $d_{c,h}$ allow the raw probability scale to differ across countries, while imposing that a high-support VAT policy change has the same endpoint effect in the pooled application. The impulse response divides A_h by the same linear-in-p denominator used in the current U.S. baseline, evaluated at $p = 1$. The estimates are therefore responses to a high-support VAT policy increase of one percent of GDP. Inference uses calendar-quarter HAC scores aggregated across countries.

6.3 Macroeconomic mechanism

A VAT policy increase changes the tax-inclusive price of consumption goods and reduces household purchasing power. The relevant margins are therefore output, prices, private consumption, investment, and the external offset through imports and net exports. Figure 9 reports the main response. The top row shows output and prices; the bottom row shows the two domestic-absorption components that are most directly relevant for the VAT mechanism. A high-support VAT policy increase of one percent of GDP lowers GDP by about 2.5 percentage points of trend GDP after four quarters and by about 4.6 percentage points after eight quarters. Inflation rises initially and then returns toward zero, consistent with a VAT policy change that combines a direct tax-inclusive price effect with a contraction in real activity.

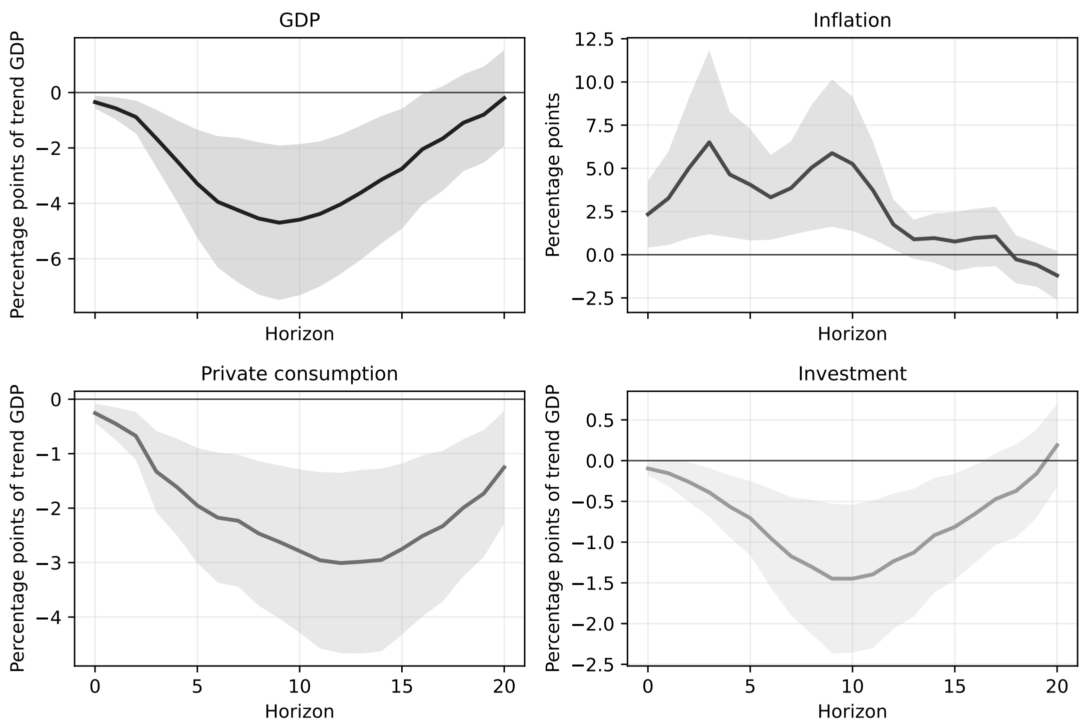


Figure 9: UK–France VAT: Output, Prices, and Domestic Absorption

Note: The figure reports the pooled high-support VAT endpoint. GDP, private consumption, and investment are expressed as $y_{c,t+h} - y_{c,t-1}$ in percentage points of trend GDP. Inflation is in percentage points. Responses are normalized by the linear-in-p denominator evaluated at $p = 1$. Shaded areas are 68 percent HAC confidence bands.

The response is concentrated in domestic absorption. Private consumption falls immediately and continues to decline through the medium run. Investment also falls, but the pooled response is smaller than the consumption response. Table 9 summarizes the same pattern at horizons 4, 8, and 12. At horizon 8, GDP is 4.6 percentage points below its pre-event trend-GDP position. Private consumption accounts for 2.5 percentage points of the decline, while investment accounts for 1.3. The result is therefore not only a negative VAT multiplier. It is a mechanism result: high-support VAT policy increases raise prices on impact and then depress real activity primarily through household demand.

Table 9: VAT Main Mechanism at Key Horizons

Outcome	$h = 4$	$h = 8$	$h = 12$
GDP deviation from trend	-2.46	-4.55	-4.04
Inflation	4.64	5.04	1.75
Private consumption response	-1.61	-2.47	-3.01
Investment response	-0.56	-1.30	-1.23

Note: GDP and expenditure components are changes from the pre-event quarter in percentage points of trend GDP. Inflation is in percentage points. Imports enter as $-M$, so a positive import contribution denotes import compression.

Appendix H also varies the treatment of quarters without recorded VAT actions. The pooled recorded-action-quarter estimates and the alternative assignments $c = 0.5$ and $c = 1$ preserve the output contraction, decline in private consumption, and initial rise in inflation. Magnitudes are more sensitive than in the U.S. application, however, and the affine normalization becomes weak for some high-c assignments.

The additional expenditure components, reported in Appendix G.1, complete the accounting interpretation. Government consumption does not explain the initial contraction. Net exports move in the opposite direction and partly offset the fall in domestic demand, mostly because imports decline. The VAT application therefore complements the aggregate U.S. evidence by showing a different transmission pattern for a different tax base: the U.S. composite endpoint is investment-heavy and disinflationary, whereas the VAT endpoint combines an output decline with an initial price response and a consumption-centered absorption channel.

A natural concern is that VAT changes may be bundled with other tax instruments. This is the central warning in Nguyen et al. (2021): a broad consumption-tax proxy can attribute package effects to indirect taxes if income-tax and consumption-tax changes are systematically linked. The same concern is relevant here, but the diagnostics point to a VAT-specific result. Same-quarter overlap is common because income-tax measures are frequent in budget quarters, yet the VAT-income correlation is small. Appendix G.2 shows that an explicit VAT/income adjustment leaves the GDP endpoint essentially unchanged and that dropping opposite-sign VAT-income package quarters does not weaken the response. By contrast, broadening the object from VAT to VAT plus duties and excises substantially attenuates the response. The result should therefore be read as a VAT-specific high-support endpoint, not as evidence about the broader average-consumption-tax-rate object studied by Nguyen et al. (2021).

7 Conclusion

This paper develops a probabilistic approach to narrative identification. Traditional narrative methods use historical evidence to classify policy actions as exogenous or endogenous and retain a subset of observations for identification. I instead interpret the same evidence as probabilistic information about whether a policy action belongs to a latent exogenous type. Variation in narrative support for exogeneity then identifies the outcome–policy moment associated with that type. This formulation allows ambiguous and mixed-motive policy actions to remain informative without requiring any individual action to be treated as perfectly exogenous.

I implement this approach using contemporaneous policy documents and apply it to tax policy. The measurement procedure concentrates judgment in explicit choices over the narrative taxonomy, calibration design, and language models; conditional on these choices, the mapping from documents to probabilities is systematic, reproducible, and auditable. The resulting probabilities are systematically related to the Romer–Romer and Cloyne classifications while retaining substantial variation within their binary categories. Applied to a broader set of U.S. federal tax proposals, the framework yields contractionary effects of high-support proposed tax increases, with the medium-run response concentrated in individual and corporate income-tax proposals and operating mainly through investment. I then combine the UK tax record with a newly constructed narrative dataset of French VAT proposals from 1960 to 2024. The VAT application provides a complementary result for indirect taxation: high-support VAT policy increases contract output primarily through private consumption and generate a short-run increase in inflation. Across these applications, probabilistic narrative evidence enters the identifying moments directly rather than first being resolved into a binary classification.

The same logic may be useful when the policy variation that is easiest to defend as exogenous is not representative of the variation of broader economic interest. Narrative studies of government spending, for example, often rely on military expenditures associated with geopolitical events, although their effects need not generalize to public investment or civilian purchases. Probabilistic narrative evidence could permit a broader set of spending actions to remain informative while allowing documentary support for exogeneity to vary across them. The approach could similarly use monetary-policy statements, minutes, and forecasts to distinguish different degrees of support for exogeneity across policy decisions.

An important extension concerns anticipation. The analysis characterizes why a policy action was taken but does not separately model when households and firms learned about it. A policy change may have strong narrative support for exogeneity while nevertheless being anticipated before implementation. Combining evidence about policy motivations with evidence about announcement and information timing would extend the framework to this additional identification problem.

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8 Appendix

A When a Two-Way Contamination Representation Is Valid

A.1 Multiple Sources of Contamination

The policy-response type $u_t = 0$ need not be homogeneous. It may combine several policy-response channels with different effects on the outcome–tax moment. The two-way representation in Section 1 does not require these channels to have the same economic content or the same sign of bias. It requires their average contribution to be stable across narrative-support levels.

Let $J_t \in \{1, \dots, K\}$ index the channel generating a tax change within the policy-response type. For each channel k, assume that the relevant moment does not additionally vary with narrative support:

$$\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 0, J_t = k, p_t = p) = r_h^k.$$

Define the average contaminated moment among observations assigned support level p as

$$\bar{r}_h^B(p) = \sum_{k=1}^{K} \Pr(J_t = k \mid u_t = 0, p_t = p)\, r_h^k, \qquad p < 1.$$

Combining this definition with the scalar calibration condition in Section 1 gives

$$m_h(p) = p\ \mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1) + (1-p)\bar{r}_h^B(p).$$

The two-way affine representation in equation (3) is therefore valid exactly when $\bar{r}_h^B(p)$ does not vary with p. This condition does not require all policy-response tax changes to have the same moment. It requires only that narrative support does not systematically sort their composition in a way that changes their average moment. This condition allows contamination with opposite signs: if two channels have different moments, their relative incidence among policy-response observations must not vary systematically with p_t.

In the tax applications, these channels can correspond to demand management, supply stimulus, short-run deficit reduction, and spending-driven tax changes. The category-level probabilities produced before aggregation into the scalar support score provide a diagnostic for whether the composition of these policy-response motives varies systematically with narrative support.

A.2 Mixed Policy Actions and an Exogenous-Share Interpretation

The main text assigns each recorded tax-policy action to one of two latent types. Under a stronger alternative interpretation, a single recorded action may contain both an exogenous component and a policy-response component. Let $s_t \in [0, 1]$ denote the latent exogenous share relevant for the outcome–tax moment, so that these two components are represented by $s_t\Delta\tau_t$ and $(1-s_t)\Delta\tau_t$. This proportional decomposition is an additional assumption; it is distinct from the aggregation of several binary policy actions considered below. The corresponding share calibration condition is

$$\mathbb{E}(s_t \mid p_t = p) = p.$$

Thus p is the average exogenous share among tax changes assigned support level p, rather than the probability that the entire tax change is exogenous.

Share calibration alone is insufficient for identification. The component associated with the exogenous share must satisfy

$$\mathbb{E}[s_t \Delta\tau_t \varepsilon_{t+h} \mid p_t = p] = 0,$$

and the contaminated covariance per unit of remaining policy-response share must be stable:

$$\frac{\mathbb{E}[(1-s_t)\Delta\tau_t \varepsilon_{t+h} \mid p_t = p]}{1-p} \quad \text{does not depend on } p, \qquad p < 1.$$

The first restriction defines the component indexed by s_t as exogenous. The second is the share-based analogue of within-type moment stability. Together, they imply that $\mathbb{E}(\Delta\tau_t \varepsilon_{t+h} \mid p_t = p)$ is proportional to $1-p$ and therefore vanishes at $p = 1$. Combined with the analogous support-indexed restriction for the squared tax change maintained in Section 1, this gives the same response evaluated at $p = 1$ as the binary model.

This interpretation does not imply that $p_t \Delta\tau_t$ is an exogenous shock. Indeed,

$$p_t \Delta\tau_t = s_t \Delta\tau_t + (p_t - s_t)\Delta\tau_t,$$

so orthogonality of $s_t \Delta\tau_t$ does not imply orthogonality of $p_t \Delta\tau_t$. Calibration only requires $\mathbb{E}(s_t - p_t \mid p_t) = 0$; it does not require the discrepancy $s_t - p_t$ to be uncorrelated with $\Delta\tau_t \varepsilon_{t+h}$. The share interpretation therefore supports the moment-sorting argument, not an observation-level construction of the exogenous shock.

A.3 Binary Policy Types and Quarterly Aggregation

The baseline model assigns each policy action a binary latent type. To distinguish the constituent policy actions from the quarterly aggregate, let $u_i = 1$ if action i is exogenous and $u_i = 0$ if it belongs to the policy-response type. For a quarter containing at least one policy action, define the realized exogenous share of gross tax variation as

$$\bar{u}_t = \frac{\sum_{i \in A_t} |\Delta\tau_i| u_i}{\sum_{i \in A_t} |\Delta\tau_i|}.$$

This definition does not make individual policy actions partially exogenous. Each u_i remains binary; $\bar{u}_t$ records only the composition of the actions assigned to the same quarter.

Marginal policy-level calibration, $\mathbb{E}(u_i \mid p_i) = p_i$, does not by itself imply calibration of the quarterly aggregate. A sufficient condition is that each policy-level score remains calibrated conditional on the tax amounts and the other scores entering the aggregation:

$$\mathbb{E}(u_i \mid \{p_j, \Delta\tau_j : j \in A_t\}) = p_i, \qquad i \in A_t.$$

Under this condition,

$$\mathbb{E}(\bar{u}_t \mid \{p_i, \Delta\tau_i : i \in A_t\}) = \frac{\sum_{i \in A_t} |\Delta\tau_i| p_i}{\sum_{i \in A_t} |\Delta\tau_i|} = p_t.$$

It follows by iterated expectations that

$$\mathbb{E}(\bar{u}_t \mid p_t = p) = p.$$

Equivalently, consider selecting one unit of gross tax variation at random from the quarter, with each action selected in proportion to $|\Delta\tau_i|$. Conditional on the information used in aggregation, p_t is the probability that this unit belongs to an exogenous action. This is not generally the probability that every action in the quarter is exogenous. At $p_t = 1$, however, calibration and $\bar{u}_t \in [0, 1]$ imply $\bar{u}_t = 1$ almost surely, so all gross tax variation belongs to exogenous actions.

The calibration result concerns the composition of quarterly policy variation; it does not by itself imply the affine outcome–tax moment. That relation additionally requires that narrative support does not systematically sort the size, sign, composition, or timing of policy actions in ways that change the within-type outcome–tax and squared-tax moments. These are the within-type moment-stability and denominator restrictions maintained in Section 1. Under those restrictions, the $p = 1$ endpoint recovers the same exogenous-policy moment as in the binary benchmark. The gross share $\bar{u}_t$ need not coincide with the structural share s_t in the preceding subsection without the additional proportional decomposition imposed there.

Finally, because $\bar{u}_t$ is constructed using absolute tax amounts while $\Delta\tau_t$ is the net quarterly tax-policy measure, $\bar{u}_t\Delta\tau_t$ need not equal the sum of exogenous action-level revenue effects when tax increases and cuts coexist. The same is true of $p_t\Delta\tau_t$. The quarterly support score is therefore used to sort moments and recover the $p = 1$ endpoint; it is not itself used to construct an observation-level exogenous tax shock.

B From Policy Documents to Narrative Support

This appendix complements the conceptual measurement discussion in Section 2 and the implementation choices in Section 3. It records the document mapping, application-specific retrieval choices, motivation-extraction settings, and grounding diagnostics used to trace each support score back to its source material.

B.1 Illustration of the mapping

B.2 Embedding Models and Distance Metric

Semantic retrieval relies on vector representations of text segments and policy descriptions. The embedding model differs across applications because the underlying document collections were assembled at different stages of the project. Table 10 records the models and retrieval depth used in each case.

Table 10: Application-Specific Retrieval Settings

Application	Embedding model	Retrieved passages K
French VAT	OpenAI `text-embedding-3-large`	10
Cloyne validation	OpenAI `text-embedding-3-small`	10
U.S. budget documents	Google `text-embedding-004`	10
Romer–Romer validation	Not applicable	Not applicable

Note: The Romer–Romer exercise starts from edited narrative entries and therefore does not require document retrieval.

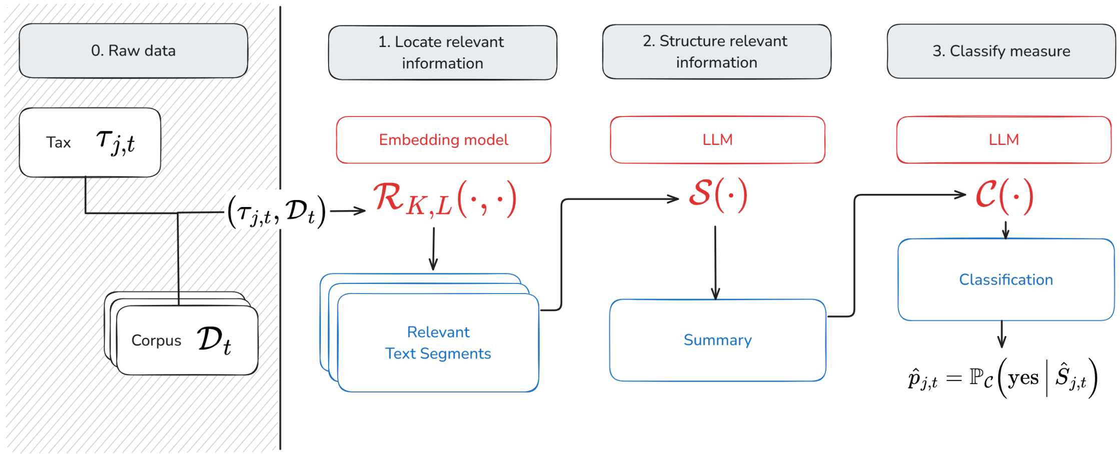


Figure 10: Mapping from policy descriptions and contemporaneous documents to narrative support. The figure illustrates how the documentary record is reduced to relevant passages, stated motivations, category probabilities, and the policy-level support score p_i, as described in Section 2.

For each policy description q_i and document segment $d_{i,k} \in \mathcal{D}_i$, the corresponding embedding vectors e_i and $v_{i,k}$ are normalized to unit length. Semantic proximity is then evaluated using the cosine distance,

$$\text{dist}_{\cos}(q_i, d_{i,k}) = 1 - \frac{e_i \cdot v_{i,k}}{\|e_i\| \, \|v_{i,k}\|},$$

so that smaller distances indicate greater semantic similarity. Because all embeddings are normalized to unit length, cosine and Euclidean distances are mathematically equivalent in this setting, yielding identical retrieval rankings under either metric. Distances are computed through standard nearest-neighbor search routines provided by the underlying vector store, which efficiently identifies the text segments closest to each policy query in the embedding space. This normalization ensures that retrieval reflects semantic similarity rather than differences in vector magnitude. The retrieval operator defined in Equation (5) in the main text therefore selects, for each policy event i, the K document segments with the smallest cosine distances to the corresponding policy query q_i.

B.3 Retrieval Performance across Parameter Settings

I use the French VAT corpus to assess retrieval quality in the fiscal-policy setting. Each of the 94 VAT measures is manually linked to the Senate-report article discussing the corresponding budget-law provision. For each candidate chunk length, I record whether the retrieved passages overlap the manually matched article. Table 11 reports top-1 retrieval, the share of measures with at least one match among the top three and top ten passages, and the average number of matches among the top ten.

Performance is high across the parameter grid. The correct article appears among the ten highest-ranked passages for between 96.8 and 100 percent of measures, while top-1 retrieval ranges from 89.4 to 93.6 percent. Although the French Senate reports differ in structure from the UK and U.S. source collections, this benchmark suggests that retrieval is not sensitive to the chunk lengths considered.

Table 11: French VAT Retrieval Performance across Chunk Lengths

Chunk length (characters)	Top-1 retrieval	Top-3 recall	Top-10 recall	Mean matches in top 10
512	0.915	0.968	1.000	6.70
1,024	0.936	0.989	1.000	6.14
2,048	0.894	0.947	0.968	5.19
4,096	0.915	0.979	1.000	4.98

Note: The sample contains 94 French VAT measures. Each measure is matched manually to the relevant article in the corresponding Senate report. Retrieval uses the policy-description query employed in the baseline construction and OpenAI's *text-embedding-3-large* model.

B.4 Motivation-Extraction Prompt and Settings

The extraction prompt asks why the tax measure was adopted rather than what the measure did. It instructs the model to rely only on the retrieved passages, ignore unrelated policies, avoid external interpretation, and return one to three ordered motivations without introductory text. The main estimates condition on the archived fixed motivation sets. For the U.S. application, the exact model used to produce the legacy baseline motivation workbook was not archived; the motivation text itself is included in the replication files, so the subsequent probability construction and estimation are reproducible conditional on these motivations. Table 12 records the settings used in the separate open-weight extraction sensitivity exercise.

Table 12: Motivation-Extraction Settings

Setting	Sensitivity design
Open-weight models	Qwen-9B, Gemma-4B, and Mistral-7B
Maximum retrieved passages	10
Maximum generated tokens	512
Temperature	0.1
Seeds and draws	Twenty archived fixed-seed draws per model
Output format	One to three ordered motivation bullets

B.5 Example details

Applying the retrieval operator $\mathcal{R}_K$ to the query q_i and corpus $\mathcal{D}_i$, I obtain the top-K most relevant passages. Here are the top-5 passages:

- **Chunk 1.** On December 31, 1976, to mitigate, by a reduction of the weight of taxes bearing on consumption, the upward movement likely to occur at the end of this phase of price freeze; — it constitutes the last stage of a reform undertaken since 1969 — the date on which the standard rate of VAT was 23.456% — which results, ultimately, in a reduction, in less than seven years, of one quarter of the standard rate and in the merging of the latter with the intermediate rate.
- **Chunk 2.** An exceptional contribution would be requested in 1977 from the wealthiest French; it would affect all of them and only them, contrary to what had been provided for in the deleted article 5. This taxation would have as its base the total of the bases corresponding to the elements of lifestyle enumerated in article 168 of the General Tax Code, with the exception of

main residences and cars of a power equal to or less than 16 horsepower. The income-equivalent threshold, calculated according to the norms of the aforementioned article 168, above which the exceptional contribution applies, is set at 60,000 F and the number of accumulated elements provoking taxation is three. Your Finance Committee proposes that you adopt the present article. VAT (Value Added Tax) – Article 6. Reduction of the standard rate and unification of the standard rate and the intermediate rate. Text. — I. — The standard rate of value-added tax is set at 17.60%. In the departments of Guadeloupe, Martinique and Réunion, it is set at 7.50II. — The upper limit of the special rebate for artisans fixed in paragraph 3 of article 282 of the General Tax Code is raised to 20,000 F. III. — The provisions of the present article enter into force on January 1, 1977. Comments. — It is proposed in the present article to lower by 2.4 points, starting from January 1, 1977, the standard rate of value-added tax which is currently 20% and, thereby, to bring it down to the level of the intermediate rate, namely 17.60%. This measure fits within a double perspective: — it is part of the program of combating inflation and is of a nature to permit, after the price freeze applied during the period from September 15 to December 31, 1976, to mitigate, by a reduction of the weight of taxes bearing on consumption, the upward movement likely to occur at the end of this phase of price freeze;

- **Chunk 3.** The number of rates applied in the matter of value-added tax, starting from next January 1, will be reduced to three instead of four; there would remain only: — the increased rate (33 %: cars, motorcycles, photography and cinematography equipment, sound and image recording devices, precious stones and precious metals, etc.); — the standard rate (17.6%: energy products, beverages, services and all transactions hitherto subject to the 20% rate, that is to say all those which were not taxed at the increased rate, nor at the intermediate rate, nor at the reduced rate); — the reduced rate (7%: essentially food products, pharmaceutical products, water and fertilizers). At the same time, it is planned to carry out a reduction of 2.5 points of the standard VAT rate and to align it with the intermediate rate in the departments of Guadeloupe, Martinique and Réunion, French Guiana not being concerned since the value-added tax regime is not applied there. In the three departments concerned, where, due to constraints inherent in the economies of developing countries, VAT rates are significantly lower than those of Metropolitan France, starting from January 1, 1977, one will record: — the increased rate: 14%; — the standard rate: 7.5%; — the reduced rate: 3.5%. In order to support the artisanal professions, it is proposed in paragraph II of the present article to raise, starting from January 1, 1977, the upper limit of the special rebate from which they benefit, raising it from 13,500 F to 20,000 F. Let us recall that, in order to lighten the burdens of small taxpayers, and in particular artisans, article 282 of the General Tax Code has provided, in their favor, special provisions resulting in a waiver or a reduction of the VAT normally incumbent upon them.

- **Chunk 4.** Standard of 6.5% per year. b) The elimination of the levy. As had been provided by the Finance Law for 1976, in the initial text presented by the Government, the levy was to cease to apply when the progression of prices was below a set norm: this one, in the governmental project, was fixed at 2.50% for the consumer price index of the group of manufactured products during a period of six consecutive months.

- **Chunk 5.** Article: — in paragraph I, the objective is to encourage business leaders to convert into capital the advances they are led to grant in the form of deposits in current accounts to the companies they manage. Up to now, in such cases, the ordinary contribution right was a proportional duty of 1%; it is planned to substitute for it a fixed duty of 220 F on the condition, however, that the sums thus incorporated into the capital have been placed at the constant disposal of the company, for a minimum period of twelve months.

B.6 Grounding Diagnostics: Marginal Narrative Contributions

B.6.1 Global Placebo Test for Summarisation Grounding

The coefficient-level placebo comparison in the main text evaluates, for each passage individually, whether its positive marginal contribution is unusually large relative to the distribution of positive placebo contributions. Because this diagnostic is applied to multiple passages within each event, some flagged cells may be mechanically expected under the null of no grounding. This appendix defines an omnibus diagnostic that aggregates evidence across passages.

Placebo mass. For each event i, let $w_{i,k}^{+} = \max\{\hat{w}_{i,k}, 0\}$ denote the positive marginal contribution of passage k, and let $\mathcal{I}_i^{\mathrm{rel}}$ and $\mathcal{I}_i^{\mathrm{pl}}$ denote the sets of relevant and placebo passages, respectively. The share of positive weight falling on placebo passages is

$$\mathrm{PM}_i = \frac{\sum_{k \in \mathcal{I}_i^{\mathrm{pl}}} w_{i,k}^{+}}{\sum_{k \in \mathcal{I}_i^{\mathrm{rel}} \cup \mathcal{I}_i^{\mathrm{pl}}} w_{i,k}^{+}} \in [0, 1],$$

which summarises, at the event level, how much of the model's total positive dependence is placed on semantically irrelevant text. Grounded behaviour is characterised by low PM_i, while ungrounded dependence yields large values.

Randomisation inference. To assess whether PM_i is unusually low, I test the sharp null that placebo labels are exchangeable with relevant-passage labels conditional on the estimated contributions. Under the null, any assignment of $|\mathcal{I}_i^{\mathrm{pl}}|$ placebo labels among the $|\mathcal{I}_i^{\mathrm{rel}}| + |\mathcal{I}_i^{\mathrm{pl}}|$ passages is equally likely. I therefore permute these labels while preserving the number of placebo passages, recompute the placebo mass $\mathrm{PM}_{i,b}$ for each permutation, and form the lower-tailed randomization p-value

$$p_i^{\mathrm{rand}} = \frac{1 + \sum_{b=1}^{B} \mathbb{1}\{\mathrm{PM}_{i,b} \le \mathrm{PM}_i\}}{1 + B}.$$

A small p_i^{rand} indicates that positive dependence is unusually concentrated on the retrieved policy-relevant passages. Failure to reject does not establish that the motivation is ungrounded; it means that the attribution pattern does not distinguish relevant from placebo passages under this diagnostic.

Interpretation. This global diagnostic complements the passage-level exercise in the main text. The passage-level comparison identifies which retrieved passages receive unusually large positive contributions, whereas the global statistic asks whether positive dependence is concentrated on relevant passages at the event level.

French VAT results. For the French VAT corpus, complete attribution outputs are available for 400 motivations associated with 140 policy actions. Averaging the OLS attribution coefficients across the Qwen and Mistral evaluators, the global test rejects exchangeability between retrieved and placebo passages for 70.5 percent of motivations. The median motivation assigns only 30.4 percent of its positive attribution weight to placebo passages, substantially below the 75 percent expected if attribution were unrelated to passage relevance. At the policy level, 132 of 140 actions contain at least one motivation that passes the test; only eight contain none. The exercise currently covers only the French VAT proposals, a subset of the tax changes studied in the paper. Extending it to all tax changes and to repeated motivation extractions across models is computationally intensive and will be completed using the hardware designated for the paper's replication package.

C Taxonomy and Probability Construction

C.1 Classification Prompts and Category Scheme

This appendix reports the prompt templates that define the eight narrative categories used in the tax applications. Each category corresponds to one of the subtypes summarized in Figure 1, following the narrative taxonomy of Cloyne (2013). The prompts should be read as the semantic dictionary of the measurement problem, not as the main probability estimator. In the baseline construction, these definitions guide the generation of labeled synthetic motivations, and hidden-state features are calibrated on that synthetic corpus. In the alternative signal in Appendix C.5, the same prompts are used directly to compute the model's probability of answering *Yes*.

Prompt templates

Each prompt is presented in a standardized format and corresponds to one category listed in Figure 1. The prompts are organized by group: *endogenous* (Demand Management 1, Supply Stimulus 2, Deficit Reduction 3, Spending-driven 4) and *exogenous* (Long-term 5, Ideological 6, External 7, Deficit Consolidation 8). Each prompt provides the task definition, inclusion and exclusion criteria, and output format. Together, they make the category definitions reproducible and provide the pivot from qualitative narrative taxonomy to synthetic calibration data.

English classification prompt (Demand Management / DM):

Task: Classify a fiscal measure based on its stated motivation. Does the measure correspond to Demand Management (DM)?

Definition of Demand Management (DM):

A measure belongs to this category when it is motivated by the need to change the short-run level or composition of aggregate demand in response to current output, inflation, unemployment, overheating, recession, or balance-of-payments conditions.

This includes:

- Tax increases intended to restrain consumption, imports, aggregate demand, or inflation.
- Tax cuts intended to support consumption, disposable income, investment demand, or economic activity during a slowdown.
- Measures addressing a balance-of-payments problem through demand restraint or import compression.
- Temporary or reversible measures justified by immediate cyclical conditions.

Do NOT classify as Demand Management if:

- The immediate objective is to close a current budget or external financing gap; classify it as Deficit Reduction.
- The measure gives short-run assistance to firms or a distressed sector through costs, liquidity, hiring, or production incentives; classify it as Supply Stimulus.
- The measure is a long-run structural reform intended to improve incentives, productivity, or competitiveness.
- The measure finances a specific spending program or implements a binding legal requirement.

Output format:

- Respond with one option:
- Yes - [short reasoning]
- No - [short reasoning]

Prompt 1: Classification Prompt for the Demand Management (DM) Category

English classification prompt (Supply Stimulus / SS):

Task: Classify a fiscal measure based on its stated motivation. Does the measure correspond to Supply Stimulus (SS)?

Definition of Supply Stimulus (SS):
A measure belongs to this category when it is motivated by short-run support for production, investment, employment, profitability, liquidity, or competitiveness, especially during a downturn or for a currently distressed sector.

This includes:

- Temporary tax relief intended to support firms, hiring, or investment during weak current conditions.
- Measures intended to protect the immediate liquidity or competitiveness of a sector.
- Tax cuts described as supporting recovery through short-run supply conditions.
- Temporary reductions in employment or business taxes intended to limit current job losses.

Do NOT classify as Supply Stimulus if:

- The justification is a durable structural or incentive reform whose purpose is long-run productivity or competitiveness; classify it as Long Run.
- The mechanism is short-run aggregate demand management rather than support for production or firms.
- The motivation is deficit reduction, spending finance, or legal compliance.
- The change is routine indexation or an administrative adjustment without a supply rationale.

Output format:

- Respond with one option:
- Yes - [short reasoning]
- No - [short reasoning]

Prompt 2: Classification Prompt for the Supply Stimulus (SS) Category

English classification prompt (Deficit Reduction / DR):

Task: Classify a fiscal measure based on its stated motivation. Does the measure correspond to Deficit Reduction or a current balance-of-payments crisis (DR)?

Definition of Deficit Reduction / current balance-of-payments crisis (DR):
A measure belongs to this category when it is motivated by a budget or external deficit caused by current conditions, an immediate need to raise or preserve revenue, a current reserve problem, or urgent government financing pressure.

This includes:

- Measures introduced to close a current budget deficit or meet an immediate Treasury financing need.
- Measures responding directly to a current external deficit or loss of reserves.
- Revenue measures responding to current investor-confidence or financial-market pressure on government financing.
- Temporary levies or restrictions of reliefs introduced to meet a documented near-term revenue shortfall.

Do NOT classify as Deficit Reduction if:

- The measure addresses a balance-of-payments problem by changing aggregate demand or imports; classify that mechanism as Demand Management.
- The measure implements an inherited or medium- or long-run fiscal consolidation strategy not forced by current conditions; classify it as Deficit Consolidation.
- The measure finances a specific new spending program; classify it as Spending-Driven.
- The stated reason is compliance with a pre-existing binding legal or institutional requirement; classify it as External.
- The document reports a revenue effect but does not state a current deficit, reserve, or financing rationale.

Output format:

- Respond with one option:
- Yes - [short reasoning]
- No - [short reasoning]

Prompt 3: Classification Prompt for the Deficit Reduction (DR) Category

English classification prompt (Spending-Driven / SD):

Task: Classify a fiscal measure based on its stated motivation. Does the measure correspond to Spending-Driven (SD)?

Definition of Spending-Driven (SD):
A measure belongs to this category when it is explicitly motivated by the need to finance, offset, or enable a specific increase in public spending or a named public program.

This includes:

- Tax increases introduced to cover the cost of a new or expanded program.
- Revenue measures described as necessary to finance a specific spending proposal.
- Exceptional levies earmarked for a stated public expenditure.
- Restrictions of tax reliefs intended to release resources for a targeted spending program.
- Charges introduced to recover the stated public cost of a specific service or program.

Do NOT classify as Spending-Driven if:

- The motivation is general current deficit reduction or medium-term consolidation.
- The measure is intended to manage aggregate demand.
- A spending program is mentioned but the tax change is not presented as financing or offsetting it.
- The stated reason is compliance with a binding legal or institutional obligation.

Output format:

- Respond with one option:
- Yes - [short reasoning]
- No - [short reasoning]

Prompt 4: Classification Prompt for the Spending-Driven (SD) Category

English classification prompt (Long Run / LR):

Task: Classify a fiscal measure based on its stated motivation. Does the measure correspond to Long Run (LR)?

Definition of Long Run (LR):

A measure belongs to this category when it is motivated by durable structural, supply-side, or incentive effects intended to improve productivity, investment, saving, competitiveness, or economic efficiency over time, rather than to offset a current shock.

This includes:

- Reforms intended to improve work, saving, investment, or entrepreneurial incentives over the long run.
- Measures intended to raise productivity, capital formation, or technological modernization.
- Durable reforms intended to improve efficiency or international competitiveness.
- Tax simplification and deregulation measures with a stated structural rationale.
- Long-term support for businesses or sectors that is not introduced in response to current distress.

Do NOT classify as Long Run if:

- The measure preserves the immediate liquidity, employment, or competitiveness of a distressed firm or sector; classify it as Supply Stimulus.
- The measure manages short-run demand, inflation, output, or employment.
- The measure responds to a current deficit or financing emergency.
- Competitiveness is mentioned only as a response to current external pressure.
- Simplification is purely administrative and no structural or incentive rationale is stated.

Output format:

- Respond with one option:
- Yes - [short reasoning]
- No - [short reasoning]

Prompt 5: Classification Prompt for the Long Run (LR) Category

English classification prompt (Ideological / IDEO):

Task: Classify a fiscal measure based on its stated motivation. Does the measure correspond to Ideological (IDEO)?

Definition of Ideological (IDEO):
A measure belongs to this category when it is motivated by a long-horizon social or political goal, a view about fairness or the proper design of the tax system, or a standing political commitment, independently of current macroeconomic performance and not to offset a current shock.

This includes:

- Measures justified by fairness, justice, social values, public health, environmental responsibility, or distributional principles.
- Reliefs or exemptions for groups described as deserving or vulnerable.
- Reforms implementing an electoral promise or standing political commitment.
- Anti-avoidance or anti-evasion measures when no current deficit, financing, or stabilization motive is stated.
- Changes intended to make the tax system fairer or more coherent as a matter of political or social philosophy.

Do NOT classify as Ideological if:

- The operative motivation is current demand management, deficit reduction, or financing pressure.
- The operative motivation is medium- or long-run fiscal consolidation.
- The measure is required by a pre-existing binding legal or institutional obligation.
- Anti-avoidance is used specifically to address a stated current revenue shortfall.
- The change is routine indexation or a neutral administrative adjustment without a stated social, political, or anti-avoidance objective.

Output format:

- Respond with one option:
- Yes - [short reasoning]
- No - [short reasoning]

Prompt 6: Classification Prompt for the Ideological (IDEO) Category

English classification prompt (External / EXT):

Task: Classify a fiscal measure based on its stated motivation. Does the measure correspond to External (EXT)?

Definition of External (EXT):
A measure belongs to this category when a pre-existing court ruling or binding legal or institutional obligation requires the government to change tax policy and leaves no contemporaneous discretion over whether to act. The operative motivation must be compliance with that requirement, not a response to current macroeconomic conditions.

This includes:

- Measures required to implement a court or judicial ruling.
- Measures required to enforce a binding EU or international directive.
- Measures required for compliance with a ratified treaty or binding international agreement when institutional compliance is the stated reason.
- Measures required by another pre-existing legal obligation that compels the tax change.

Do NOT classify as External if:

- The measure is merely enacted in a statute or finance bill; ordinary legislation does not by itself make a discretionary tax change External.
- The measure responds to a balance-of-payments problem, exchange-rate pressure, reserve loss, or an external financing shortage.
- The measure supports exports, international competitiveness, foreign trade, or investor confidence.
- The measure responds to a contemporaneous foreign or domestic macroeconomic shock.
- An international institution is mentioned but the operative motivation is demand management, current deficit reduction, or long-run reform.

Boundary examples:

- A tax change required to comply with a binding EU directive is External.
- A tax increase intended to reduce imports during a balance-of-payments crisis is Demand Management.
- A tax increase introduced to meet an immediate external financing shortfall is Deficit Reduction.
- A tax reform intended to improve international competitiveness over time is Long Run.

Output format:

- Respond with one option:
- Yes - [short reasoning]
- No - [short reasoning]

Prompt 7: Classification Prompt for the External (EXT) Category

English classification prompt (Deficit Consolidation / DC):

Task: Classify a fiscal measure based on its stated motivation. Does the measure correspond to Deficit Consolidation (DC)?

Definition of Deficit Consolidation (DC):
A measure belongs to this category when it is motivated by an inherited fiscal imbalance, a standing fiscal philosophy, or a medium- or long-run strategy to strengthen public finances, reduce borrowing, or lower public debt, rather than by a financing problem created by current conditions.

This includes:

- Measures implementing a multi-year fiscal consolidation plan.
- Structural revenue measures intended to improve the fiscal balance over several years.
- Measures addressing an inherited deficit or debt burden.
- Measures intended to offset the future fiscal cost of current policy actions.

Do NOT classify as Deficit Consolidation if:

- The measure responds to a current budget deficit, external deficit, reserve problem, or immediate financing pressure; classify it as Deficit Reduction.
- The revenue is introduced to finance a specific spending program; classify it as Spending-Driven.
- The measure is a short-run demand intervention or response to current output or inflation conditions.
- The stated reason is compliance with a pre-existing binding legal or institutional requirement.

Output format:

- Respond with one option:
- Yes - [short reasoning]
- No - [short reasoning]

Prompt 8: Classification Prompt for the Deficit Consolidation (DC) Category

C.2 Synthetic Calibration Design

The calibration corpus is generated from the prompts defining the eight narrative categories above. For each category, I generate motivations under four complementary designs: canonical, institutional, boundary, and mixed-motive examples. The generated observations are self-contained, between 14 and 90 words, and exclude category names, classifier terminology, headings, and repeated templates. Boundary and mixed-motive examples retain potentially confounding or secondary considerations in the text, while a designated primary motivation determines the category label. The four generation designs are summarized below.

Table 13: Generation Designs for the Synthetic Calibration Samples

Design	Generation instruction
Canonical	Write clear positive examples for the target category in natural policy prose, without headings, category names, or repeated openings.
Institutional	Use the style of extracted U.S., Romer–Romer, or Cloyne policy motivations, including institutional references and policy language, while keeping every example synthetic.
Boundary	Include one misleading secondary cue from a nearby category while keeping the target category as the primary reason for the policy.
Mixed motive	Combine one primary target category with one secondary motivation in language resembling an actual policy document.

The following observations are taken verbatim from the stored calibration corpus and illustrate the differences across designs:

Canonical, long run. The Budget Statement framed the reduction in the corporate rate as a way to lift productivity by encouraging firms to renew plant, adopt newer methods, and commit capital with greater confidence over the long run.

Institutional, spending driven. The committee report states that the excise increase on tobacco products was adopted to provide dedicated receipts for the new federal health-insurance program beginning in fiscal 1974. The estimated yield was described as the financing source for premiums and administrative costs, not as a general budget device.

Boundary, demand management. The temporary cut in the sales levy was advanced to support household consumption while retail orders were weakening and factory shifts were being trimmed. Ministers noted that the measure could be withdrawn once inflation settled and private demand recovered. A separate note acknowledged the smaller receipts, but the immediate aim was to keep tills moving through the next few quarters.

Mixed motive, external. Following the Constitutional Court's ruling that the inheritance tax unduly discriminated between direct descendants and more distant relatives, the government is obliged to amend the schedule of rates and exemptions within the timeframe set by the judgment. Ministers note that, although the reform will incidentally reduce planning opportunities and stabilize receipts, the core objective is to bring the law into conformity with constitutional equality guarantees and avoid further successful litigation.

For each design, I generate three independent samples using seeds 11, 29, and 47. Each sample contains 70 motivations per category, or 560 observations, yielding 6,720 synthetic motivations across the twelve calibration samples. The complete texts, category labels, secondary-category fields, generation design, and associated metadata are retained in the replication materials.

C.3 Probability-Map Estimation

This subsection provides the estimation details behind the probability map summarized in Section 3.2. Each of the twelve synthetic sets contains 70 motivations in each of the eight categories, for a balanced sample of 560 observations. Every synthetic motivation has one primary category, which provides its label. Examples may include secondary considerations to reproduce mixed policy language, but these are not treated as fractional labels.

For language model m and synthetic set k, let $h^{(m)}(M)$ denote the hidden-state representation of motivation M. I estimate a ridge-penalized multinomial logit mapping from this representation to the eight narrative categories:

$$\widehat{P}_{m,k}(c \mid M) = \frac{\exp\bigl(\widehat{\alpha}_{c,m,k} + h^{(m)}(M)'\widehat{\gamma}_{c,m,k}\bigr)}{\sum_{c' \in \mathcal{G}} \exp\bigl(\widehat{\alpha}_{c',m,k} + h^{(m)}(M)'\widehat{\gamma}_{c',m,k}\bigr)}, \qquad c \in \mathcal{G}.$$

The shrinkage parameter is selected by five-fold stratified cross-validation within each synthetic set. Applying the estimated map to historical motivation M_{ij} gives

$$p_{ij}^{(m,k)} = \sum_{c \in \mathcal{G}_{\text{exo}}} \widehat{P}_{m,k}(c \mid M_{ij}),$$

where $\mathcal{G}_{\text{exo}}$ contains the categories associated with exogenous policy variation.

For each set–model pair, motivation-level probabilities are averaged within policy action:

$$p_i^{(m,k)} = \frac{1}{L_i} \sum_{j=1}^{L_i} p_{ij}^{(m,k)}.$$

The baseline policy-level probability averages over the twelve independently generated synthetic sets and the three language models:

$$p_i = \frac{1}{36} \sum_{m=1}^{3} \sum_{k=1}^{12} p_i^{(m,k)}.$$

Set- and model-specific estimates are retained to assess how strongly the resulting support score depends on the synthetic sample and language-model representation. The associated diagnostics are reported below.

C.4 Model Replication Details

The hidden-state representation stage and the open-weight motivation-extraction sensitivity exercise use the same pinned model versions. The exact language-model identity in these exercises is the pair consisting of the Hugging Face repository identifier and the resolved revision SHA: Runs used Python 3.11.5, PyTorch 2.10.0+`cu128`, and a single NVIDIA L40S GPU.

Table 14: Language-model identifiers used for representation and extraction sensitivity

Short name	Hugging Face repository	Resolved revision
Gemma-4B	`google/gemma-4-E4B-it`	`d6436b3d62967e1af08bbb046c6300b2a9ae8e85`
Mistral-7B	`mistralai/Mistral-7B-Instruct-v0.2`	`63a8b081895390a26e140280378bc85ec8bce07a`
Qwen-9B	`Qwen/Qwen3.5-9B`	`c202236235762e1c871ad0ccb60c8ee5ba337b9a`

C.5 Alternative LLM Signal: Incremental Log Probabilities

The main text uses hidden-state calibration as the baseline way to extract category probabilities from language models. A simpler alternative is to use the same semantic prompts more directly. For motivation $M_{i\ell}$ and category $c \in \mathcal{G}$, let X_c be the category-specific prompt reported in Appendix C.1. Define the incremental log-probability signal

$$s_{i\ell,c}^{\log} := \log \Pr_{\text{LM}}(\text{Yes} \mid X_c \cup M_{i\ell}) - \log \Pr_{\text{LM}}(\text{Yes} \mid X_c). \tag{13}$$

The first term measures the model's support for category c after observing the motivation; the second removes the prompt-specific baseline tendency to answer "Yes." Collect these signals in

$$Z_{i\ell}^{\log} = \left(s_{i\ell,c}^{\log}\right)_{c\in\mathcal{G}}.$$

Because token-probability signals generated by different prompts need not be on a common probability scale, they should be treated as features to be calibrated rather than as final category probabilities. The same synthetic motivations used in the baseline calibration can provide the training sample: each synthetic motivation is generated from the category definitions in Appendix C.1, so its category label is known by construction. One can therefore estimate, on this calibration sample, a multinomial mapping

$$\Pr(C_{i\ell} = c \mid Z_{i\ell}^{\log}) := \frac{\exp(\alpha_c + Z_{i\ell}^{\log\,\prime}\beta_c)}{\sum_{c'\in\mathcal{G}} \exp(\alpha_{c'} + Z_{i\ell}^{\log\,\prime}\beta_{c'})}, \qquad c \in \mathcal{G}. \tag{14}$$

The resulting fitted probabilities can then be aggregated exactly as in Section 2.1: sum the calibrated category probabilities over $\mathcal{G}_{\text{exo}}$ to obtain motivation-level support and average motivation-level support within the policy action to obtain p_i. This construction is useful as an alternative signal extraction method that keeps the prompt templates directly in the measurement equation. The main application instead uses hidden-state calibrated probabilities, reported in Table 1, because the hidden-state route avoids relying on the model's final Yes/No token probabilities as the primary signal.

D Calibration Diagnostics

This appendix reports diagnostics for the synthetic calibration exercise used to construct the narrative-support probabilities. The external Romer–Romer and Cloyne exercises in Section 4 assess the economic content of the score; the diagnostics here assess the construction of the synthetic corpus and the stability of the resulting probability scale across calibration datasets and language-model representations.

D.1 Synthetic construction diagnostics

I audit the synthetic calibration corpus before using it to train the hidden-state classifier. These checks are construction diagnostics: they are designed to detect mechanical artifacts in the synthetic examples, not to validate the empirical identifying assumptions.

First, I verify exact balance. Each calibration set must contain the same eight motivation categories and exactly 70 examples per category. This preserves the row-block structure used by the hidden-state extraction workflow and prevents category imbalance from mechanically affecting the multinomial mapping.

Second, I scan each motivation for prompt artifacts. The audit searches for banned prompt residues and template phrases such as "demand management motive", "program funding motive", "recovery through incentives", "values-based reform", "structural reform motive", and related heading-like expressions. These checks are intentionally narrower than a general search for economically meaningful words: terms such as "long-term" or "deficit" may be legitimate content, while prompt headings or classifier language would indicate leakage from the generation prompt.

Third, I check for duplicate and near-duplicate examples. Exact duplicates are computed after lowercasing and normalizing punctuation and whitespace. Near duplicates are computed within each set and category using Jaccard similarity over normalized token sets, with the configured threshold equal to 0.72. This check is stricter than simply looking for identical strings and is meant to ensure that a category block is not populated by lightly reworded copies of the same example.

Fourth, I check for repeated openings. For each motivation, the audit records the first seven words and counts repeated openings within each set-category cell. The construction gate flags openings that occur more than three times within the same category and set. This prevents the classifier from exploiting formulaic first clauses rather than the substantive motivation.

Fifth, I estimate simple lexical shortcut diagnostics. For each synthetic set, I train cross-validated TF–IDF multinomial classifiers to predict the eight categories. One classifier uses the full motivation text; the other uses only the opening phrase. High full-text separability is expected in a synthetic category-labeled corpus and is reported only as a diagnostic. The more important shortcut check is the opening-only classifier: if the first few words alone predict category too well, the set may encode category through templates rather than motivation content.

Finally, I compare the synthetic texts to real extracted motivations from the extension and Romer–Romer/Cloyne replication files. This is a coverage diagnostic, not a training step. The audit computes TF–IDF nearest-neighbor similarities between real motivations and synthetic examples, reports average and lower-tail similarities, and stores the least-covered real motivations for inspection. Real labeled observations are not added to the synthetic training set in this version of the calibration exercise.

For the main robust synthetic corpus, all twelve generated sets are exactly balanced. The audit finds zero banned prompt-artifact rows, zero repeated-opening violations, zero exact duplicates, and zero near-duplicate pairs above the configured threshold. The opening-only classifier remains much less predictive than the full-text classifier, although a small number of generated sets are flagged as mild shortcut warnings. These warnings are retained as diagnostics and motivate the stability analysis across independently generated calibration sets.

D.2 Synthetic Holdout Performance

For each synthetic calibration set and each language model, I train the multinomial hidden-state decoder using five-fold stratified cross-validation within the synthetic corpus. The resulting diagnostics evaluate whether the category distinctions embedded in the synthetic taxonomy can be recovered from the hidden-state representations.

Table 15 reports average holdout performance across the twelve independently generated calibration sets. Performance is uniformly strong across models. Brier scores and log-loss values are low, classification accuracy is high, and the binary exogeneity classification implied by the taxonomy is almost perfectly separated in the synthetic environment.

Table 15: Synthetic Holdout Performance

Model	Brier	Log-loss	Accuracy	AUC	8-category log-loss
Gemma-4B	0.026	0.094	0.968	0.995	0.259
Mistral-7B	0.008	0.031	0.989	0.999	0.068
Qwen-9B	0.007	0.025	0.992	0.999	0.061

Note: Entries average five-fold stratified holdout performance across the twelve balanced synthetic calibration samples.

These results should be interpreted with caution. Synthetic examples are necessarily cleaner and more category-separated than real policy language. Strong synthetic performance therefore demonstrates that the category definitions are internally coherent and recoverable from hidden-state representations, but does not by itself establish external validity.

D.3 Stability Across Calibration Sets

The calibration design deliberately uses twelve independently generated datasets. Each dataset is used to train a separate decoder, producing twelve independent probability estimates for every policy action. This design makes it possible to assess how sensitive the final support score is to the particular synthetic examples used during calibration.

Table 16 reports agreement across the twelve set-specific probabilities. Correlations are generally high, although meaningful variation remains in some samples. The median standard deviation across set-trained decoders ranges from roughly 0.08 to 0.11, indicating that calibration uncertainty is non-negligible but substantially smaller than the overall variation in support probabilities. The 90th percentile of the set-level standard deviation captures the uncertainty in harder-to-classify observations.

Table 16: Agreement Across Calibration Samples

Sample	Mean Pearson	Mean abs. diff.	Median set s.d.	P90 set s.d.
U.S. RR replication	0.901	0.118	0.105	0.150
U.S. extension	0.788	0.133	0.108	0.186
France VAT	0.766	0.105	0.082	0.147
France non-VAT	0.747	0.095	0.076	0.136

Note: Pairwise agreement is computed across the twelve calibration-sample-specific support estimates after averaging over representation models. The last two columns summarize within-policy dispersion across samples.

The baseline support probability averages over these twelve calibration datasets. The remaining dispersion is retained as a calibration-uncertainty measure and reported alongside the final support scores.

D.4 Stability Across Models and Measurement Constructions

The baseline specification also averages probabilities across three open-weight language models: Gemma-4B, Mistral-7B, and Qwen-9B. This averaging is intended to reduce dependence on any particular hidden-state geometry.

Table 17 reports agreement across the model-specific probabilities after averaging over calibration samples. Pairwise correlations are generally high across applications, although the French samples exhibit somewhat greater dispersion than the U.S. samples. The lower panel separates variation across extraction models from stochastic extraction variation and reports the joint U.S. budget construction.

Table 17: Agreement Across Models and Measurement Constructions

Sample or source	N / variants	Mean Pearson	Mean abs. diff.	Median s.d.
U.S. RR replication	50	0.817	0.130	0.095
U.S. extension	1,188	0.806	0.129	0.091
France VAT	140	0.679	0.155	0.114
France non-VAT	1,020	0.753	0.138	0.103
Extraction model, draw-averaged	3	0.797	0.108	0.077
Stochastic draw within extraction model	20	0.926	0.050	0.042
All varied choices jointly	2,160	0.525	0.214	0.202

Note: The first four rows vary the language model used to represent a fixed motivation after averaging over calibration samples. The remaining rows use the U.S. budget application and vary motivation extraction. The joint construction combines extraction model, extraction draw, representation model, and calibration sample; its variants share components and are not independent draws.

The separate construction choices remain positively related, but their joint variation is materially larger than variation along any single dimension.

D.5 Dimensionality of the Hidden-State Representation

Each synthetic calibration set contains 560 observations, whereas the hidden-state representations contain 2,560 coordinates for Gemma-4B and 4,096 coordinates for Mistral-7B and Qwen-9B. The baseline multinomial logit addresses this dimensional imbalance through coefficient shrinkage selected by cross-validation. I additionally examine whether the resulting probabilities depend on the full dimension of the hidden-state representation.

For each calibration set and language model, I re-estimate the probability mapping after retaining either 32 or 64 principal components. Standardization and principal-component estimation are performed separately within each cross-validation training fold. Held-out observations are therefore transformed using a principal-component basis estimated without them. After selecting the shrinkage parameter, I re-estimate the mapping on the complete synthetic set and apply it to the historical motivations. This procedure is repeated for all twelve calibration sets and three language models, and the resulting probabilities are aggregated exactly as in the baseline.

Table 18 shows that dimensionality reduction preserves the ranking and levels of the historical support scores. The correlations with the baseline exceed 0.97 in every sample, and the external-

Table 18: Robustness to Hidden-State Dimensionality

Panel A: Agreement with the full hidden-state probability

Sample	Specification	Pearson	Spearman	MAE	P90 abs. diff.
U.S. RR replication	PCA 64	0.995	0.991	0.022	0.044
	PCA 32	0.991	0.987	0.034	0.064
U.S. budget application	PCA 64	0.991	0.990	0.022	0.047
	PCA 32	0.986	0.985	0.028	0.061
Cloyne validation	PCA 64	0.990	0.984	0.017	0.039
	PCA 32	0.982	0.974	0.023	0.053
France VAT	PCA 64	0.989	0.985	0.019	0.041
	PCA 32	0.980	0.977	0.024	0.049
France non-VAT	PCA 64	0.988	0.985	0.019	0.040
	PCA 32	0.978	0.971	0.027	0.055

Panel B:

Separation in external narrative benchmarks

	Romer–Romer		Cloyne	
Specification	Exo–endo gap	AUC	Exo–endo gap	AUC
Full hidden state	0.176	0.699	0.101	0.655
PCA 64	0.166	0.712	0.096	0.659
PCA 32	0.152	0.700	0.089	0.652

Note: Panel A compares policy-level probabilities from each reduced-dimensional specification with the full-hidden-state baseline. Panel B reports the mean-support gap between externally classified exogenous and endogenous observations. Standardization and principal-component estimation are performed within each calibration fold.

benchmark gaps remain close to those obtained from the full hidden states. Figure 11 carries the same exercise through the support-indexed GMM estimator. The GDP and inflation responses are similar across the three representations.

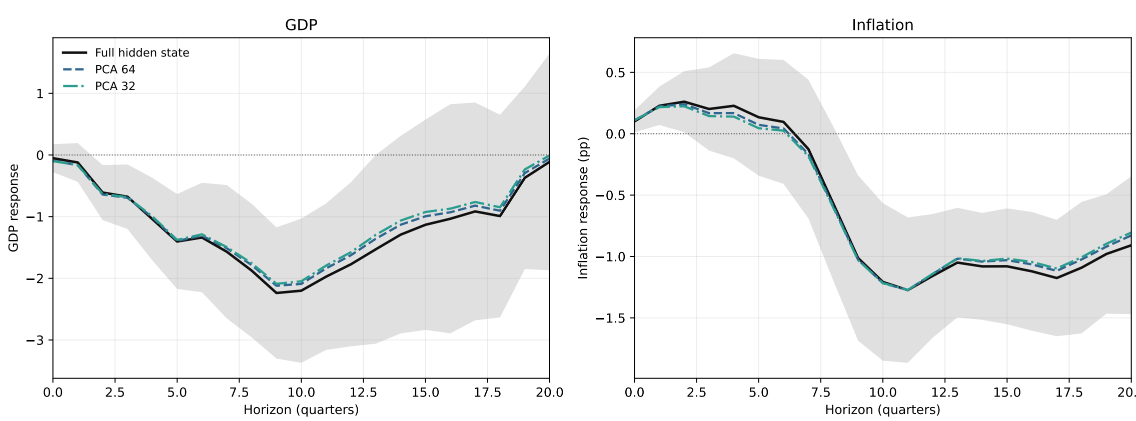


Figure 11: U.S. Budget Responses under Alternative Hidden-State Dimensions

Note: The black line reports the baseline affine $p = 1$ endpoint and the shaded area its 68% confidence interval. The dashed and dash-dotted lines re-estimate the same support-indexed GMM specification using probabilities constructed from 64 and 32 principal components, respectively.

This exercise isolates the role of representation dimensionality while holding fixed the synthetic motivations, category labels, language models, and aggregation rule. It does not test whether the synthetic categories fully span the historical language. Rather, it shows that the empirical results do not depend on retaining idiosyncratic directions in the full hidden-state representation.

E Estimation and Inference for Support-Indexed Moments

This appendix provides the complete sample-moment representation and inference procedure for the high-support endpoint reported in Subsection 3.4. The estimator uses the full quarterly sample. In quarters without a recorded tax action, I set both the tax change and narrative support equal to zero. These observations therefore enter the support-indexed moment relation at $p_t = 0$; this is an estimation convention and does not define a separately observed policy class.

Throughout the appendix, the constructed support scores are treated as observed regressors. The sampling distribution therefore describes uncertainty in the macroeconomic estimator conditional on the document-to-support mapping; it does not include an additional first-stage variance component from reconstructing the support scores.

E.1 Population Target and Sample Estimator

Fix a response horizon h, let $\mathcal{T}_h$ denote the ordered quarterly estimation sample, and let $n_h = |\mathcal{T}_h|$. The dimension of the control vector $W_{t,h}$ is fixed as $n_h \to \infty$. The residual functions, parameter vector η_h, and complete stacked moment contribution $g_{t,h}(\eta_h)$ are defined in Section 3.4 and equation (6). The corresponding sample-average moment is

$$\overline{g}_h(\eta_h) = \widehat{E}_h[g_{t,h}(\eta_h)] = \frac{1}{n_h} \sum_{t \in \mathcal{T}_h} g_{t,h}(\eta_h).$$

The estimator $\widehat{\eta}_h$ solves

$$\overline{g}_h(\widehat{\eta}_h) = 0. \tag{15}$$

Because the system is just identified, the GMM weighting matrix does not affect the point estimate. The solution is equivalent to the sequential implementation in the main text. In particular,

$$\widehat{\pi}_{y,h} = \arg\min_{\pi} \sum_{t \in \mathcal{T}_h} \left(y_{t+h} - \pi' W_{t,h}\right)^2, \qquad \widehat{\pi}_{\tau,h} = \arg\min_{\pi} \sum_{t \in \mathcal{T}_h} \left(\Delta\tau_t - \pi' W_{t,h}\right)^2.$$

Writing

$$\widetilde{y}_{t,h} = \widetilde{y}_{t,h}(\widehat{\pi}_{y,h}), \qquad \widetilde{\Delta\tau}_{t,h} = \widetilde{\Delta\tau}_{t,h}(\widehat{\pi}_{\tau,h}),$$

the numerator and denominator observations are

$$x_{t,h} = \widetilde{y}_{t,h} \widetilde{\Delta\tau}_{t,h}, \qquad z_{t,h} = \widetilde{\Delta\tau}_{t,h}^{2}.$$

For compactness in the derivations below, define

$$R_t = (1, p_t)', \qquad \theta_{N,h} = (a_h, b_h)', \qquad \theta_{D,h} = (a_{D,h}, b_{D,h})', \qquad \iota = (1, 1)'.$$

The two coefficient vectors simply regroup the last four entries of η_h. Provided that $\sum_{t\in\mathcal{T}_h} R_t R_t'$ is nonsingular,

$$\widehat{\theta}_{N,h} = \left(\sum_{t\in\mathcal{T}_h} R_t R_t'\right)^{-1} \sum_{t\in\mathcal{T}_h} R_t x_{t,h}, \qquad \widehat{\theta}_{D,h} = \left(\sum_{t\in\mathcal{T}_h} R_t R_t'\right)^{-1} \sum_{t\in\mathcal{T}_h} R_t z_{t,h}. \tag{16}$$

The reported high-support response is

$$\widehat{\beta}_h = \frac{\iota'\widehat{\theta}_{N,h}}{\iota'\widehat{\theta}_{D,h}} = \frac{\widehat{a}_h + \widehat{b}_h}{\widehat{a}_{D,h} + \widehat{b}_{D,h}}. \tag{17}$$

E.2 Identification of the Sample Moments

The stacked system is locally identified under transparent rank conditions. Let $d_{W,h}$ denote the dimension of $W_{t,h}$, and write

$$\Sigma_{W,h} = \operatorname{plim} \frac{1}{n_h} \sum_{t\in\mathcal{T}_h} W_{t,h} W_{t,h}', \qquad \Sigma_{R,h} = \operatorname{plim} \frac{1}{n_h} \sum_{t\in\mathcal{T}_h} R_t R_t'.$$

The population Jacobian has the block lower-triangular form

$$G_h = -\begin{pmatrix} \Sigma_{W,h} & 0 & 0 & 0 \\ 0 & \Sigma_{W,h} & 0 & 0 \\ G_{xy,h} & G_{x\tau,h} & \Sigma_{R,h} & 0 \\ 0 & 2G_{xy,h} & 0 & \Sigma_{R,h} \end{pmatrix}, \tag{18}$$

where

$$G_{xy,h} = \mathbb{E}\left[R_t \widetilde{\Delta\tau}_{t,h} W_{t,h}'\right], \qquad G_{x\tau,h} = \mathbb{E}\left[R_t \widetilde{y}_{t,h} W_{t,h}'\right].$$

These off-diagonal blocks capture the effect of the residualization coefficients on the numerator and denominator moments. Their values do not affect the rank argument because the Jacobian is block lower triangular. If $\Sigma_{W,h}$ and $\Sigma_{R,h}$ are positive definite, then G_h is nonsingular. The first condition is the usual full-rank requirement for the controls. The second requires nondegenerate variation in narrative support; because $R_t = (1, p_t)'$, it is equivalent to a positive limiting variance of p_t.

This rank condition concerns identification of the statistical affine projection. In the full-calendar implementation, quarters without a recorded policy action satisfy $p_t = 0$ by convention, so variation between event and non-event quarters may contribute to $\Sigma_{R,h}$. The structural interpretation of the high-support endpoint additionally relies on variation in narrative support over policy variation, together with the calibration and within-type moment-stability restrictions in Section 1.4. In particular, the corresponding affine restriction must hold for both the numerator moment and the squared-tax denominator. Absent these restrictions, equation (17) remains a well-defined affine projection ratio evaluated at $p = 1$, but need not equal the structural response β_h. Appendix H assesses directly how the fitted response at $p = 1$ changes when the no-action completion is varied or when no-action quarters are excluded from the support-indexed projections.

E.3 Regularity Conditions

The asymptotic results are pointwise in the response horizon: h is held fixed while $n_h \to \infty$. This is the relevant approximation for the pointwise confidence bands reported in the paper. Let

$$g_h^0(\eta_h) = \lim_{n_h \to \infty} E[\bar{g}_h(\eta_h)]$$

whenever this limit exists.

Assumption E.1 (Fixed-horizon regularity)**.** For each fixed horizon h, the following conditions hold.

1. The parameter space is compact, $\eta_{h,0}$ lies in its interior, and $\eta_{h,0}$ is the unique zero of $g_h^0(\eta_h)$.

2. For some $\delta > 0$,
$$\sup_{t \in \mathcal{T}_h} E\Big[(1 + |y_{t+h}| + |\Delta\tau_t| + \|W_{t,h}\|)^{4+\delta}\Big] < \infty,$$
and $p_t \in [0, 1]$. These bounds ensure that the stacked scores and their first derivatives have the moments required below.

3. A uniform law of large numbers holds for the stacked moments and their first derivatives in a neighborhood of $\eta_{h,0}$. The sample Jacobian converges uniformly in probability to a finite matrix G_h.

4. The rank conditions
$$\Sigma_{W,h} \succ 0, \qquad \Sigma_{R,h} \succ 0$$
hold. Hence G_h is nonsingular by equation (18).

5. The score sequence satisfies the central limit theorem
$$\frac{1}{\sqrt{n_h}} \sum_{t \in \mathcal{T}_h} g_{t,h}(\eta_{h,0}) \Rightarrow \mathcal{N}(0, \Omega_h), \tag{19}$$
where Ω_h is finite and positive semidefinite.

6. The population endpoint denominator is bounded away from zero:
$$\iota' \theta_{D,h,0} \geq \underline{q}_h > 0. \tag{20}$$

Assumption E.1 is stated at a high level to allow for changes in the unconditional distribution of the macroeconomic variables over the historical sample. It does not require strict stationarity. Standard weak-dependence conditions for time-series arrays, together with the displayed moment bound, provide primitive sufficient conditions for the uniform laws and central limit theorem. The substantive requirements for this application are that the quarterly score sequence obeys a central limit theorem, that the controls and support index retain nondegenerate variation, and that the endpoint denominator is separated from zero.

E.4 Asymptotic Distribution

Under Assumption E.1, standard GMM arguments imply

$$\sqrt{n_h}\left(\widehat{\eta}_h - \eta_{h,0}\right) \Rightarrow \mathcal{N}\big(0, G_h^{-1}\Omega_h G_h^{-1\prime}\big)\,. \tag{21}$$

Define the response map

$$\beta_h(\eta_h) = \frac{\iota'\theta_{N,h}}{\iota'\theta_{D,h}}.$$

The estimator and target are obtained by evaluating this map at $\widehat{\eta}_h$ and $\eta_{h,0}$, respectively. Then

$$\sqrt{n_h}\left[\widehat{\beta}_h - \beta_h\right] \Rightarrow \mathcal{N}\big(0, \sigma^2_{\beta,h}\big)\,, \tag{22}$$

where

$$\sigma^2_{\beta,h} = \nabla\beta_h(\eta_{h,0})' G_h^{-1}\Omega_h G_h^{-1\prime}\nabla\beta_h(\eta_{h,0}). \tag{23}$$

Equation (21) follows from a mean-value expansion of the sample moments, the score central limit theorem, and Slutsky's theorem. The denominator condition (20) makes $\beta_h(\eta_h)$ continuously differentiable near $\eta_{h,0}$, so equations (22) and (23) follow from the delta method.

E.5 Long-Run Covariance Estimation

Let

$$\widehat{g}_{t,h} = g_{t,h}(\widehat{\eta}_h), \qquad \widehat{G}_h = \left.\frac{\partial \overline{g}_h(\eta)}{\partial \eta'}\right|_{\eta=\widehat{\eta}_h}.$$

Write the ordered sample as $\mathcal{T}_h = \{t_1 < \cdots < t_{n_h}\}$. For $\ell \geq 0$, define

$$\widehat{\Gamma}_{\ell,h} = \frac{1}{n_h}\sum_{r=\ell+1}^{n_h} \widehat{g}_{t_r,h}\widehat{g}'_{t_{r-\ell},h}.$$

The Bartlett-kernel HAC estimator is

$$\widehat{\Omega}_h = \widehat{\Gamma}_{0,h} + \sum_{\ell=1}^{L_{n,h}}\left(1 - \frac{\ell}{L_{n,h}+1}\right)\left(\widehat{\Gamma}_{\ell,h} + \widehat{\Gamma}'_{\ell,h}\right). \tag{24}$$

For the asymptotic sequence, set

$$L_{n,h} = \max\left\{4, h, \left\lfloor n_h^{1/4}\right\rfloor\right\}. \tag{25}$$

For fixed h, this rule satisfies $L_{n,h} \to \infty$ and $L_{n,h}/n_h^{1/2} \to 0$.

Assumption E.2 (Long-run variance estimation)**.** For each fixed h, the weak-dependence and moment conditions governing the score sequence are sufficient for the Bartlett estimator in equation (24), with bandwidth (25), to satisfy

$$\widehat{\Omega}_h \xrightarrow{p} \Omega_h. \tag{26}$$

Assumption E.2 is the standard HAC consistency requirement. It is satisfied, for example, under conventional mixing or near-epoch-dependence conditions with summable dependence and the moment bound in Assumption E.1. The bandwidth in (25) grows slowly enough to estimate increasingly distant covariance terms while remaining asymptotically negligible relative to the sample size. In the samples used in the paper, $\lfloor n_h^{1/4} \rfloor \leq 4$, so the rule coincides with the implemented choice $L_h = \max\{4, h\}$. Thus the asymptotic sequence does not alter the reported estimates or standard errors. Robustness checks vary the minimum bandwidth and its rate of increase.

The covariance estimator for the stacked parameter vector is

$$\widehat{V}_{\eta,h} = \frac{1}{n_h} \widehat{G}_h^{-1} \widehat{\Omega}_h \widehat{G}_h^{-1\prime}. \tag{27}$$

Under Assumptions E.1 and E.2, $\widehat{V}_{\eta,h}$ consistently estimates the pointwise asymptotic covariance of $\widehat{\eta}_h$.

E.6 Inference for the High-Support Response

The gradient of the response map has zero direct entries with respect to the residualization coefficients and satisfies

$$\nabla_{\theta_N} \beta_h(\eta_h) = \frac{\iota}{\iota' \theta_{D,h}}, \qquad \nabla_{\theta_D} \beta_h(\eta_h) = -\frac{\iota' \theta_{N,h}}{(\iota' \theta_{D,h})^2} \iota.$$

Equivalently,

$$\nabla \beta_h(\eta_h) = \begin{pmatrix} 0_{d_{W,h}} \\ 0_{d_{W,h}} \\ \iota / (\iota' \theta_{D,h}) \\ -(\iota' \theta_{N,h}) \iota / (\iota' \theta_{D,h})^2 \end{pmatrix}. \tag{28}$$

The estimated variance of the response is

$$\widehat{V}_{\beta,h} = \nabla \beta_h(\widehat{\eta}_h)' \widehat{V}_{\eta,h} \nabla \beta_h(\widehat{\eta}_h). \tag{29}$$

Although the response map does not depend directly on $\pi_{y,h}$ or $\pi_{\tau,h}$, uncertainty from estimating these projections enters through the off-diagonal blocks of $\widehat{V}_{\eta,h}$. The resulting standard error therefore accounts jointly for residualization, the numerator evaluated at $p = 1$, and the denominator evaluated at $p = 1$.

The same variance can be written using the estimated influence score

$$\widehat{\psi}_{t,h} = -\nabla \beta_h(\widehat{\eta}_h)' \widehat{G}_h^{-1} \widehat{g}_{t,h}. \tag{30}$$

Applying equation (24) to $\{\widehat{\psi}_{t,h}\}_{t \in \mathcal{T}_h}$ yields the scalar long-run variance entering (29).

The delta-method approximation requires the denominator evaluated at $p = 1$ to be well separated from zero. Accordingly, the estimation diagnostics report $\widehat{a}_{D,h} + \widehat{b}_{D,h}$, its standard error, and the fitted denominator over the empirical support of p_t. Delta-method bands should not be interpreted at horizons for which this condition is empirically doubtful; such cases require inference designed for a weakly identified ratio.

E.7 Corrected Moments for Generated Support

To distinguish the generated score from latent narrative support in this subsection, write

$$p_t = p_t^* + e_t, \qquad E(e_t \mid p_t^*, x_{t,h}, z_{t,h}) = 0, \qquad \omega_t = E(e_t^2 \mid p_t^*, x_{t,h}, z_{t,h}).$$

The classical-error benchmark also requires the residual measurement error to be orthogonal to the structural residuals in the numerator and denominator projections. Let

$$\vartheta_h = (a_h, b_h, a_{D,h}, b_{D,h})'.$$

Conditional on the residualization coefficients, the four corrected affine moments are

$$g_{t,h}^{\mathrm{ME}}(\vartheta_h) = \begin{pmatrix} x_{t,h} - a_h - b_h p_t \\ (x_{t,h} - a_h - b_h p_t)\, p_t + b_h \omega_t \\ z_{t,h} - a_{D,h} - b_{D,h} p_t \\ (z_{t,h} - a_{D,h} - b_{D,h} p_t)\, p_t + b_{D,h} \omega_t \end{pmatrix}, \qquad E\big[g_{t,h}^{\mathrm{ME}}(\vartheta_{h,0})\big] = 0. \tag{31}$$

This is the componentwise form of the compact correction in equation (8). The first and third entries are the intercept moments; the second and fourth are the support-weighted moments affected by measurement error in p_t.

The correction follows directly from the quadratic expansion used by Evdokimov and Zeleneev (2022). For example, the second numerator moment has the form

$$g_x(p) = (x_{t,h} - a_h - b_h p)\, p, \qquad g_x''(p) = -2b_h.$$

The first-order term generated by e_t has mean zero. The second-order term shifts the naive moment by $-b_h \omega_t$, which is removed by the correction in equation (31). Because $g_x(p)$ is quadratic, the expansion is exact under the maintained additive classical-error assumptions. The same argument applies to the denominator support moment.

In the implementation, let $p_i^{(m,k)}$ denote the support assigned to policy action i by representation model $m \in \{1, 2, 3\}$ and calibration set $k \in \{1, \ldots, 12\}$. Each construction is first aggregated to the quarterly frequency using the baseline absolute-tax weights,

$$p_t^{(m,k)} = \frac{\sum_{i \in A_t} |\Delta\tau_i| p_i^{(m,k)}}{\sum_{i \in A_t} |\Delta\tau_i|}, \qquad p_t = \frac{1}{36} \sum_{m=1}^{3} \sum_{k=1}^{12} p_t^{(m,k)}.$$

Thus the dispersion calculation is applied to the same quarterly support objects that enter the estimator. For each quarter, I use the two-way decomposition

$$p_t^{(m,k)} = \mu_t + \xi_{t,m} + \zeta_{t,k} + r_{t,mk}$$

where $\xi_{t,m}$ and $\zeta_{t,k}$ are the model and calibration-set components, and estimate the variance of the averaged score as

$$\widehat{\omega}_t = \frac{\widehat{\sigma}^2_{\mathrm{model},t}}{3} + \frac{\widehat{\sigma}^2_{\mathrm{calibration},t}}{12} + \frac{\widehat{\sigma}^2_{\mathrm{residual},t}}{36}. \tag{32}$$

The variance components are the usual two-way method-of-moments estimates, truncated at zero. This construction preserves variation shared within a representation model or calibration set instead of treating the 36 model-by-calibration probabilities as independent measurements. In quarters without a recorded tax action, both p_t and $\widehat{\omega}_t$ are set to zero.

E.8 Approximate-Moment Sensitivity

The approximate-moment exercise is complementary to the classical correction. In the reported application it is centered on the baseline four-dimensional affine moment block in equation (7) and allows those moments to deviate locally from zero:

$$E\big[g^S_{t,h}(\eta_{h,0};p_t)\big] = \frac{B_h\delta_h}{\sqrt{n_h}}, \qquad \|\delta_h\|_2 \le M. \tag{33}$$

Here, B_h is diagonal, with entries equal to the HAC standard deviations of the four moment contributions. Thus each coordinate of δ_h describes a possible violation in units of the sampling variability of the corresponding moment. The vector δ_h determines how the violation is distributed across the numerator and denominator moments, while M bounds its joint magnitude. The case $M = 0$ imposes exact moment validity. For $M > 0$, the confidence interval accounts for the worst-case endpoint bias over all directions satisfying equation (33), following Armstrong and Kolesár (2021).

The sensitivity radius can also be related to a concrete perturbation of the support scores. Let Δ_h denote the resulting displacement of the four population moments. Its equivalent standardized magnitude is

$$M^{\mathrm{eq}}_h = \sqrt{n_h}\,\left\|B_h^{-1}\Delta_h\right\|_2. \tag{34}$$

For classical error with average variance $\bar{\omega}$, the leading displacement of the two support moments is proportional to $-b_h\bar{\omega}$ and $-b_{D,h}\bar{\omega}$. More generally, equation (34) can translate the moment changes induced by an alternative language model, calibration set, or motivation extraction into the M scale used in the sensitivity figures.

The equivalent radius measures the norm of the complete standardized displacement, not its effect on the endpoint. To make this distinction explicit, let

$$G^S_h = E\left[\frac{\partial g^S_{t,h}(\eta_{h,0};p_t)}{\partial\vartheta_h'}\right]$$

denote the Jacobian of the four support moments with respect to $\vartheta_h = (\theta'_{N,h}, \theta'_{D,h})'$, conditional on the residualization coefficients, and let $\beta_h(\vartheta_h)$ denote the response ratio evaluated at $p = 1$. Define

$$\kappa_h = -\left(G^S_h\right)^{-\prime}\nabla\beta_h(\vartheta_{h,0}).$$

For a small observed displacement Δ_h, the resulting first-order endpoint change is $\kappa_h'\Delta_h$. Among all displacements with equivalent radius M^{eq}_h, the largest first-order endpoint change is

$$\frac{M^{\mathrm{eq}}_h}{\sqrt{n_h}}\left\|B_h'\kappa_h\right\|_2.$$

The directional alignment reported in the application is therefore

$$\rho_h^{\text{align}} = \frac{|\kappa_h' \Delta_h|}{M_h^{\text{eq}} \|B_h' \kappa_h\|_2 / \sqrt{n_h}}. \tag{35}$$

It lies between zero and one. A value near one means that the observed perturbation points toward the most consequential direction for the endpoint; a value near zero means that a potentially large moment displacement is largely orthogonal to that direction. Direct re-estimation under the alternative support construction provides the corresponding nonlinear endpoint comparison.

E.9 Pooled Applications

The pooled UK–France application allows the support slope to differ across countries while imposing a common high-support endpoint. With a fixed number C of countries, define

$$R_{ct}^P = (1, \mathbf{1}\{c=1\}(p_{ct}-1), \ldots, \mathbf{1}\{c=C\}(p_{ct}-1))'. \tag{36}$$

The first coefficient is the common value of the fitted moment at $p_{ct} = 1$; the remaining coefficients are country-specific support slopes. Replacing R_t by R_{ct}^P in the numerator and denominator blocks of equation (6) gives the pooled estimator used in the application.

For inference, let $g_{ct,h}^P(\eta_h)$ denote the country-level moment contribution and form the quarter-level score

$$g_{t,h}^P(\eta_h) = \sum_{c=1}^{C} g_{ct,h}^P(\eta_h). \tag{37}$$

The HAC estimator is then applied to the ordered sequence of quarter-level scores $\{g_{t,h}^P(\widehat{\eta}_h)\}$. This permits arbitrary contemporaneous dependence across countries and serial dependence across quarters. Because C is fixed, the fixed-horizon argument in Subsection E.4 applies after replacing the single-country score by equation (37).

F Replication of Romer and Romer

F.1 Probability Benchmark Settings

F.1.1 Narrative extraction

This appendix documents the benchmark procedure behind Section 4.1. The edited RR narratives are used to evaluate the support construction rather than for a direct binary relabeling exercise. Each example below shows the portion removed before the narrative is passed to the summarization stage. As in the main text, the deleted block includes the explicit RR classification sentence and the two immediately preceding sentences. Removing these passages prevents the model from seeing the authors' bottom-line judgment while leaving enough narrative detail for the summarizer to recover multiple stated motivations from the remaining text.

Example 1: an endogenous measure

The motivation for the Revenue Act of 1951 was again the increase in spending related to the Korean War. The *1951 Economic Report* stated: "These new taxes are required to finance the defense effort; and to help keep total spending within the capacity of current production, so that inflation does not reduce the purchasing power" (p. 17). This same sentiment was echoed in a number of presidential speeches. On February 2, 1951, Truman said, "If we do not tax ourselves enough to pay for defense expenditures, the Government will ... add to total purchasing power and inflationary pressures" (*Special Message to the Congress Recommending a "Pay as We Go" Tax Program*, p. 2). These quotations make it clear that the tax increase was designed to keep growth normal.

It stated: "The military action in Korea, coupled with the general threat to world peace, has made it necessary to provide extraordinary increases in revenues to meet essential national defense expenditures" (82d Congress, 1st Session, House of Representatives Report No. 586, 6/18/51, p. 1). It also stressed the temporary nature of the tax, saying: "The $12\frac{1}{2}$-percent flat across-the-board increase [in individual income tax rates] was selected as the form of increase, since it is not intended that this increase will be permanent, and therefore, it was desired to provide an increase which would not become an integral part of the rate structure" (p. 8). Because the tax increase was designed to counteract the effects of higher military spending and thereby keep growth normal, I classify it as an endogenous, spending-driven action.

Truman's speech upon signing the bill stated that it would raise about $5.5 billion of additional revenues at an annual rate (*Statement by the President Upon Signing the Revenue Act of 1951*, 10/20/51, p. 1). Both the *1952 Economic Report* (p. 135) and the *1951 Treasury Annual Report* (p. 44) gave the figure as $5.4 billion. I follow my usual practice and use the number from the *Economic Report*.

The increase in corporate taxes was retroactive to April 1, 1951, while the remainder took effect on November 1, 1951 (*1951 Treasury Annual Report*, p. 51). Because the action took place before the middle of the quarter, I date the tax increase in 1951Q4. Approximately $2.3 billion of the tax increase was from corporate taxes (p. 501). Therefore, the retroactive portion of the bill increased taxes at an annual rate of $2.3 billion · 2 extra quarters, or $4.6 billion. Adding this to the steady-state effect of $5.4 billion yields a tax increase of $10 billion in 1951Q4. The return to the steady-state level of tax increase implies a tax cut of $4.6 billion in 1952Q1. If one ignored the retroactive part of the legislation, there would be just a tax increase of $5.4 billion in 1951Q4.

The tax increase largely took the form of an increase in marginal rates. The act also raised the capital gains tax, the tax on corporate profits, and some excise taxes (*1951 Treasury Annual Report*, pp. 50–52).

Most of the changes were explicitly temporary. The individual income tax increases were legislated to expire on January 1, 1954; the corporate and excise tax increases were to continue until March 31, 1954 (*1951 Treasury Annual Report*, pp. 51–52).

Example 2: an exogenous measure

This bill was a significant change of the income tax system. It was the first comprehensive revision of the Internal Revenue Code in 75 years (*Statement by the President Upon Signing Bill Revising the Internal Revenue Code*, 8/16/54, p. 1).

This bill was passed primarily to increase the fairness and efficiency of the tax system. It had been under discussion since 1952 (Holmans, 1961, p. 219), well before the 1953–54 recession. At the signing ceremony, Eisenhower stressed: "It is a law which will help millions of Americans by giving them fairer tax treatment than they now receive." He also said, "economic growth will be fostered by such provisions as more flexible depreciation and better tax treatment of research and development costs, thus encouraging all business—large and small—to modernize and expand" (*Statement by the President Upon Signing Bill Revising the Internal Revenue Code*, 8/16/54, p. 1). The Report of the Committee on Ways and Means to Accompany H.R. 8300 (Internal Revenue Code of 1954) repeated this motivation. It stated: "In general, the purpose of these changes has been to remove inequities, to end harassment of the taxpayer and to reduce tax barriers to future expansion of production and employment" (83d Congress, 2d Session, House of Representatives Report No. 1337, 3/9/54, p. 1).

From this perspective, the reduction in revenues at a given level of income was the result of a belief that the bill would stimulate growth and of a desire to make the reform roughly revenue neutral after accounting for the growth effects. The *1954 Economic Report* stated, "A number of additional tax measures should now be taken in order to strengthen the forces of growth in employment and production. These measures, which involve some immediate sacrifices of revenue, contain the seeds of important future revenue gains to be reaped from the economic growth they will stimulate" (p. 79). The *1955 Economic Report* described the motivations for the bill as "to stimulate competitive enterprise, to strengthen the floor of security for the individual, and to curb tendencies toward either depression or inflation," and referred to "the loss of 1.4 billion dollars as a result of enacting structural tax changes" as almost a side effect of the reform (pp. 19–20).

The possible countercyclical benefits of the bill were certainly mentioned frequently. As discussed above, the administration favored some additional tax relief in mid-1954 because of further declines in defense spending. And, much of the Congressional debate focused on the need for additional stimulus. Indeed, several amendments were offered that involved increasing the personal exemption and cutting tax rates (Holmans, 1961, pp. 229–233). However, once the Excise Tax Reduction Act of 1954 was passed, the Administration argued strenuously against further tax cuts, especially the increase in the personal exemption.

[1] On March 15, 1954, Eisenhower said: "at this time economic conditions do not call for an emergency program that would justify larger Federal deficits and further inflation through large additional tax reductions" (*Radio and Television Address to the American People on the Tax Program*, 3/15/54, p. 4; see also *Annual Budget Message to the Congress: Fiscal Year 1955*, 1/21/54, p. 8). Also, the House report on the bill explicitly downplayed the countercyclical aspects, saying: "This bill is a long overdue reform measure which is vitally necessary regardless of the momentary economic conditions and should not be confused with other measures which may be, or might become, appropriate in the light of a particular

short-run situation" (House Report No. 1337, pp. 1–2). Thus, it seems clear that at least some of the tax cuts contained in the Excise Tax Reduction Act of 1954 and the Internal Revenue Code of 1954 should be classified as exogenous.

[2] Thus, it seems clear that at least some of the tax cuts contained in the Excise Tax Reduction Act of 1954 and the Internal Revenue Code of 1954 should be classified as exogenous. There was a cut beyond what the administration thought was justified by the decline in spending and the state of the economy. Because the excise tax reduction came first and had no obvious other motivation, I choose to classify it as endogenous and the Internal Revenue Code of 1954 as an exogenous, long-run action.

[3] There was a cut beyond what the administration thought was justified by the decline in spending and the state of the economy. Because the excise tax reduction came first and had no obvious other motivation, we choose to classify it as endogenous and the Internal Revenue Code of 1954 as an exogenous, long-run action. We feel this makes sense given that the Internal Revenue Code of 1954 clearly had a substantial exogenous component and indeed was passed after it was widely understood that the trough of the recession (May 30, 1954) had been turned (Holmans, 1961, p. 233).

[4] Because the excise tax reduction came first and had no obvious other motivation, we choose to classify it as endogenous and the Internal Revenue Code of 1954 as an exogenous, long-run action. We feel this makes sense given that the Internal Revenue Code of 1954 clearly had a substantial exogenous component and indeed was passed after it was widely understood that the trough of the recession (May 30, 1954) had been turned (Holmans, 1961, p. 233). But, at some level, which of the two tax cuts is classified as endogenous and which as exogenous is somewhat arbitrary.

The act was signed in mid-August 1954, so I date the initial tax change in 1954Q3. My sources give a figure of –\$1.4 billion for the revenue effects at an annual rate (for example, *1955 Economic Report*, p. 20; *1954 Treasury Annual Report*, p. 44). The bill was retroactive to January 1, 1954 (*1954 Treasury Annual Report*, p. 286). It thus reduced taxes by $3 \cdot \$1.4$ billion, or \$4.2 billion at an annual rate, in 1954Q3. The return to the steady-state reduction of \$1.4 billion in 1954Q4 implies a tax increase of \$2.8 billion in 1954Q4. If one chose to neglect the retroactive element, there would be just a tax cut of \$1.4 billion in 1954Q3.

The changes consisted largely of the removal of obvious inequities, complications, ambiguities, and opportunities for tax avoidance, and of provisions designed to stimulate investment. All the changes were legislated to be permanent.

F.1.2 Overlap Between the RR Series and Budget-Based Measures

The Romer–Romer series and the budget-based data cover different parts of the federal tax-policy process and are therefore not expected to coincide observation by observation. Romer and Romer record enacted federal tax changes, whereas the budget data record tax proposals made by the administration to Congress in the annual *Budget of the United States Government*. Some overlap nevertheless arises. A proposal reported in the Budget may subsequently be enacted and enter the Romer–Romer series, and later budgets may propose extensions, accelerations, reductions, or other modifications of provisions associated with earlier Romer–Romer episodes.

Relating the two sources therefore requires more than matching law titles. A single Romer–Romer episode may contain several tax provisions that appear separately in the budget data, while a budget proposal may concern only one component of a broader piece of legislation. In other cases, a statute appears in a later Budget because the administration proposes to modify one of its provisions rather than because the Budget records the original tax change. I therefore compare the two sources at the level of individual tax provisions, taking account of the tax instrument, proposed effective date, and reported revenue effect.

A preliminary comparison of the 50 Romer–Romer episodes with the budget-based measures identifies a small number of relatively clear overlaps. Two examples are reported in Table 19.

Table 19: Examples of RR–Budget Overlap

RR episode	**Budget-based measure**	**Comparison**
Revenue and Expenditure Control Act of 1968	The budget data include the temporary 10 percent surcharge on individual and corporate income taxes, together with related excise-tax extensions.	The same broad tax action appears in both sources, although the budget data record the individual provisions separately.
Omnibus Budget Reconciliation Act of 1993	The fiscal-year 1994 Budget includes the proposed high-income individual income-tax increase associated with the legislation.	The individual-income-tax component can be matched relatively cleanly. Other components of OBRA 1993 are less straightforward to reconcile.

Several other episodes have plausible but less direct counterparts. For example, provisions associated with the Economic Growth and Tax Relief Reconciliation Act of 2001 appear in later budget proposals through accelerations of previously legislated tax changes. These observations are related to the same legislation but do not correspond to the original Romer–Romer timing. The Economic Recovery Tax Act of 1981, the Tax Equity and Fiscal Responsibility Act of 1982, the Revenue Act of 1978, and OBRA 1990 also have related entries in the budget documents, although the correspondence is often limited to particular provisions or revenue components. For the Revenue Acts of 1950 and 1951, the candidate budget entries are not sufficiently close to the enacted provisions to treat them as direct matches.

For this reason, I treat the budget-based data as a separate source of tax-policy variation rather than as observations that can simply be appended to the Romer–Romer series. Constructing a combined series would require a provision-level crosswalk linking each Romer–Romer action to the corresponding budget proposal, where such a counterpart exists.

F.2 Romer and Romer replication Robustness checks

F.2.1 Probabilistic validation against the RR classification

As a complementary check, I estimate a simple probabilistic model for whether the original Romer–Romer label is exogenous. Let

$$D_i^{\mathrm{RR}} = \mathbf{1}\{\text{episode } i \text{ is classified as exogenous by Romer–Romer}\}.$$

I fit the one-variable calibration model

$$\Pr(D_i^{\mathrm{RR}} = 1 \mid p_i) = \Lambda\left(a + b\log\frac{p_i}{1-p_i}\right),$$

where $\Lambda(\cdot)$ is the logistic cdf. The null $b = 0$ corresponds to no monotone predictive content in the policy-level probability. This exercise is deliberately exploratory: the RR classification is used as an external benchmark, not as a definitive latent truth, and the sample contains only 50 policy episodes.

Table 20: Exploratory Logit Validation Against the Original RR Classification

Statistic	Estimate
Number of RR episodes	50
Coefficient on $\log(p/(1-p))$	0.712
Standard error	0.291
Wald p-value	0.014
Likelihood-ratio p-value	0.006
McFadden pseudo-R^2	0.109
In-sample AUC	0.699
Leave-one-out AUC	0.660

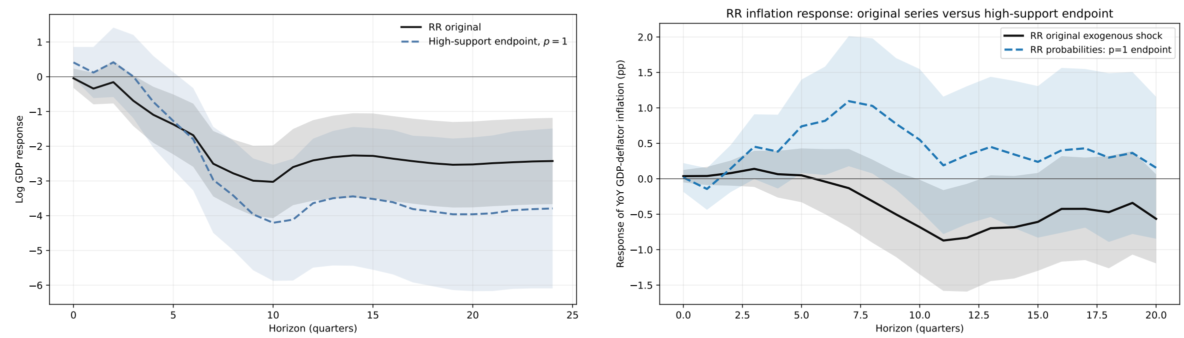
Note: The dependent variable equals one for episodes classified as exogenous by Romer and Romer. The leave-one-out AUC refits the one-variable logit after dropping each episode in turn.

The positive and statistically significant slope indicates that higher policy-level probabilities are systematically associated with the original RR exogenous classification. At the same time, the AUC remains far below one. This is consistent with the interpretation in the main text: the probability contains meaningful narrative information, but ambiguous policy episodes are not perfectly separable by a single continuous score.

F.2.2 Exact Romer–Romer Distributed-Lag Macro Check

As a complementary benchmark, Figure 12 reports the exercise closest to the published Romer–Romer dynamic specification. Quarterly real GDP growth is regressed on the contemporaneous tax shock and twelve lags of that shock, together with eleven lags of GDP growth, over the 1950Q1–2007Q4 sample. The figure reports the simulated cumulative GDP response implied by this distributed-lag equation after a one-percent-of-GDP tax innovation. The inflation panel uses the same original-versus-high-support comparison in the baseline inflation projection. This appendix check is useful because it asks whether the probabilities preserve the canonical RR macro content when the macro equation is held fixed, but it is not the paper's main validation exercise because the main applications use the support-indexed moment estimator.

The original RR series reaches a peak contraction of roughly 3 log points around quarter 10, in



(a) GDP

(b) Inflation

Figure 12: Romer–Romer Responses: Original Series versus High-Support Endpoint

Note: Panel (a) reports the exact Romer–Romer GDP specification. Panel (b) reports the corresponding inflation projection using year-over-year GDP-deflator inflation. The black line is the original RR exogenous shock series. The colored line is the affine two-type estimator evaluated at $p = 1$. Shaded areas denote 68% confidence bands.

line with the original Romer–Romer result. The two-type $p = 1$ endpoint uses the probability ranking to estimate the high-support endpoint of an affine mixture, rather than converting the continuous score back into a binary rule. The resulting response remains close to the canonical RR response and much farther from the attenuated response obtained by treating all tax changes as equally exogenous. Figure 14 reports the corresponding low- and high-support endpoint diagnostic.

F.2.3 Threshold Sensitivity in the Support-Indexed Moment

Figure 13 reports a binary-style version of the support-indexed moment validation. Instead of estimating the affine endpoint in continuous p_i, the exercise keeps only episodes whose policy-level support exceeds a cutoff $\bar{p}$. This is useful as an interpretive diagnostic because it asks what happens if the continuous support uncertainty is resolved into a hard included/excluded rule.

The hard-threshold version is informative but fragile. At horizon 8, the estimates are -0.67 for $\bar{p} = 0.5$ and -1.98 for $\bar{p} = 0.6$, compared with -2.04 for the original RR exogenous indicator in the same moment specification. Raising the cutoff to $\bar{p} = 0.7$ leaves about 20 event quarters at short horizons and reverses the horizon-8 estimate to 2.09, with wider uncertainty. The sequence is not evidence that increasingly stringent cutoffs progressively isolate a cleaner version of the same response. Each cutoff changes the composition of the retained historical episodes, and the economic conclusion can reverse when narratively ambiguous but influential observations are removed. Appendix F.2.5 documents these episodes. The sensitivity is the reason the main text does not choose a single threshold as the headline validation object.

F.2.4 Two-type endpoint diagnostic

Figure 14 unpacks the two-type estimator used in the exact-RR distributed-lag exercise. The left panel reports the two endpoint responses, $p = 0$ and $p = 1$. The right panel shows the implied linear mapping at horizon $h = 10$, where the exact-RR benchmark reaches its peak contraction in the baseline specification. Figure 15 reports the corresponding diagnostic for inflation using the baseline inflation projection.

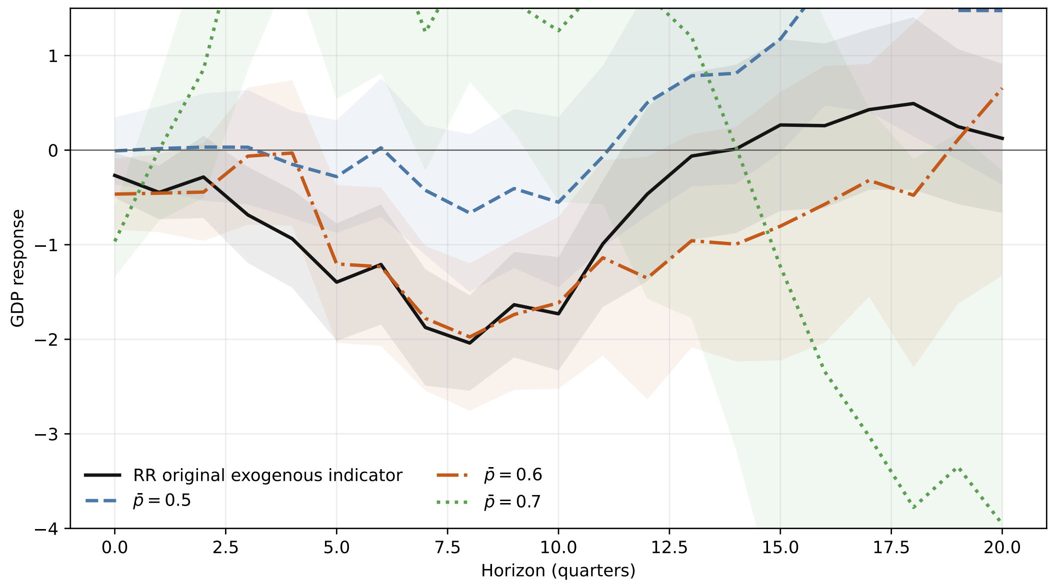


Figure 13: Romer–Romer Moment Validation Under Hard Support Thresholds

Note: The figure reports the support-indexed moment estimator after replacing continuous support with hard threshold rules. The benchmark line uses the original RR exogenous indicator in the same moment estimator. Threshold rules keep episodes with quarter-level support at least $\bar{p}$. Shaded areas denote 68% HAC confidence bands.

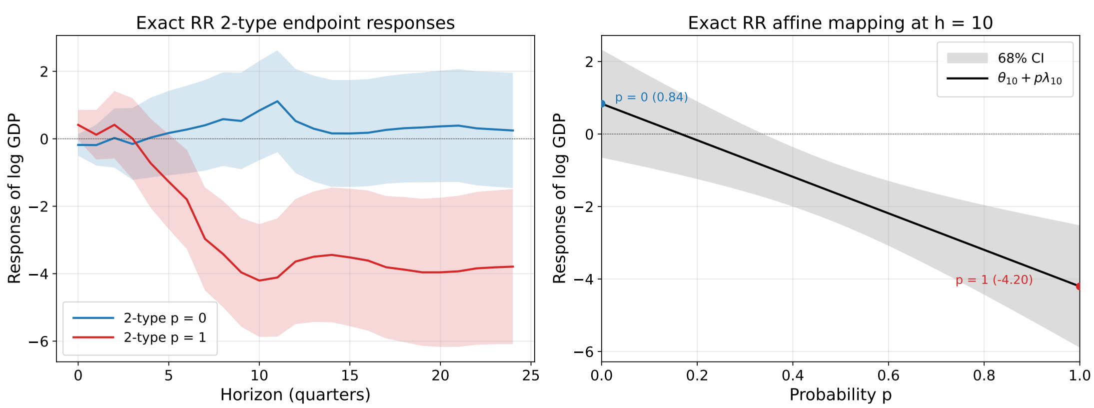


Figure 14: Two-Type RR Benchmark: Endpoint IRFs and the Mapping at $h = 10$

Note: The left panel reports the exact-RR two-type endpoint responses for $p = 0$ and $p = 1$, each with 68% confidence bands. The right panel plots the fitted affine response as a function of p at horizon $h = 10$, together with its 68% confidence band.

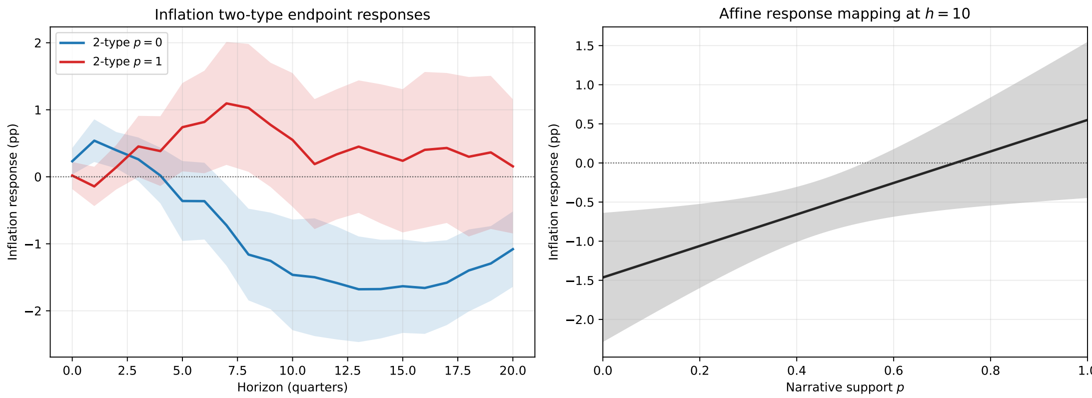


Figure 15: Inflation Two-Type RR Benchmark: Endpoint IRFs and the Mapping at $h = 10$

Note: The left panel reports the inflation two-type endpoint responses for $p = 0$ and $p = 1$, each with 68% confidence bands. The right panel plots the fitted affine response as a function of p at horizon $h = 10$, together with its 68% confidence band. This is an inflation projection analogue of the exact-RR GDP diagnostic, not the published Romer–Romer GDP distributed-lag specification.

F.2.5 Narrative Ambiguity in the Romer–Romer Probability Benchmark

This appendix provides a qualitative audit of the policy-level probabilities used in the Romer–Romer benchmark. The exercise uses the edited narrative texts described in Appendix F.1.1: the model observes the policy narrative after the explicit Romer–Romer classification statement and the immediately adjacent classification cues have been removed. The purpose is not to mechanically reproduce the original binary label in Romer and Romer (2010). Instead, the diagnostic asks whether intermediate values of $\hat{p}_i$ correspond to genuine ambiguity in the narrative record, or whether they reveal cases where the estimated support appears too high or too low relative to the semantic object used in the narrative-tax literature.

The key distinction is that the Romer–Romer label combines several judgment margins. Some episodes are excluded because the tax change responds to current or prospective macroeconomic conditions; others because it is tied to spending or benefit changes. Conversely, some deficit-driven actions are retained as exogenous when they are not judged to respond to current cyclical conditions. The probability measure applies the paper's common taxonomy instead: current-macro responses, spending-package dependence, and fiscal financing motives can contribute different amounts of policy-response support. Differences from the RR label are therefore informative about category boundaries rather than, by themselves, evidence of measurement failure. Table 21 summarizes the main cases.

Clean alignment. When the narrative evidence is relatively clear, the probability moves in the expected direction. The Tax Reform Act of 1986 receives very high support, $\hat{p}_i = 0.950$. This is consistent with the Romer–Romer narrative, where the motivation is tax reform, fairness, simplicity, and long-run incentives. The narrative does mention economic growth, but the relevant growth motive is long-run reform rather than contemporaneous stabilization.

At the other end, the expiration of the Korean-War temporary tax increases receives very low support, $\hat{p}_i = 0.306$. Romer–Romer treat the expiration as a deliberate policy choice after the wartime fiscal environment changed. The motivation is tied to the end of Korean-War spending and to the

Table 21: Romer–Romer Policy Narratives and Estimated Support

Policy	RR category	$\widehat{p}_i$	Interpretation
Expiration of Excess Profits Tax and of Temporary Income Tax Increases	Endogenous, spending-driven	0.306	The low support is consistent with the original narrative: the tax reductions followed the end of Korean-War spending and a changed postwar macro-fiscal environment.
Social Security Amendments of 1958	Endogenous, spending-driven	0.651	The RR exclusion is about benefit financing and actuarial balance, not a direct response to output or inflation. The intermediate support is sensible if p_i primarily captures macro exogeneity rather than all package contamination.
Revenue Act of 1964	Exogenous, long-run	0.448	A boundary case. RR emphasize long-run growth and explicitly reject a quick recession-prevention tax cut. The same text, however, invokes unemployment, below-capacity production, and fiscal drag.
Revenue and Expenditure Control Act of 1968	Endogenous, countercyclical	0.208	The narrative is directly macro-stabilization oriented: overheating, inflation pressure, fiscal restraint, and external pressure on the dollar.
Tax Reform Act of 1969	Endogenous, countercyclical	0.640	The probability is high relative to a strict RR rule. The narrative contains inflation-control language, but also fairness, equity, poverty relief, and tax-reform motives.
Crude Oil Windfall Profit Tax Act of 1980	Exogenous, long-run	0.464	RR treat the tax as unrelated to macro stabilization. The intermediate support reflects oil-price decontrol, energy policy, and consumer-assistance uses of revenue.
Social Security Amendments of 1983	Exogenous, deficit-driven	0.690	Support remains below one because the narrative is strongly tied to trust-fund solvency, Social Security financing, and deficit reduction.
Tax Reform Act of 1986	Exogenous, long-run	0.950	A clean alignment case: tax reform, fairness, simplicity, and long-run incentives dominate the narrative, with little role for current macro conditions.
Omnibus Budget Reconciliation Act of 1990	Exogenous, deficit-driven	0.619	RR classify the act as deficit-driven, but the narrative acknowledges recession timing and weak expected growth.
Omnibus Budget Reconciliation Act of 1993	Exogenous, deficit-driven	0.657	The act is not a short-run stabilization measure, but deficit reduction is the central state-contingent motive.
Taxpayer Relief Act of 1997 and Balanced Budget Act of 1997	Endogenous, spending-driven	0.713	The RR exclusion reflects budget-package contamination. The tax-side rationale is less clearly cyclical, so estimated support is relatively high.
Economic Growth and Tax Relief Reconciliation Act of 2001	Endogenous, countercyclical	0.515	The intermediate support reflects mixed motives: surplus return and long-run incentives on one side, explicit slowdown and short-run stimulus language on the other.

Note: RR category denotes the original Romer–Romer classification. Probabilities use the baseline taxonomy and are averaged across calibration samples and representation models.

macro-fiscal context in which temporary taxes were allowed to lapse. The low probability is therefore not a failure to recover a label; it reflects the same identifying concern that motivates the original endogenous, spending-driven classification.

The 2001 Economic Growth and Tax Relief Reconciliation Act receives intermediate support, $\hat{p}_i = 0.515$. The act combines an argument about returning projected budget surpluses and improving long-run incentives with an explicit short-run stimulus motive during a slowdown. The probability records both exogeneity-supporting and policy-response elements.

Boundary exogenous cases. The Revenue Act of 1964 is the central boundary case. Romer–Romer classify it as exogenous and long-run because the stated objective was to raise the economy's longer-run growth path rather than to prevent an immediate recession. In their narrative, the administration explicitly distinguished the proposal from a "quick, temporary tax cut" and emphasized longer-run economic challenges. At the same time, the surrounding text contains substantial language about unemployment, below-capacity production, and the removal of fiscal drag. The resulting probability, $\hat{p}_i = 0.448$, records partial rather than complete documentary support for the exogenous interpretation.

This case is also empirically important because it helps explain the non-monotone threshold diagnostic. With support below 0.5, the 1964 tax-cut episode is excluded at each reported cutoff, even though it is central to the original Romer–Romer contractionary response. Other medium-support episodes enter or leave as the cutoff changes. A higher cutoff therefore does not simply remove uniformly less credible observations; it changes the composition of the historical episodes identifying the response.

The Crude Oil Windfall Profit Tax Act of 1980 illustrates a different kind of boundary case. Romer–Romer classify the act as exogenous because the tax was motivated by capturing windfall profits from oil-price decontrol rather than by managing output or inflation. The probability is nevertheless only 0.464. The policy is tied to oil-price regulation, energy policy, consumer assistance, and the use of earmarked revenues. Those features are not the same as current-macro endogeneity, but they make the tax change less like an isolated exogenous tax change.

Deficit and Social Security cases. Deficit-driven and Social Security cases reveal the main boundary between the two classification objects. Romer–Romer retain some deficit-driven tax increases as exogenous when the stated motivation is not to offset current cyclical conditions. Under the paper's taxonomy, an action can nevertheless receive policy-response support when it is explicitly motivated by trust-fund financing, a fiscal imbalance, or a spending package. Intermediate probabilities in these cases reflect this broader set of possible links to the policy rule.

The Social Security Amendments of 1983 make this tension clear. Romer–Romer classify the episode as exogenous, deficit-driven, but the narrative is organized around Social Security solvency, trust-fund financing, and deficit reduction. The estimated support is $\hat{p}_i = 0.690$. This is above one half but remains well below full support despite the original exogenous classification. The tax change is presented as a response to a program-financing problem rather than as an isolated tax-policy experiment. The RR label is natural under a non-countercyclical criterion; the paper's broader taxonomy assigns partial policy-response support to the financing motive.

The 1990 and 1993 budget acts show the same issue. OBRA 1990 receives $\hat{p}_i = 0.619$. The original classification is deficit-driven, but the narrative also notes recession timing and weak expected growth.

OBRA 1993 receives $\hat{p}_i = 0.657$. The motive is deficit reduction to improve long-run macroeconomic performance, not short-run demand management. Still, deficit reduction is a state-contingent fiscal motive. The medium probabilities therefore reflect the fact that RR's deficit-driven convention is not the same object as full narrative support for an orthogonal tax shock.

The Social Security Amendments of 1958 point in the opposite direction. They are classified as endogenous, spending-driven, because higher payroll taxes financed higher benefits and improved actuarial balance. The estimated support is 0.651, which is high relative to the hard RR label. The narrative concerns benefit financing rather than output, inflation, or stabilization. The probability may therefore be too high for the full RR exclusion rule but remains consistent with support for macro exogeneity.

Departures that reveal category boundaries. The Tax Reform Act of 1969 is a case where the probability may be high relative to a strict reading of the RR taxonomy. Romer–Romer classify the act as endogenous, countercyclical, partly because the administration accepted a period of slower growth to reduce inflation. The extracted motivations also include fairness, poverty relief, and political tax relief. The probability, $\hat{p}_i = 0.640$, places substantial weight on the non-cyclical components. Under a strict exclusion of policies motivated by current or prospective macro conditions, the inflation-control motive would instead call for lower support.

The 1997 Taxpayer Relief Act and Balanced Budget Act provide a complementary example. Romer–Romer classify the tax cuts as endogenous, spending-driven, because they are tied to a broader budget package in which spending reductions made the tax cuts feasible. The estimated support, $\hat{p}_i = 0.713$, is high relative to that hard label. The reason is that the tax-side motive is not clearly a short-run stabilization response. Under the RR exclusion rule, however, the package structure is itself a source of contamination. A probability focused on macro exogeneity may therefore under-penalize spending-package dependence unless that distinction is made explicit in the measurement target.

Takeaway. Estimated support is not a noisy approximation to the Romer–Romer hard label. High-support cases are dominated by tax reform and long-run motives, whereas low-support cases feature war spending, macroeconomic stabilization, or explicit short-run stimulus. Departures arise mainly where the RR convention and the paper's taxonomy treat deficit financing, social-insurance financing, or spending-package dependence differently. The probability measures documentary support for exogeneity under the common taxonomy; it is not an automated replication of the original classification.

F.2.6 Probabilistic validation against the Cloyne classification

I apply the same one-variable probabilistic check to the Cloyne validation sample. Let

$$C_i = \mathbf{1}\{\text{policy row } i \text{ is classified as exogenous by Cloyne}\}.$$

I estimate

$$\Pr(C_i = 1 \mid p_i) = \Lambda\left(a + b \log \frac{p_i}{1 - p_i}\right),$$

where the sample consists of Cloyne policy rows with usable canonical labels and complete probability inputs. Because many policy rows belong to the same narrative event, I report both row-level and event-clustered uncertainty, and I also evaluate prediction after holding out entire narrative events.

Table 22: Exploratory Logit Validation Against the Cloyne Classification

Statistic	Estimate
Number of Cloyne policy rows	2,101
Coefficient on $\log(p/(1-p))$	0.579
Row-level standard error	0.048
Row-level Wald p-value	< 0.001
Event-clustered standard error	0.087
Event-clustered Wald p-value	< 0.001
Likelihood-ratio p-value	< 0.001
McFadden pseudo-R^2	0.059
In-sample AUC	0.655
Leave-one-event-out AUC	0.641

Note: The dependent variable equals one for policy rows classified as exogenous by Cloyne. Event-clustered inference groups policy rows by narrative event. The held-out AUC refits the one-variable logit after dropping each event in turn.

The Cloyne probability is therefore statistically predictive of the external expert label: the slope is positive and remains precisely estimated after clustering by narrative event. The predictive strength is still modest, however. The AUC is 0.65 in sample and 0.64 when entire narrative events are held out. This is consistent with the main-text table: the probability ranks Cloyne-exogenous rows above Cloyne-endogenous rows on average, but the two distributions overlap substantially. I interpret this as evidence of monotone external-label validity, not as evidence that the procedure mechanically reproduces Cloyne's binary taxonomy.

F.2.7 Training-data leakage robustness check

To assess whether the language model has memorized specific source material, I implement a sentence-reconstruction test taken from Golchin and Surdeanu's (2024) contamination detection framework. The intuition is the following: if a model has encountered a document during training, it may be able to reproduce missing content from that document with unusual precision—especially when provided with cues pointing to the original source. To implement this test, I systematically remove sentences from the Romer and Romer narratives and asks the model to reconstruct them under two conditions: one relying solely on the surrounding context, and the other explicitly guiding the model by referencing the source paper and section. If the model performs significantly better under guided prompting, this may suggest prior exposure to the original text. To quantify and evaluate the reconstruction quality, I use both semantic (embedding cosine similarity) and lexical (ROUGE-L) metrics, followed by a bootstrap-based significance test to flag cases of potential memorization.

Using both metrics, and evaluating with `gpt-4o-mini` and `gemini-2.0-flash`, no narrative is flagged as a potential case of memorization. This result strengthens confidence that the model was not trained on Romer and Romer (2010), and supports the validity of the replication exercise by reducing concerns about training-data leakage as a source of bias in the model's classifications.

Procedure.

1. **Multiple draws & prompts.** For each narrative d I perform $T = 10$ draws. In draw t, one sentence $s_{d,t}$ is chosen uniformly at random, removed, and replaced by the token `[MISSING SENTENCE]`. I then issue two prompts a general prompt, and a guided prompt (Prompts 9).

General prompt:
You are provided with a passage in which one sentence has been removed and replaced with `[MISSING SENTENCE]`. Based ONLY on the context provided, generate a plausible sentence that could fill the gap. You do not need to match any specific source text — just ensure the result is coherent and contextually appropriate.

`{narrative text from Romer and Romer}`

Reconstructed sentence:

Guided prompt:
You are provided with a passage from the scientific paper *A Narrative Analysis of Postwar Tax Changes* by Christina D. Romer and David H. Romer, published in 2010. The passage comes from the section titled '`{title}`" in the main body of the paper.
One sentence has been removed and replaced with `[MISSING SENTENCE]`. Based **only** on the context provided and your knowledge of this specific paper, reconstruct the missing sentence exactly as it appeared.

`{narrative text from Romer and Romer}`

Reconstructed sentence:

Prompt 9: General and Guided Prompt

The model returns two reconstructions $\hat{s}^{\text{gen}}_{d,t}$ and $\hat{s}^{\text{guide}}_{d,t}$.

2. **Similarity metrics.** For each original $s_{d,t}$ and each reconstruction, compute:
   - *Embedding cosine similarity* $\rho^{\text{gen}}_{d,t} = \cos\big(s_{d,t}, \hat{s}^{\text{gen}}_{d,t}\big)$, $\rho^{\text{guide}}_{d,t} = \cos\big(s_{d,t}, \hat{s}^{\text{guide}}_{d,t}\big)$, using `text-embedding-3-large`.
   - *ROUGE-L F1 score* $r^{\text{gen}}_{d,t}$ and $r^{\text{guide}}_{d,t}$ between the original and each reconstruction.

3. **Paired bootstrap test.** For each narrative d, define the mean differences

$$\Delta\rho_d \;=\; \frac{1}{T}\sum_{t=1}^{T}\big(\rho^{\text{guide}}_{d,t} - \rho^{\text{gen}}_{d,t}\big), \quad \Delta r_d \;=\; \frac{1}{T}\sum_{t=1}^{T}\big(r^{\text{guide}}_{d,t} - r^{\text{gen}}_{d,t}\big).$$

   We perform a nonparametric paired bootstrap with $B = 1{,}000$ resamples of the T pairs $\{(\rho^{\text{guide}}_{d,t}, \rho^{\text{gen}}_{d,t})\}$ (and similarly for $\{(r^{\text{guide}}_{d,t}, r^{\text{gen}}_{d,t})\}$), yielding bootstrap distributions of $\Delta\rho_d$ and Δr_d. Each indicator $\mathbf{1}\{\Delta\rho^{*(b)}_d \le 0\}$ (or $\mathbf{1}\{\Delta r^{*(b)}_d \le 0\}$) counts a bootstrap sample in which the similarity under the guided prompt is no better than—or worse than—that under the general prompt; that is, the model failed to improve reconstruction quality despite being explicitly cued about the source. The empirical p-values are

$$p^{\rho}_d \;=\; \frac{1}{B}\sum_{b=1}^{B}\mathbf{1}\{\Delta\rho^{*(b)}_d \le 0\}, \quad p^{r}_d \;=\; \frac{1}{B}\sum_{b=1}^{B}\mathbf{1}\{\Delta r^{*(b)}_d \le 0\}.$$

Decision rule. A narrative d is flagged as potentially memorized if its bootstrap p-value for either similarity metric falls below 0.05, i.e. if guided prompting produces significantly closer reconstructions than general prompting.

F.2.8 Manually Investigating Legacy Binary Misclassifications

The misclassification pattern, mostly exogenous policies misclassified as endogenous, can possibly be attributed to an asymmetry inherent in macroeconomic narrative analysis: while it is relatively straightforward to construct narratives supporting the endogeneity of a policy intervention, establishing its exogeneity typically requires more stringent evidence. In the absence of sufficiently strong signals of such independence LLMs may default to a conservative interpretation, erring on the side of endogeneity.

Steering growth above normal levels. A recurring source of systematic misclassification involves tax cuts enacted with the aim of boosting growth above normal levels. In at least three instances, the LLM interprets these as countercyclical, and thus endogenous, due to the presence of growth-oriented motivation. However, according to Romer and Romer's decision rule, such motivations qualify as exogenous, as they are not designed to return output to trend but to exceed it. This kind of mislabeling can be mitigated by explicitly including this rule in the prompt design or few-shot examples, highlighting that aspirational growth policies, unless tied to negative output gaps, are to be treated as exogenous.

Conflicting motivations. Another set of misclassifications arises from cases with mixed or conflicting motivations. For example, one narrative discusses a tax increase intended primarily for deficit reduction and solvency of the Highway Trust Fund, both exogenous motives, but also references inflation concerns and contemporaneous spending increases. While the latter suggest endogenous intent, the original classification deems the measure exogenous with noted ambiguity. Such nuanced judgment is difficult for LLMs to replicate without additional guidance on prioritizing primary over secondary motivations.

"Package" related measure. Finally, there are highly specific edge cases that hinge on bespoke decision rules. A salient example is the Tax Adjustment Act of 1966. Despite being implemented to counteract high growth following an exogenous tax cut, Romer and Romer classify it as exogenous, arguing that treating it as endogenous would yield nonsensical results. This type of reasoning, context-specific and precedent-based, poses a significant challenge for LLMs, particularly in the absence of similar examples in the prompt.

F.3 Affine GMM Endpoint Diagnostic

The main text reports the fitted response at $p = 1$ under the baseline affine GMM estimator. Figure 16 reports the corresponding fitted value at $p = 0$ as a diagnostic. This contrast is useful for describing the affine specification: the contractionary medium-run response is concentrated at $p = 1$, while the fitted response at $p = 0$ is much flatter and eventually positive. The latter should not be read as a separately observed class of purely endogenous policies. Under the baseline convention, it is anchored jointly by no-action quarters assigned $p_t = 0$ and by recorded policy quarters receiving little support for exogeneity. Appendix H shows how the fitted response at $p = 1$ changes when no-action quarters are placed elsewhere on the support scale or excluded from the support-indexed projections.

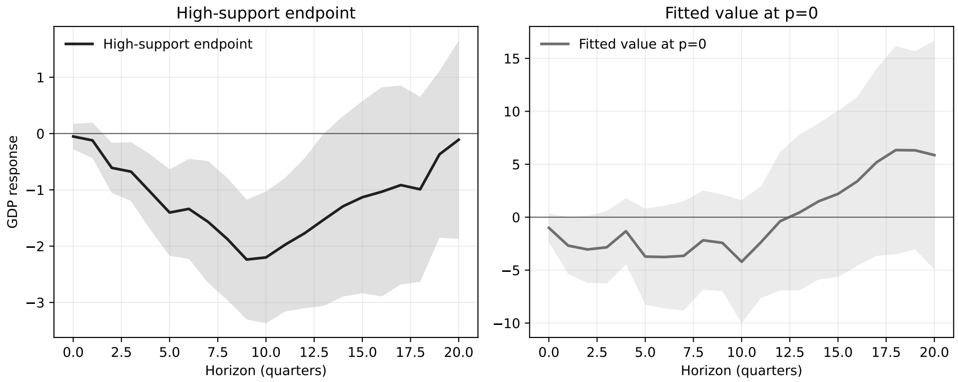
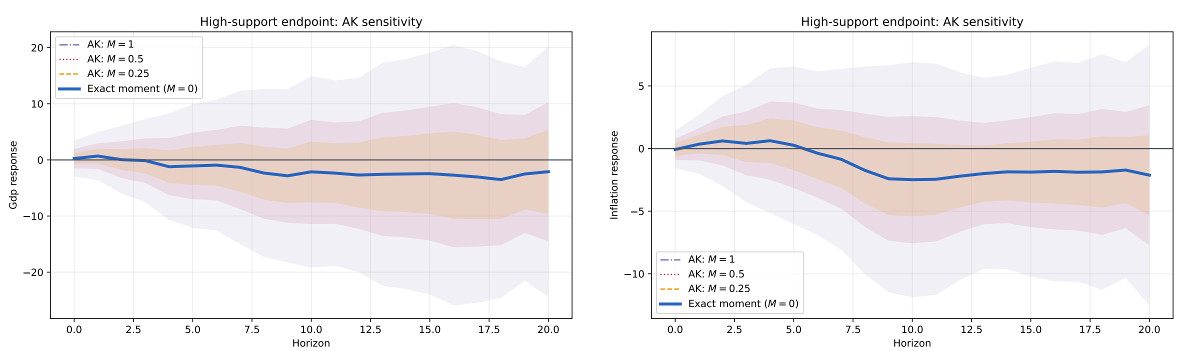


Figure 16: Affine GMM Endpoint Diagnostic: $p = 0$ and $p = 1$

Note: The figure reports the U.S. budget affine GMM fitted values for real GDP at $p = 1$ and $p = 0$. Both use the same baseline estimator, residualization, and fitted squared-tax denominator evaluated at the corresponding value of p, with 68% HAC confidence bands. Under the baseline convention, the $p = 0$ fitted value is anchored by both no-action quarters and recorded policy quarters with little support for exogeneity; it is not a separately observed policy class.

F.4 Measurement Diagnostics for the U.S. Budget Endpoint

This appendix reports four diagnostics for the generated support score used in the U.S. budget application. The first asks how sensitive the GDP and inflation endpoints are to direct violations of the projected moment condition. The second asks whether the endpoints are driven by one representation model. The third varies the model and stochastic draw used to extract motivations from the retrieved passages. The fourth translates the separate and joint support-score perturbations into the standardized sensitivity scale used by the approximate-moment exercise.



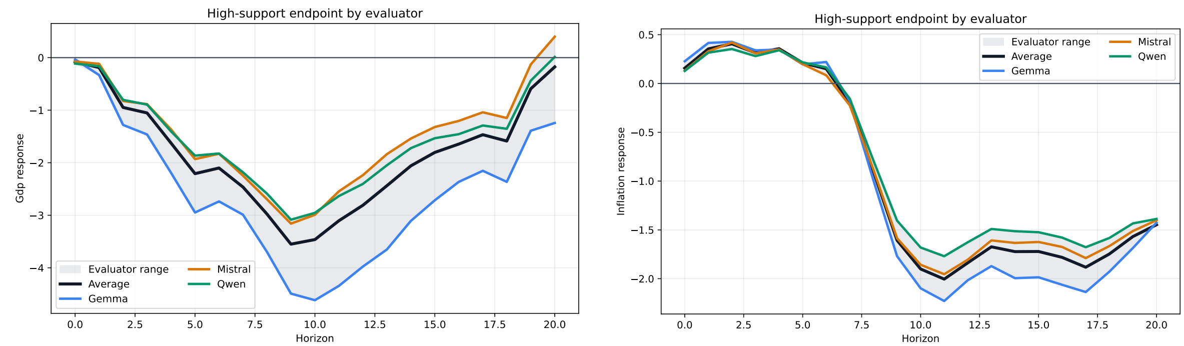
(a) GDP

(b) Inflation

Figure 17: U.S. Budget Endpoints: Approximate-Moment Sensitivity

Note: Each panel reports bias-aware endpoint intervals under approximate moment validity. $M = 0$ corresponds to exact projected moment validity. Larger values of M allow direct moment violations measured relative to the Newey–West sampling-error scale.

Figure 17 shows that moderate approximate-moment violations widen the bands. The GDP contraction remains visible under small violations, while the inflation endpoint is more sensitive once larger direct moment failures are allowed. This exercise should be read as a sensitivity diagnostic, not as a correction for a specific measurement-error model.



(a) GDP (b) Inflation

Figure 18: U.S. Budget Endpoints: Representation-Model Stability

Note: Each line recomputes the same endpoint using support scores from one representation-model family. The exercise isolates sensitivity to the language-model representation used to construct the support score.

Figure 18 shows that the model-specific endpoints are not identical, but they preserve similar medium-run patterns for both GDP and inflation. This diagnostic indicates that the main pattern is not driven by a single representation model.

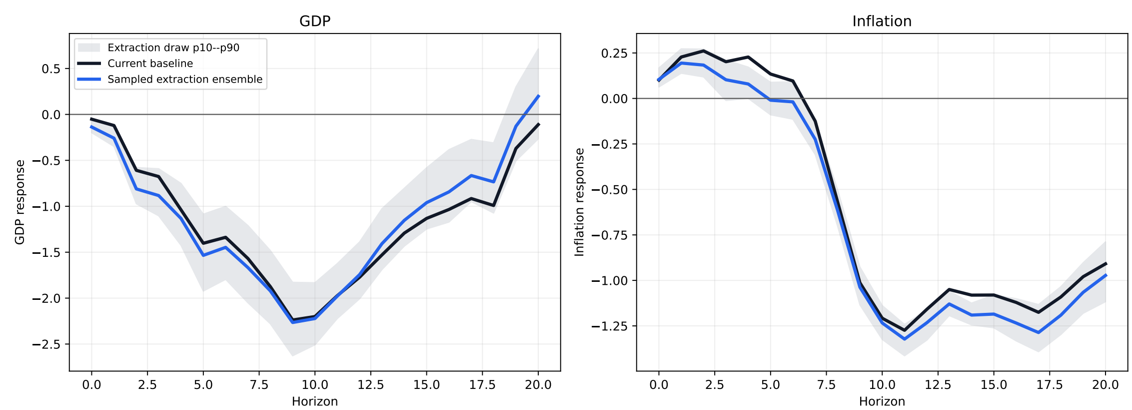


Figure 19: U.S. Budget Endpoints: Motivation-Extraction Sensitivity

Note: The black line uses the baseline support score. The blue line uses the support score averaged across alternative motivation extractions. The gray region is the 10th–90th percentile range of endpoint estimates across 60 constructions: 20 stochastic motivation draws from each of Qwen-9B, Gemma-4B, and Mistral-7B. Retrieval is held fixed, and each extraction is passed through the same evaluator-model and calibration-set construction.

Figure 19 shows that varying the motivations extracted from the same retrieved passages changes the magnitude of some estimates but preserves the contractionary GDP response and the medium-run decline in inflation. The draw-averaged extraction-model probabilities have a mean Pearson correlation of 0.80 with one another; stochastic draws within an extraction model have a mean correlation of 0.93.

Figure 20 gives the sensitivity radius an empirical scale for one-at-a-time perturbations. Across the main horizons, calibration-sample perturbations have median equivalent radii around $M = 0.5$,

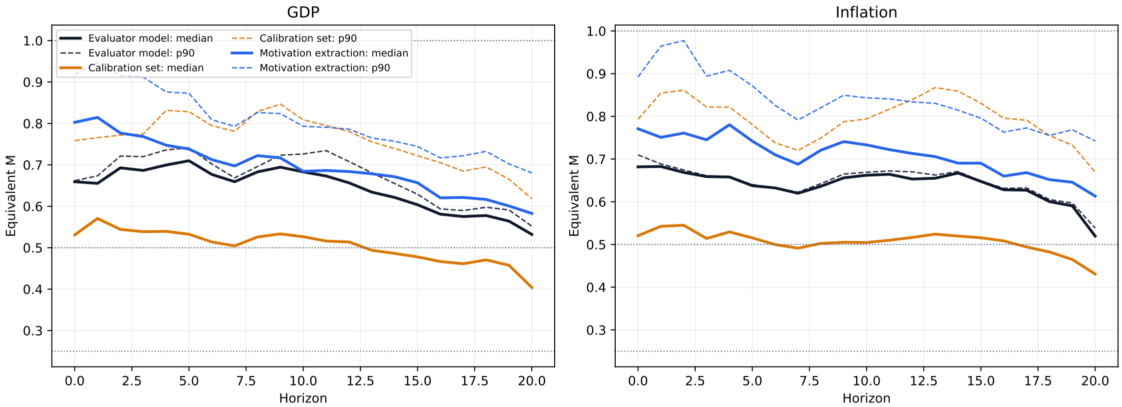


Figure 20: Observed Probability Perturbations in Equivalent Sensitivity Units

Note: For each alternative evaluator model, calibration set, or motivation-extraction draw, the figure evaluates the four affine moments at the baseline parameter and replaces only the support sequence. The resulting moment displacement is scaled by the baseline HAC variability according to equation (34). Solid lines report the median equivalent M within each family; dashed lines report the 90th percentile.

representation-model perturbations around $M = 0.7$, and motivation-extraction perturbations around $M = 0.7$. These values should not be interpreted as estimates of structural misspecification: they report how far the implemented moments move when observed measurement choices are changed. The equivalent radius M^{eq} summarizes the overall magnitude of the change in the four identifying moments. Only part of this change, however, occurs in directions that materially affect the estimated response. Table 23 reports both the equivalent radius and this directional relevance. The joint perturbations generate median equivalent radii around one, while their median directional relevance is 0.19 for GDP and 0.18 for inflation. Thus, although alternative constructions of narrative support can generate sizable changes in the identifying moments, most of this variation occurs in directions to which the estimated response is relatively insensitive.

Table 23: Magnitude and Endpoint Relevance of Observed Support Perturbations

Outcome	Source	Median M^{eq}	P90 M^{eq}	Median relevance	Median $\lvert\Delta\widehat{\beta}\rvert$
GDP	Representation model	0.68	0.71	0.15	0.114
GDP	Calibration sample	0.53	0.83	0.11	0.098
GDP	Extraction model	0.73	0.74	0.27	0.256
GDP	Stochastic extraction draw	0.09	0.15	0.58	0.067
GDP	All choices jointly	0.84	2.16	0.19	0.268
Inflation	Representation model	0.65	0.66	0.07	0.023
Inflation	Calibration sample	0.52	0.82	0.18	0.097
Inflation	Extraction model	0.72	0.73	0.14	0.061
Inflation	Stochastic extraction draw	0.09	0.16	0.59	0.032
Inflation	All choices jointly	0.80	2.17	0.18	0.145

Note: Entries summarize perturbations at horizons 4, 8, and 12. M^{eq} is the standardized radius containing the observed displacement of the four support moments. Relevance is the absolute linearized endpoint change relative to the largest change possible at the same radius. The final column reports the absolute endpoint change after re-estimation.

F.5 Component Diagnostics and Romer–Romer Comparison

Figure 21 reports the remaining component panels corresponding to Figure 14 in Romer and Romer (2010). The comparison is useful for separating the mechanism that carries over from the original RR exercise from the margins that are less stable in the document-based budget application. Romer–Romer find that durable consumption falls much more than nondurables and services, that both nonresidential and residential fixed investment decline, and that net exports improve, with exports rising and imports falling. In the support-indexed GMM estimates, the domestic investment channel remains the clearest common element: nonresidential fixed investment falls strongly and persistently. The remaining margins are less aligned. Durables fall early but later rebound, nondurables and services move little, residential investment turns positive at longer horizons, and the external-sector responses go in the opposite direction of the original RR pattern. I therefore treat these panels as component diagnostics rather than as the central mechanism evidence.

Figure 21: Additional RR-Style Component Responses Under the Baseline GMM Estimator

Note: The figure reports affine $p = 1$ endpoint responses using the same budget-measure probabilities, quarterly tax shocks, endpoint denominator, and HAC inference as Figure 5. Components are log quantity indexes, so responses are percent changes. The panels correspond to Panels B–D in Figure 14 of Romer & Romer (2010).

Tables 24–26 decompose the mechanism outcomes using the same additive object as Table 7. The pattern is consistent with the main-text interpretation. The medium-run decline in inflation and the large fall in investment are mainly associated with income-tax components, especially individual income taxes. Consumption moves in the same direction but less sharply. The nonstandard consumption/excise residual continues to behave differently from a clean indirect-tax object, reinforcing the decision to treat the UK–France VAT evidence as the sharper consumption-tax contrast.

Table 24: Composition and Additive Inflation Contributions to the U.S. Budget Endpoint

Instrument	Event share	Gross tax share	Mean p	Inflation $h = 4$	Inflation $h = 8$	Inflation $h = 12$
Individual income	30.6%	37.0%	0.72	0.23 (0.38)	-0.34 (0.46)	-0.82 (0.48)
Corporate income	32.2%	22.7%	0.71	0.20 (0.20)	-0.03 (0.15)	-0.40 (0.24)
Clean standard excise	8.9%	11.2%	0.52	-0.05 (0.06)	-0.13 (0.12)	-0.14 (0.10)
Nonstandard consumption/excise	11.1%	15.5%	0.53	-0.31 (0.24)	-0.23 (0.21)	0.11 (0.12)
Other low-support taxes	17.2%	13.6%	0.50	0.15 (0.11)	0.16 (0.17)	0.10 (0.15)

Note: Entries are additive contributions to the aggregate affine $p = 1$ budget endpoint; parentheses report HAC standard errors. The rows use the aggregate support index and common endpoint denominator and therefore sum to the aggregate response up to rounding.

Table 25: Composition and Additive Consumption Contributions to the U.S. Budget Endpoint

Instrument	Event share	Gross tax share	Mean p	Consumption $h = 4$	Consumption $h = 8$	Consumption $h = 12$
Individual income	30.6%	37.0%	0.72	-1.12 (0.90)	-1.37 (1.06)	-1.04 (1.16)
Corporate income	32.2%	22.7%	0.71	-0.47 (0.40)	-0.53 (0.47)	-0.24 (0.40)
Clean standard excise	8.9%	11.2%	0.52	0.03 (0.12)	-0.09 (0.16)	-0.03 (0.12)
Nonstandard consumption/excise	11.1%	15.5%	0.53	0.77 (0.54)	1.01 (0.64)	1.17 (0.68)
Other low-support taxes	17.2%	13.6%	0.50	0.03 (0.11)	-0.13 (0.22)	-0.34 (0.26)

Note: Entries are additive contributions to the aggregate affine $p = 1$ budget endpoint; parentheses report HAC standard errors. The rows use the aggregate support index and common endpoint denominator and therefore sum to the aggregate response up to rounding.

Table 26: Composition and Additive Investment Contributions to the U.S. Budget Endpoint

Instrument	Event share	Gross tax share	Mean p	Investment $h = 4$	Investment $h = 8$	Investment $h = 12$
Individual income	30.6%	37.0%	0.72	-4.42 (1.83)	-7.45 (3.36)	-7.51 (3.13)
Corporate income	32.2%	22.7%	0.71	-2.09 (1.27)	-3.12 (1.88)	-2.44 (1.48)
Clean standard excise	8.9%	11.2%	0.52	0.03 (0.59)	-0.36 (0.54)	0.06 (0.40)
Nonstandard consumption/excise	11.1%	15.5%	0.53	2.33 (1.12)	3.43 (2.03)	3.46 (1.88)
Other low-support taxes	17.2%	13.6%	0.50	0.28 (0.76)	-0.17 (0.86)	-0.88 (1.07)

Note: Entries are additive contributions to the aggregate affine $p = 1$ budget endpoint; parentheses report HAC standard errors. The rows use the aggregate support index and common endpoint denominator and therefore sum to the aggregate response up to rounding.

G VAT Shocks and Macro Transmission in the United Kingdom and France

G.1 Additional VAT Channel Diagnostics

Figure 22 reports the fuller expenditure-side decomposition. The figure keeps the same outcome transformation as the main mechanism figure: each component is expressed as a change from the pre-event quarter in percentage points of trend GDP. The main text focuses on private consumption and investment because they are the central domestic-absorption channels for a VAT shock. The additional panels show that government consumption is muted at short horizons, while net exports partly offset the contraction.

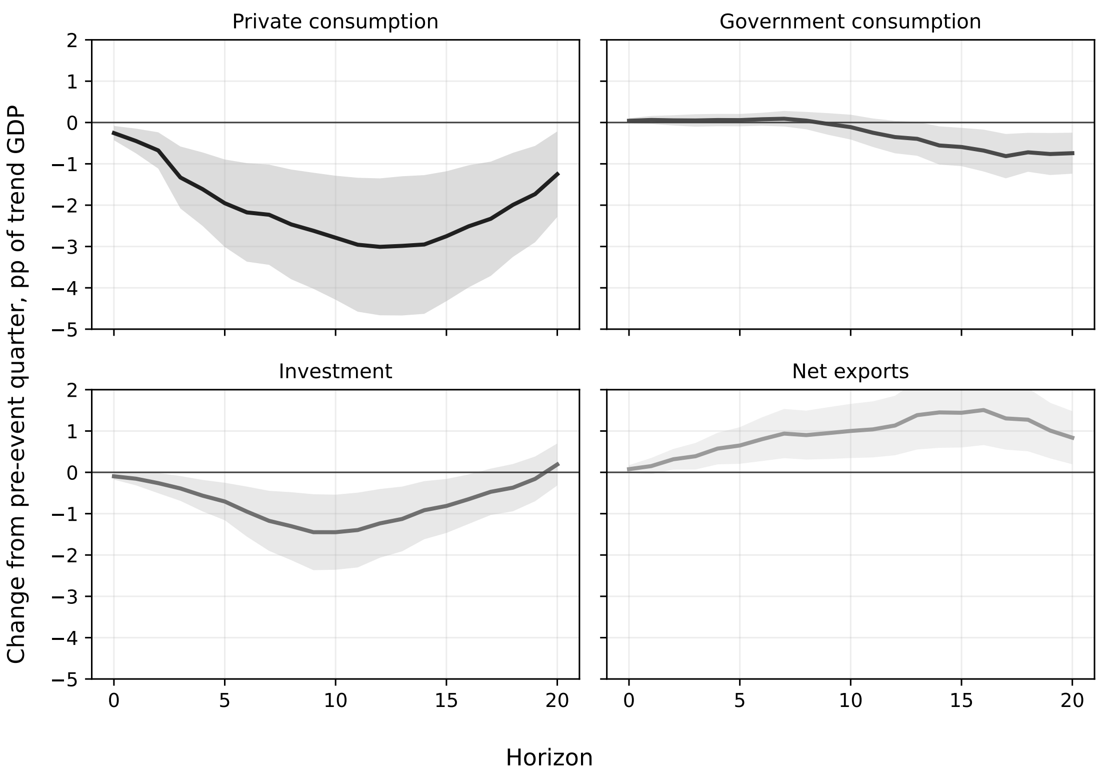


Figure 22: UK–France VAT: Additional Expenditure-Side Responses

Note: Outcomes are expressed as $y_{c,t+h} - y_{c,t-1}$ in percentage points of trend GDP. The figure reports the pooled VAT endpoint for private consumption, government consumption, investment, and net exports on a common vertical scale. Shaded areas are 68 percent HAC confidence bands.

Table 27 reports the corresponding key-horizon estimates. Decomposing net exports shows that the external offset is mostly import compression rather than a robust export boom: exports are small and imprecise, while the import contribution is positive. Since imports enter the accounting decomposition as $-M$, a positive import contribution means that imports fall.

Table 27: VAT Additional Channels at Key Horizons

Outcome	$h = 4$	$h = 8$	$h = 12$
GDP deviation from trend	-2.46	-4.55	-4.04
Private consumption response	-1.61	-2.47	-3.01
Government consumption response	0.06	0.05	-0.35
Investment response	-0.56	-1.30	-1.23
Exports response	0.17	-0.37	0.16
Imports contribution ($-M$)	0.41	1.27	0.97
Net exports response	0.58	0.90	1.13

Note: GDP and expenditure components are changes from the pre-event quarter in percentage points of trend GDP. Inflation is in percentage points. Imports enter as $-M$, so a positive import contribution denotes import compression.

G.2 VAT Packages and the Nguyen–Onnis–Rossi Concern

Nguyen et al. (2021) emphasize that UK income-tax and consumption-tax changes frequently occur together. In their proxy-SVAR, the broad consumption-tax object is an average consumption tax rate built from VAT and consumption duties. They find modest effects for this broad object once income-tax and consumption-tax shocks are jointly identified, while single-proxy specifications can make consumption taxes look much more powerful. This appendix asks whether the same concern jeopardizes the VAT endpoint reported in the main text.

The checks keep the UK–France sample fixed at 1955Q1–2007Q4. They vary only the tax actions entering the VAT object. The first diagnostic is a co-movement audit. Same-quarter overlap is high, but the VAT-income correlation is small: the pooled correlation between the VAT shock and the income-like shock is 0.02 over all quarters and 0.04 over policy quarters. An explicit two-instrument denominator diagnostic gives nearly identical GDP endpoints. At horizons 4, 8, and 12, the naive VAT endpoints are -2.39, -4.43, and -3.92, while the VAT/income adjusted endpoints are -2.40, -4.44, and -3.94.

Table 28 reports the main sensitivity estimates. Dropping UK 1979Q3, the VAT unification episode highlighted by Nguyen et al. (2021), weakens precision but preserves the negative GDP response. Dropping both UK 1973Q2 and UK 1979Q3 weakens the result more substantially, which shows that large UK VAT transition episodes are empirically important. By contrast, dropping VAT actions in quarters where income-like taxes move in the opposite direction and are at least half as large as the VAT move makes the response stronger, not weaker. Thus the VAT result is not driven by compensating VAT-income packages. The main attenuation occurs when the VAT object is broadened to include duties and other excise-like measures, closer to the Nguyen–Onnis–Rossi average-consumption-tax-rate object.

Table 28: VAT Sensitivity to Tax Packages and Broad Consumption-Tax Definitions

Specification	$h = 4$	$h = 8$	$h = 12$
Baseline VAT	-2.46 (0.095)	-4.55 (0.100)	-4.04 (0.111)
Drop UK 1979Q3 VAT	-2.32 (0.081)	-3.45 (0.084)	-2.80 (0.116)
Drop UK 1973Q2 and 1979Q3 VAT	-1.72 (0.273)	-3.46 (0.185)	-4.44 (0.119)
Drop compensating VAT–income packages	-3.02 (0.107)	-5.08 (0.121)	-4.92 (0.106)
Drop full compensating-package quarters	-3.06 (0.113)	-5.17 (0.110)	-4.76 (0.087)
VAT plus duties/excise-like measures	-0.69 (0.111)	-1.57 (0.108)	-1.57 (0.142)

Note: Entries are pooled GDP endpoints at $p = 1$, in percentage points of trend GDP; p-values are in parentheses. The sample is 1955Q1–2007Q4 in every row. The broad row combines VAT with duties and excise-like measures.

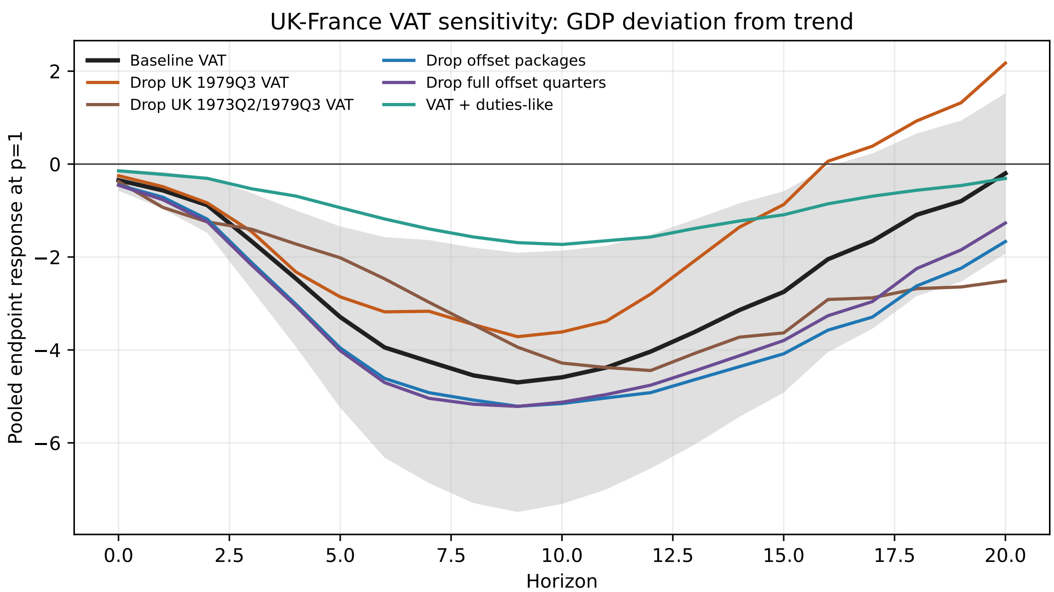


Figure 23: VAT GDP Response Under Package Sensitivity Checks

Note: The figure reports pooled GDP endpoint responses under the same sensitivity variants as Table 28. The shaded area is the 68 percent HAC band for the baseline VAT endpoint.

G.3 Country-Specific VAT Diagnostics

Figure 24 reports the same output and price endpoints separately for France and the United Kingdom on the common comparison sample. These country-specific estimates are supporting diagnostics rather than the main estimand. They show that the pooled GDP response is not averaging two opposite country patterns: GDP falls in both countries after high-support VAT shocks. The country-specific inflation responses are less precisely estimated, but both countries display a positive short-run price response.

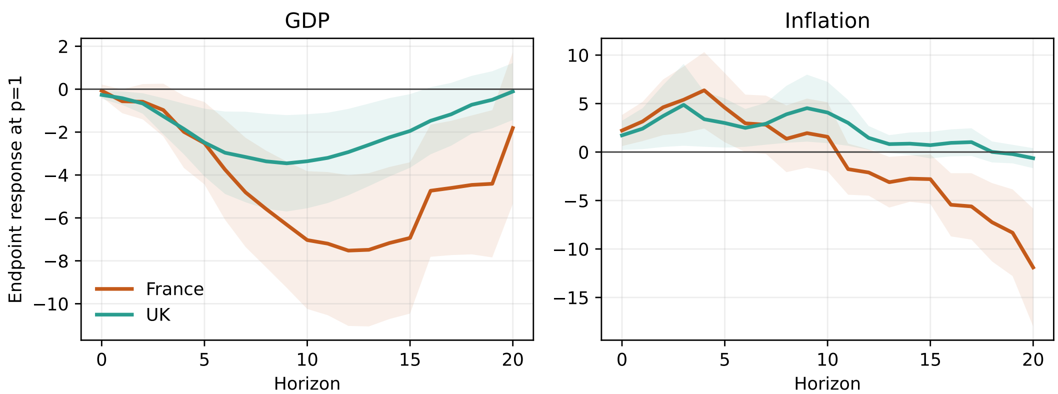


Figure 24: Country-Specific VAT Endpoint Responses

Note: The figure estimates the VAT endpoint separately for France and the United Kingdom on the common comparison sample, 1955Q1–2007Q4. GDP is expressed in percentage points of trend GDP; inflation is in percentage points. Responses are normalized by the country-specific linear-in-p denominator evaluated at $p = 1$. Shaded areas are 68 percent HAC confidence bands.

Table 29: Country-Specific VAT Endpoint Responses at Key Horizons

Country	Outcome	$h = 4$	$h = 8$	$h = 12$
France	GDP deviation from trend	-2.52	-7.44	-9.91
France	Inflation	6.37	1.37	-2.11
United Kingdom	GDP deviation from trend	-2.64	-4.10	-4.10
United Kingdom	Inflation	3.39	3.89	1.46

Note: Entries are country-specific high-support VAT endpoints. GDP is in percentage points of trend GDP and inflation is in percentage points.

G.4 VAT Adoption Timing

Table 30: VAT (or GST-equivalent) Adoption by Year in Developed Economies

Year	**Countries**
1954	France
1967	Denmark
1968	Germany
1969	Netherlands; Sweden
1970	Luxembourg; Norway
1971	Belgium
1972	Ireland
1973	Austria; Italy; United Kingdom
1977	South Korea
1980	Mexico
1985	Turkey
1986	New Zealand
1989	Japan
1990	Iceland
1991	Canada
1994	Finland
1995	Switzerland
2000	Australia

H Treatment of Quarters without Recorded Policy Actions

The baseline specification retains all calendar quarters and assigns $\Delta\tau_t = p_t = 0$ when no policy action is recorded. The tax change is then observed to be zero, while the value assigned to narrative support is a completion used for estimation rather than an economic support value. This appendix assesses the empirical relevance of that completion.

Let

$$I_t = \mathbb{1}\{\text{at least one policy action is recorded in quarter } t\}.$$

For $c \in \{0, 0.1, \ldots, 1\}$, define

$$p_t(c) = \begin{cases} p_t, & I_t = 1, \\ c, & I_t = 0. \end{cases}$$

The value c is a sensitivity parameter, not an estimate of narrative support in quarters without a policy action. The indicator is based on recorded policy rows, so quarters with offsetting actions and a zero net tax change satisfy $I_t = 1$ and retain their measured support. For each value of c, I re-estimate the affine numerator and denominator projections while holding the tax changes, support in recorded-action quarters, controls, horizons, and outcome transformations fixed. A second specification retains the same controls, transformations, and residualization procedure but estimates both the residualization equations and the affine numerator and denominator projections using recorded-action quarters only. The original quarterly spacing is retained when constructing HAC standard errors.

H.1 Main U.S. application

Figure 25 reports the U.S. GDP and inflation responses at horizons 4, 8, and 12. GDP remains negative at every value of c and under the recorded-action-quarter specification. The medium-run inflation response also retains its baseline sign. The estimates can change in magnitude as the no-action mass is moved along the support scale, but the contractionary output response and medium-run disinflationary pattern are preserved.

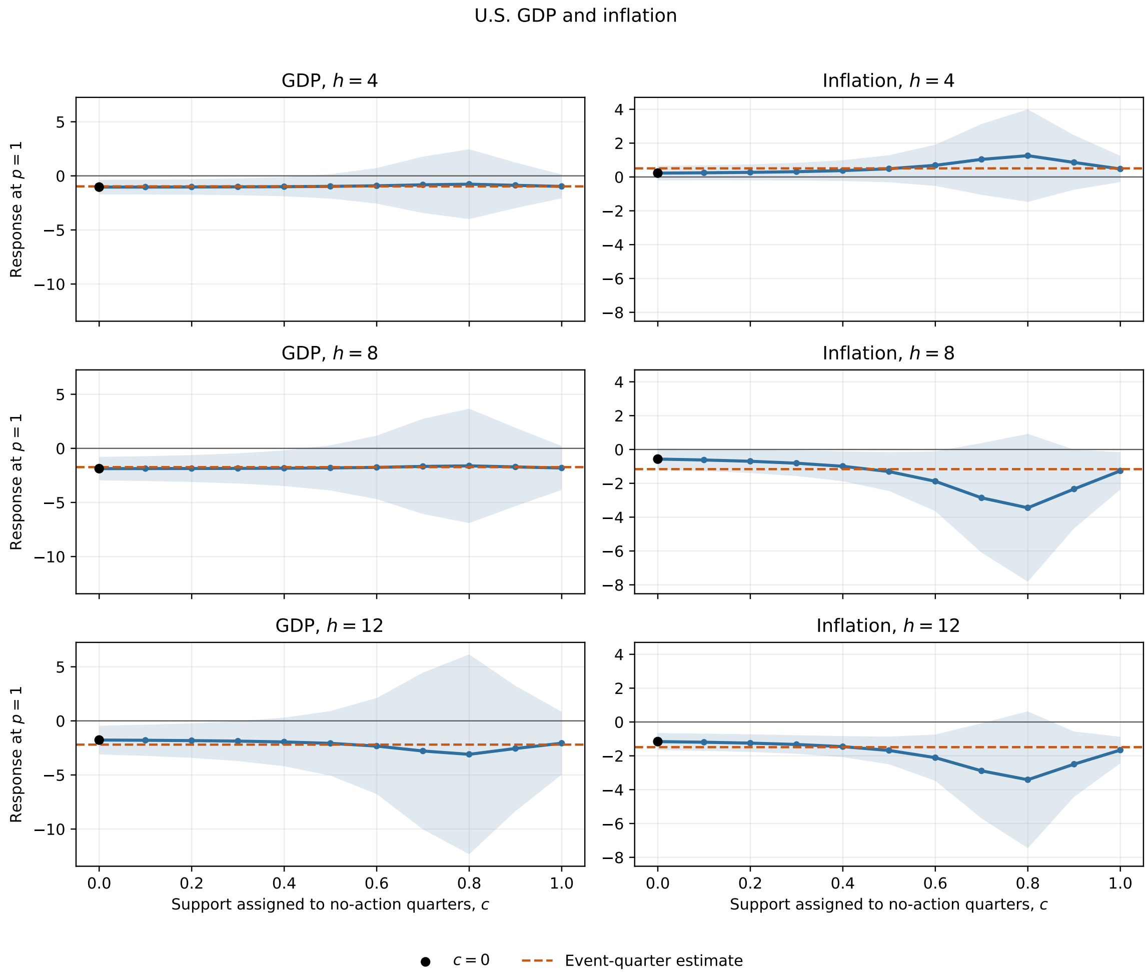


Figure 25: U.S. Responses under Alternative Placements of No-Action Quarters

Note: Each panel reports the response evaluated at $p = 1$ after assigning the common sensitivity value c to quarters without a recorded budget action. The black marker is the baseline $c = 0$ estimate. The orange dashed line is the estimate using recorded-action quarters only. Shaded areas are pointwise 68% HAC confidence bands.

Table 31 reports the principal mechanism variables under selected completions and recorded-action-quarter estimation. U.S. consumption is negative in each displayed specification. The investment contraction remains large at horizons 8 and 12 throughout the full c-grid, supporting the investment-heavy interpretation in the main text.

Table 31: Mechanism Responses under Alternative Treatments of No-Action Quarters

Application	Outcome and horizon	$c = 0$	$c = 0.5$	$c = 1$	Recorded actions
U.S. budget	Consumption, $h = 4$	-1.42	-1.32	-1.32	-1.45
	Consumption, $h = 8$	-1.87	-1.45	-1.47	-1.53
	Consumption, $h = 12$	-1.22	-0.78	-0.80	-0.92
	Investment, $h = 4$	-3.52	-2.63	-2.67	-2.60
	Investment, $h = 8$	-6.50	-8.02	-7.93	-7.55
	Investment, $h = 12$	-6.05	-9.99	-9.82	-10.41
UK–France VAT	Private consumption, $h = 4$	-1.61	-1.87	-0.68	-1.08
	Private consumption, $h = 8$	-2.47	-2.83	-1.02	-1.61
	Private consumption, $h = 12$	-3.01	-3.41	-1.23	-1.90

Note: The table reports responses evaluated at $p = 1$. "Recorded actions" restricts the support-indexed projections to quarters containing at least one recorded policy action while preserving calendar spacing for HAC inference. For the pooled VAT estimates, the fitted country-specific endpoint denominators fail the maintained positivity condition at $c = 0.9$, so no pooled response is reported at that value.

H.2 Pooled VAT application and supporting diagnostics

Figure 26 reports the corresponding pooled UK–France VAT paths. The recorded-action-quarter estimates and the completions $c = 0.5$ and $c = 1$ preserve the output contraction and the initial rise in inflation. Table 31 shows that private consumption also remains negative. Magnitudes are more sensitive than in the U.S. application, and the fitted country-specific endpoint denominators fail the maintained positivity condition at $c = 0.9$. The pooled VAT evidence therefore supports the qualitative mechanism, while also showing that its normalization is less stable under some high-support completions.

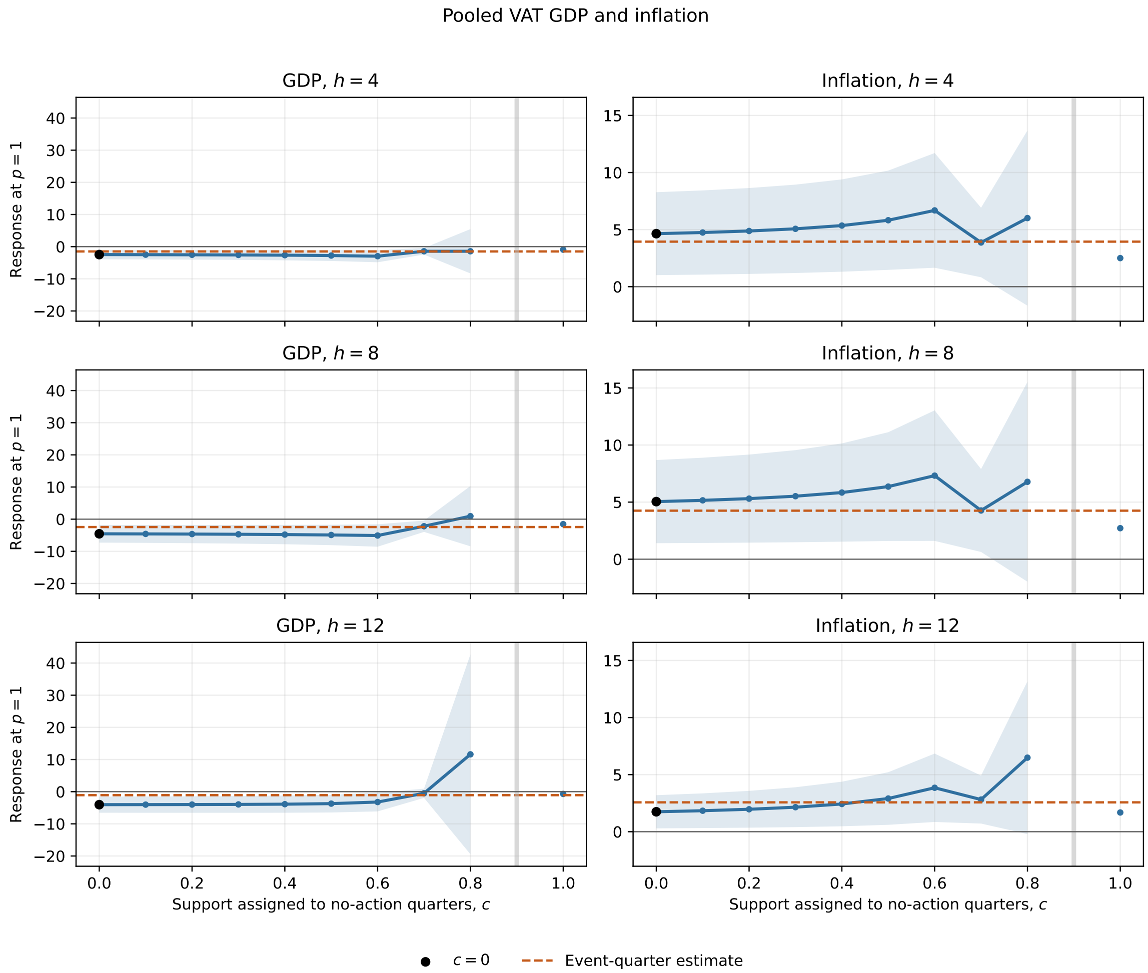


Figure 26: Pooled VAT Responses under Alternative Placements of No-Action Quarters

Note: Each panel reports the pooled UK–France response evaluated at $p = 1$ after assigning the common sensitivity value c to country-quarters without a recorded VAT action. The black marker is the baseline $c = 0$ estimate, and the orange dashed line is the estimate using recorded-action quarters only. Shaded areas are pointwise 68% HAC confidence bands. Grey markers identify values of c for which the fitted endpoint denominator fails the maintained positivity condition.

The sparse Romer–Romer and France-only exercises do not provide the same denominator stability. In the Romer–Romer sample, the fitted endpoint denominator is nonpositive over several interior values of c. Under recorded-action-quarter estimation, it is small relative to observed event-quarter tax variation, producing very imprecise estimates. For France alone, the fitted endpoint denominator is nonpositive under recorded-action-quarter estimation. I therefore treat these results as supporting diagnostics rather than as robustness evidence comparable to the main U.S. and pooled VAT applications.